

Judge IV Designs with Multiple Treatments and Instruments

Myunghyun Song*

This draft: July 28, 2026

[Latest version](#)

Abstract

This paper proposes a novel two-step procedure for estimating margin-specific treatment effects in judge instrumental variable (IV) designs under an unordered latent-return model of treatment assignment. The key feature of the first-stage model is that judge heterogeneity enters through interactions between observed case characteristics and judge-specific coefficients, while the distribution of the unobserved returns is left unspecified. We show that the judge-specific coefficients are identified without restricting this distribution, and establish that their sieve maximum likelihood estimator is semiparametrically efficient. In the second stage, we introduce a varying-coefficient outcome regression whose coefficients vary with the treatment propensity and the relative judge index, and pair it with a local instrumental-variables correction that recovers causally interpretable margin-specific effects. Revisiting a recent application of the judge IV design to recidivism using Virginia court records, we find that noncarceral conviction raises both noncarceral and carceral reoffending, with effects smaller than previously reported, while incarceration acts at least as strongly and channels reoffending almost entirely into new incarceration. The effects also vary with case characteristics, and for violent offenses incarceration appears to reduce recidivism outright.

*Department of Economics, Columbia University. Email: ms6347@columbia.edu.

Keywords: judge instrumental variable design, multivalued treatments, marginal treatment effects, sieve maximum likelihood, semiparametric efficiency.

JEL codes: C14, C21, C25, C26.

1 Introduction

Many real-world policies involve multiple treatments rather than a simple binary choice. Criminal courts, for instance, decide whether to dismiss a case, impose a noncarceral conviction, or incarcerate the defendant, while caseworkers choose among a range of social-service interventions. Policymakers often want to separate the effects of these treatments in order to understand both what each intervention accomplishes and the mechanisms through which it operates. Yet such questions are difficult to answer with observational data, because treatment assignment typically remains nonrandom even after conditioning on observed covariates. Researchers have increasingly exploited the quasi-random assignment of cases to decision makers such as judges as a source of exogenous variation in treatment. In criminal justice settings, this design, commonly known as the judge leniency design, has become a leading tool for studying the effects of carceral and noncarceral dispositions on offender rehabilitation and reentry.

This paper develops an econometric framework for estimating margin-specific treatment effects in judge instrumental variable (IV) designs. Our exposition focuses on settings with three mutually exclusive and exhaustive treatments, that is, dismissal, noncarceral conviction, and incarceration. The framework extends to settings with more than three alternatives. The parameter of interest is the effect of receiving one treatment rather than another for cases at the margin between the two. Identifying such an effect requires exogenous variation that lets us separate movements across one margin from movements across the others. The judge design does not supply such variation directly: judge identity is a single discrete instrument. Standard practice converts it into two leniency measures, the assigned judge’s propensities to convict and to incarcerate. However, neither measure shifts one margin alone. For example, a judge more prone to incarcerate typically draws cases into incarceration from dismissal and from noncarceral conviction at once. Two-stage least squares (2SLS) can use these measures in two ways, but neither escapes this difficulty. Instrumenting both treatment indicators together except the base treatment yields coefficients that are weighted sums of case-level treatment effects, in which the weights vary across cases, can be negative, and can attach to the effects on the other margin. [Bhuller and Sigstad \(2024\)](#) show that ruling out negative and mixed weights requires strong conditions on judge decisions. Instrumenting one treatment with its own propensity while controlling for another, the specification of [Humphries et al. \(2025\)](#), recovers a margin-specific

effect only if the propensity is margin-specific, a condition they test and reject in their data.

Our approach is constructive. Rather than assuming instruments with the required properties, we build them from the data. The inputs are the case characteristics that researchers observe and the random assignment of judges. Neither input is treatment specific on its own. An excluded regressor, a variable that shifts the appeal of a single treatment and leaves the others unchanged, is rarely available in judge IV settings, because each case characteristic bears on every disposition. We generate such variation by interacting the common covariates with the identity of the randomly assigned judge. The construction rests on a model of judge decisions. When deciding a case, a judge weighs what each disposition would accomplish, such as deterrence, the prospect of rehabilitation, and the cost of custody to the defendant and to society. We summarize this assessment by assigning each treatment a latent return. We assume the judge acts as if selecting the treatment with the highest return. Each return consists of a systematic component and an idiosyncratic component. The systematic component is an index of observed case characteristics whose coefficients vary with the judge and the treatment, which we call the *judge index* for that treatment. The idiosyncratic component, which we call the unobserved return, is a case-specific draw for each treatment. The joint distribution of these draws is left unrestricted. The first stage of our procedure estimates every judge’s coefficients and thereby recovers the judge indices. The procedure thus begins with covariates excluded from nothing and ends with an excluded index for each treatment.

The recovered judge indices have the properties that identification of margin-specific effects demands, and we refer to them as *judge index instruments*. They are exogenous conditional on case characteristics since an index varies only through the randomly assigned judge once the case characteristics are fixed. They are treatment-specific because each index enters the return to its own treatment alone. Raising one index thereby draws marginal cases into its treatment from the other two alternatives and moves no case between them. Section 2 shows that this property is equivalent to unordered partial monotonicity (Mountjoy, 2022), the condition that observed judge propensities generally fail.

The second stage uses these properties to form the margin-specific effects. For each margin, it runs an outcome regression within each of the two treatment groups, conditioning on case characteristics, the propensity of that group’s treatment, and

the index of that treatment relative to the treatment off the margin. Because cases select into treatment on their unobserved returns, these regressions are not causal on their own. A local instrumental variables argument, extending the representation of [Carneiro and Lee \(2009\)](#) to multiple treatments, corrects them for this selection. One requirement remains, and it is shared by designs that pursue marginal treatment effects with judge instruments: the correction calls for continuous variation in the instrument given case characteristics, which no instrument built from finitely many judges can supply. The varying-coefficient specification of the regressions trades this requirement for rich variation in the characteristics given the index instrument, a trade [Section 3](#) makes precise.

Two features of the first-stage model deserve emphasis, and both bear on the causal interpretation of the estimated effects. First, the systematic returns interact case characteristics with judge-specific slope coefficients rather than confining judge heterogeneity to intercepts. Judges may then differ not only in overall stringency but also in how they weight the characteristics of a case. Some judges are stringent on drug offenses yet lenient on property crimes, while others display the opposite pattern. The ranking of judges by leniency can accordingly change from case to case. The cases drawn across a given margin change with the judge, a substitution pattern that no specification in which judges differ only by intercepts can generate. The interactions also enrich the variation available to the second stage. Conditional on case characteristics, the cross-judge variation in the indices differs across cases, which spreads the propensities and relative indices over a wider domain than judge-specific intercepts alone would reach.

Second, the joint distribution of the unobserved returns is left unspecified and is estimated together with the coefficients. The identifying restrictions then fall on how observed characteristics enter the returns, where they are transparent and open to economic scrutiny, rather than on the shape of an unobservable distribution. The distribution nonetheless plays a substantive role: it translates the observed treatment probabilities into the locations of the margins. A parametric family such as the multinomial logit gains tractability by fixing this translation in advance. When the family is misspecified, the fitted model misplaces the margins, distorting both the constructed instruments and the selection-bias correction built on them. The empirical application gauges the practical extent of this concern.

The second stage is specified to match the parameter it estimates, a marginal

treatment effect on each margin. This is the average effect for cases indifferent between the two treatments, and it varies both with where the margin lies and with case characteristics. We therefore estimate margin-specific selected-outcome regressions with a varying-coefficient specification, in which the coefficients on case characteristics vary flexibly with the focal-treatment propensity and the relative judge index. The estimated effect is a function of both arguments: along the propensity it traces the marginal-treatment-effect curve on each margin, and across case characteristics it shows how the effect differs across cases. This is also where the approach differs from pooled 2SLS, which compresses the effect heterogeneity into a single coefficient averaged with weights the researcher does not control. The semiparametric second stage instead recovers the effect function itself and leaves the averaging to the researcher, who can weight it by the margin locations and cases that a policy question singles out. Our strategy also estimates a single derivative along the propensity rather than a ratio of two derivatives along the index, which avoids the instability a small denominator can introduce.

We apply the framework to the setting of [Humphries et al. \(2025\)](#), felony cases in the Virginia Circuit Courts. They report a sizable and persistent increase in recidivism from noncarceral conviction relative to dismissal, and a short-run reduction from incarceration relative to noncarceral conviction, which they attribute to incapacitation. They read these estimates as evidence that noncarceral conviction is an overlooked driver of reoffending. Our margin-specific estimates refine this picture. The three margins induce reoffending of different severity. A first conviction, relative to dismissal, quickly raises new charges and new noncarceral convictions, reoffending at the bottom of the severity ladder. Incarceration, relative to noncarceral conviction, barely changes whether the marginal offender is reconvicted, but it changes the form that reconviction takes. Noncarceral reconvictions fall while re-incarcerations rise slowly over years, a revolving door in which incarceration redirects reoffending into renewed incarceration rather than reducing it. Relative to dismissal, incarceration raises specifically re-incarceration, beyond a first-year dip consistent with incapacitation. The two stages contribute to these findings separately: the first stage determines where the margins lie and thus the size of the effects, while the second stage traces how the effects vary with case characteristics.

Recovering the judge-specific coefficients is the central identification problem of the paper, and existing results do not solve it. The judge is a single discrete instrument,

and no treatment carries an alternative-specific or excluded regressor. The standard routes to identifying a multinomial choice model therefore do not apply. [Lewbel \(2000\)](#) requires a special regressor of large support, and [Matzkin \(1993\)](#) requires alternative-specific regressors or errors that are i.i.d. across alternatives. Both secure identification through an exclusion among the regressors or a restriction on the distribution of the unobservables, and the judge design supplies neither. We rely instead on a shape restriction that the latent-return model itself implies, the symmetry of the propensity Jacobian. [Section 4](#) sets this argument against the remaining approaches, including multi-index results that identify the indices only up to their span.

We recover the coefficients by combining two kinds of variation. Within-judge variation across cases, through an equality of cross-partial derivatives of the propensity surface, pins down each judge’s coefficients up to scale and location. Cross-judge variation then places different judges on a common scale through an adjacency condition. Identification succeeds from two continuous covariates, fewer than the multi-index approaches require. Because the cross-partial argument is local, it identifies the coefficients from bounded covariates without requiring large support. This is meaningful in practice, where covariates are typically bounded.

The procedure is implemented in two stages, a maximum-likelihood first stage and a least-squares second stage, and we establish the statistical guarantees of each. The first stage estimates every judge’s coefficients by maximum likelihood, which yields the judge-index instruments. The complication is that the likelihood depends on the distribution of unobserved returns only through the propensity function, the map from the two indices to the treatment probabilities, and this function is unknown once the distribution is left unspecified. We therefore estimate the coefficients and the propensity function jointly, approximating the latter by a sieve, a finite-dimensional family whose flexibility grows with the sample. The sieve rests on a construction that, to our knowledge, is new and of independent interest, which we call the gradient B-spline because it represents the two propensities as the gradient of a single B-spline surface. Every function of this form satisfies the Jacobian symmetry that drives identification, so the restriction need not be imposed in estimation, and the remaining restrictions the model places on the propensities enter as linear constraints. The resulting estimator is consistent for the judge coefficients and, under mild conditions, attains the semiparametric efficiency bound at the parametric rate even though the distribution of unobserved returns is left unspecified. The second stage estimates the

selected-outcome regressions by least squares, approximating the varying coefficients by another sieve in the estimated propensity and relative index. The estimated indices and propensities enter these regressions as generated regressors, and the asymptotic theory accounts for the first-stage error they carry. We show that the margin-specific effect function is estimated uniformly consistently.

The first-stage estimator contributes to the literature on semiparametric estimation of discrete-choice and index models. That literature offers two routes to estimation with an unrestricted error distribution, neither of which applies in the judge design. One route separates the indices by reserving at least one covariate for each (Ichimura et al., 1991), whereas every case characteristic here enters the returns to both treatments. The other estimates the choice probabilities nonparametrically at a fixed value of the coefficients and then searches over the coefficients (Klein and Spady, 1993), which is impractical when the search runs over the coefficients of every judge at once. Its multinomial extension in Lee (1995) also identifies the coefficients through excluded regressors, and is therefore subject to the first constraint as well. The gradient B-spline sieve addresses both constraints. The symmetry built into the sieve separates the two indices without reserving a covariate for either, and updating the propensity function and the coefficients together avoids re-estimating the probabilities at each candidate value. No existing estimator handles this combination: an unrestricted error distribution, indices that share every covariate, and the coefficients of every judge estimated at once.

The paper contributes to the judge IV literature in criminal justice, which mostly collapses the dispositions into incarceration against everything else and therefore leaves the margins unseparated. Mueller-Smith (2015) studies incarceration against that composite, interacting judge identity with case characteristics to make monotonicity of the binary instrument credible rather than to separate margins. Among the papers that retain all three alternatives, Arteaga (2023) imposes a sequential structure on the decision and point-identifies the incarceration-conviction margin, while Kamat et al. (2024) reach the remaining margins under a latent-monotonicity condition that delivers bounds. We work instead with an unordered latent-return model. Rivera (2023), the direct antecedent of the interactive specification in our first stage, models the choice as ordered in the restrictiveness of supervision with one-dimensional unobserved heterogeneity, where ours is two-dimensional. Humphries et al. (2025) also work with an unordered latent-return model and are closest in overall design. Their first stage

confines judge heterogeneity to intercepts where we let the slopes vary and assumes a mixed logit where we leave the distribution unrestricted, and their second stage holds covariates fixed where we deliver effect functions that vary with them. Frandsen et al. (2023) test whether observed judge instruments satisfy the monotonicity and exclusion conditions, where we begin with the conditions and construct instruments to meet them.

The paper also speaks to the identification of treatment effects under multivalued treatments, where we follow the logic of Lee and Salanié (2018), who take the selection thresholds as known. The other routes to these effects rely on devices the judge design does not supply, the margin-specific continuous instruments of Mountjoy (2022) and the knowledge of next-best alternatives together with treatment-specific instruments in Kirkeboen et al. (2016). We recover the thresholds ourselves in the first stage, which extends the framework to designs in which no margin-specific instrument is observed.

The rest of the paper is organized as follows. Section 2 presents the latent-return model of judge decisions, defines the margin-specific effects of interest, and establishes their identification through a local instrumental-variables argument. Section 3 develops the two-step procedure, the sieve maximum-likelihood estimation of the first stage that recovers the judge indices and the varying-coefficient second stage that estimates the margin-specific effects. Section 4 establishes identification of the first-stage parameters. Section 5 establishes the large-sample properties of both stages. Section 6 applies the framework to felony cases in the Virginia Circuit Courts. Section 7 concludes. All technical proofs are collected in the Appendix.

2 Econometric Framework

We use the following notation. Let i index cases and j index judges. Each case is randomly assigned to a single judge, who then decides on the treatment for that case. We focus on settings with three mutually exclusive and exhaustive dispositions (treatments) $t \in \{d, c, n\}$, where d , c , and n denote dismissal, noncarceral conviction, and incarceration, respectively. Let $T_i \in \{d, c, n\}$ denote the observed treatment for case i , W_i the vector of observed case characteristics with $W_{1,i} \equiv 1$, J_i the judge assigned to case i , and Y_i the outcome of interest. We have the data $(Y_i, T_i, W_i, J_i)_{i=1}^N$ for N cases handled by J judges. For $t \in \{d, c, n\}$, we denote the corresponding treatment indicator by $T_i^t = \mathbb{1}\{T_i = t\}$.

Throughout the analysis, we assume that judges are randomly assigned across cases according to a common distribution. In practice, this complete randomization assumption may be plausible only within court or court-by-time cells. In such settings, the researcher can account for this structure by including court or court-by-time fixed effects in the specification. For simplicity, we maintain the complete randomization assumption in what follows.

2.1 Treatment Assignment Model in the First Stage

Let $T_i(j)$ be the treatment that case i would receive when assigned to judge j , so that the realized treatment is observed as $T_i = T_i(J_i)$. We posit a return-maximizing structure on treatment assignment, in which each treatment yields a latent return that depends on observed case characteristics and unobserved heterogeneity, and the assigned treatment is determined by the one with the highest return. Let $R_i^t(j)$ denote the latent return to treatment $t \in \{d, c, n\}$ under judge j . Then the treatment under j can be expressed as

$$T_i(j) = \arg \max_{t \in \{d, c, n\}} R_i^t(j).$$

We assume that the latent returns under judge j take the following form:

$$\begin{aligned} R_i^d(j) &= 0, \\ R_i^c(j) &= W_i' \gamma_j^c + U_i^c, \\ R_i^n(j) &= W_i' \gamma_j^n + U_i^n, \end{aligned} \tag{1}$$

where γ_j^t is a vector of parameters entering the t -treatment return that may vary with j , and $U_i = (U_i^c, U_i^n)$ is a vector of unobserved treatment-specific heterogeneity of case i invariant across judges. The return to treatment d is normalized to zero for all cases and judges by setting $\gamma_j^d = 0$ for all j and $U_i^d = 0$ for all i . This does not cause any loss of generality. We maintain this convention throughout the paper unless otherwise noted. The normalization also clarifies what the model parameterizes. A margin between two treatments is defined by their relative return. With three treatments, only two relative returns are free, since any one of them is determined by the other two. Under the normalization, $R_i^c(j)$ and $R_i^n(j)$ coincide with the relative returns of c and n against d , and the relative return between c and n is their difference. The model therefore parameterizes the margins themselves, a point we use when isolating

individual margins in Section 2.2.

The net return to treatment $t \in \{c, n\}$ consists of the systematic component $z_i^t(j) := W_i' \gamma_j^t$ and the idiosyncratic component U_i^t . The systematic component depends on the observed case characteristics through its interaction with the judge-specific loadings. We refer to this component as the *judge index* for treatment t , which captures the judge's inclination toward treatment t for case i . Analogously, we refer to the idiosyncratic component U_i^t as the *unobserved return* to treatment t , and to the vector U_i as the unobserved heterogeneity of case i . We assume that judge indices are linear in observed case characteristics with judge-specific intercepts and slope coefficients. We flag the scope of this assumption: linearity restricts only the systematic component of the returns, while the variation of the coefficients across judges and the distribution of the unobserved returns are left unrestricted. The intercept $\gamma_{j,1}^t$ captures the judge's baseline stringency level, while the slope coefficients $\gamma_{j,-1}^t = \gamma_j^t \setminus \{\gamma_{j,1}^t\}$ capture the judge's differential responsiveness to case characteristics. Let $\gamma_j = (\gamma_j^c, \gamma_j^n)$ collect judge-specific parameters across treatments c and n . Together, these vectors define the judge's latent type relevant for treatment decisions and are treated as unknown parameters to be estimated from the data.

The judge interactive-effects specification accommodates multidimensional heterogeneity in how judges evaluate cases as a function of observed characteristics. The judge fixed-effects specification is the special case in which only the intercepts $\gamma_{j,1}^t$ vary across judges while the slope coefficients are common. The first stage of [Humphries et al. \(2025\)](#) takes the fixed-effects form. Judge heterogeneity enters through treatment-specific thresholds, the analogue of our intercepts, while the coefficients on case characteristics are common across judges. Under common slopes, the ranking of judges by their inclination toward a treatment is the same for every case. Under interactive effects, this ranking may differ across cases. Which judge is more likely to convict or to incarcerate then depends on the case at hand. Similar interaction structures have been used in prior work to make instrument monotonicity more credible. See [Mueller-Smith \(2015\)](#) and [Rivera \(2023\)](#).

Two restrictions on the unobserved heterogeneity are built into this specification. We state them explicitly. First, U_i carries no judge subscript. The unobserved component of the returns therefore does not change with the judge who hears the case. What the analysis requires is in fact slightly weaker. Judge-specific heterogeneity U_{ij} can be accommodated provided the distribution of U_{ij} is the same for every judge j .

Because each case is decided by a single randomly assigned judge, only these judge-level marginal distributions can affect the observed data. The dependence of $(U_{ij})_j$ across judges is never restricted. The subscript j can then be dropped without loss. The substantive requirement is that judges face a common distribution of unobserved returns, not that the realizations coincide judge by judge.

Second, we assume that U_i is independent of the observed case characteristics W_i , and of the assigned judge J_i , the latter being implied by random assignment. Full independence between U_i and W_i is a substantive restriction. Interpreting U_i as the residual of the latent returns after conditioning on W_i motivates mean independence at most, whereas full independence further requires that no feature of the distribution of U_i , such as its dispersion, varies with case characteristics. We impose it because of what it delivers. Under independence, the treatment probabilities conditional on the case and the judge depend on (W_i, J_i) only through the judge indices, as shown in (2) below. The judge indices then form the sole systematic channel of treatment assignment. Continuous variation in W_i traces out the treatment propensities, while conditional on W_i the judge indices vary only through the random assignment of the judge and take finitely many values. This index sufficiency is what lets the first stage generate excluded regressors, treatment-specific return shifters, out of covariates that enter the returns to all treatments. In addition, we assume that U_i possesses a density, which is commonly assumed in discrete-choice settings to ensure that the probability of ties is zero. We denote the joint density of U_i by f , which is sufficiently smooth with moderate tails but otherwise unrestricted. In this respect the model departs from [Humphries et al. \(2025\)](#) on a second front: they specify a mixed-logit distribution for the unobserved returns, whereas we leave f unrestricted.

The dimension of U_i is what separates this first stage from its closest antecedent in the judge IV literature. [Rivera \(2023\)](#) models the judge’s choice among pretrial release, electronic monitoring, and detention as ordered in the restrictiveness of supervision. A scalar unobservable ranks the three alternatives and two judge-specific thresholds partition its range, estimated by probit regressions in which judge dummies interact with case characteristics. Under one-dimensional heterogeneity the map from thresholds to choice probabilities is a univariate monotone link, so its inversion is robust to the shape assumed for it and taking that shape to be normal costs little. Under two-dimensional heterogeneity no monotonicity protects the inversion. How the observed propensities map back to the judge indices depends on the joint distribution

of U_i , and a misspecified family relocates the margins. This is why f is left unrestricted here, and the same two-dimensionality creates the identification problem taken up in Section 4, which the ordered model’s single crossing avoids.

Once the judge-specific parameters are estimated, the judge indices can be reconstructed for any case and judge. The recovered judge indices serve as treatment-specific return shifters, whose variation reflects the random assignment of judges given observed case characteristics.

We exploit the panel structure of the data, in which each judge handles many cases, to identify γ_j for each judge j . The identification logic differs substantially from the standard method in discrete-choice settings, in which treatment-specific regressors are typically observed and exploited. Those regressors serve as numeraires for each t -treatment payoff, which by comparison identify the rate at which other regressors are traded off against them. In the judge-IV settings, by contrast, such observed numeraires are unavailable since all covariates enter the returns to all treatments through interactions with judge-specific coefficients. The absence of such treatment-specific regressors therefore calls for a new identification argument, especially when unobserved heterogeneity is unrestricted and nonparametrically specified.

2.1.1 Connection to Unordered Partial Monotonicity

The latent-return model in (1) is the primitive of our analysis. This subsection relates it to unordered partial monotonicity (UPM; Mountjoy, 2022), a set of monotonicity conditions on potential treatment assignment with respect to a pair of indices. We first state UPM and then show that it is equivalent to the return-maximizing structure, so that the economic content of the model, its argmax structure with additively separable returns, is captured exactly by UPM together with one further condition discussed below. The linearity of the systematic returns in W_i is a separate econometric restriction and falls outside the equivalence.

We postulate an idealized system of judge indices as primitive objects governing treatment assignment. We characterize the conditions under which it yields the latent-return representation in (1). Let $z^c(j)$ and $z^n(j)$ denote judge j ’s indices for treatments c and n , respectively. Ideally, these indices are treatment-specific in the following sense. Switching to a judge with higher $z^c(j)$ while holding $z^n(j)$ fixed shifts all cases weakly toward treatment c and does not shift any cases toward the remaining treatments, and analogously for $z^n(j)$. UPM formalizes this idea by imposing monotonicity conditions

on potential treatment assignment with respect to the judge indices.

Let $T_i(z) \in \{d, c, n\}$ be the potential treatment of case i under a judge with index vector $z = (z^c, z^n)$, and let $T_i^t(z) = \mathbb{1}\{T_i(z) = t\}$ denote the corresponding indicator for treatment $t \in \{d, c, n\}$.

Assumption 1. (Unordered Partial Monotonicity). *The following hold for all cases i .*

(i) *For any $z_2^c > z_1^c$ and z^n , it holds*

$$T_i^d(z_2^c, z^n) \leq T_i^d(z_1^c, z^n), T_i^c(z_2^c, z^n) \geq T_i^c(z_1^c, z^n), \text{ and } T_i^n(z_2^c, z^n) \leq T_i^n(z_1^c, z^n).$$

(ii) *For any $z_2^n > z_1^n$ and z^c , it holds*

$$T_i^d(z^c, z_2^n) \leq T_i^d(z^c, z_1^n), T_i^c(z^c, z_2^n) \leq T_i^c(z^c, z_1^n), \text{ and } T_i^n(z^c, z_2^n) \geq T_i^n(z^c, z_1^n).$$

Part (i) says that raising z^c while holding z^n fixed moves cases only into treatment c and only out of d and n , never into either. Part (ii) is the symmetric statement for z^n . Each index is thus specific to its own treatment. Raising it increases the return to that treatment while leaving the ranking among the remaining treatments unchanged. Treatment specificity does not, however, amount to margin specificity. Raising z^c moves cases across both margins involving treatment c , and since any relative return is determined by the other two, no shifter moves cases across a single margin alone. Isolating one margin instead requires holding the relevant relative index fixed, which is how the identification strategy of Section 2.2 proceeds. The treatment-specificity intuition can be formalized as an equivalence, which we record as a lemma.

Lemma 1 (Equivalence of UPM and Return Maximization). *A potential treatment assignment $T_i(z)$ satisfies Assumption 1 if and only if it is observationally equivalent to the solution to the return-maximization problem*

$$T_i(z) = \arg\max_{t \in \{d, c, n\}} R_i^t(z^t),$$

where $R_i^d(\cdot) \equiv 0$ is normalized without loss of generality, and $R_i^c(\cdot)$ and $R_i^n(\cdot)$ are case-specific increasing functions in their respective index arguments.

The nontrivial direction is due to [Loeser \(2024\)](#), who shows that any treatment assignment rule satisfying Assumption 1 admits a random-utility representation in which the ‘net utility’ of treatment c (resp. n) is an increasing function of z^c (resp. z^n).¹ The converse is immediate. Any such return-maximization rule with increasing $R_i^c(\cdot)$ and $R_i^n(\cdot)$ satisfies Assumption 1, since raising z^t raises only the return to treatment t .

To obtain the additive latent-return representation in (1), additional conditions are needed beyond UPM. [Loeser \(2024\)](#) shows that these conditions can be formulated in three equivalent ways: as an identical-compliers condition, a no-defiers condition, or an additive-random-utility-model (ARUM) condition.

For our purposes, these alternative formulations are useful for different reasons. The identical-compliers condition is closely related to the causal interpretation of the LIV estimands discussed in Section 2.2, because it ensures that LIV objects with respect to different indices recover mean potential outcomes for the same complier group under unrestricted outcome heterogeneity. The no-defiers condition has the most transparent behavioral interpretation. Restricted to cases that switch between c and n , it reduces to the standard monotonicity condition of [Imbens and Angrist \(1994\)](#). The ARUM condition, in turn, yields the additively separable form $R_i^t(z^t) = R^t(z^t) + U_i^t$, where $R^t(z^t)$ is a common function of z^t and U_i^t is an additive stochastic term. Since each common function $R^t(z^t)$ is increasing, it preserves the ordering of z^t . Relabeling $R^t(z^t)$ as the judge index therefore leaves treatment assignment unchanged and yields the representation in (1).

In sum, UPM together with any of the three equivalent conditions allows us to express the margin between treatments t and s as a single-index crossing of the relative judge-specific index $z^t - z^s$ and the case-specific threshold $U_i^s - U_i^t$. This pairwise single-crossing structure underlies the identification of margin-specific treatment effects, because it makes the relevant marginal comparison well-defined and amenable to the LIV strategy described in Section 2.2. Viewed against the judge IV literature, the equivalence also clarifies what the first stage contributes. It constructs, from common covariates and random judge assignment, the treatment-specific monotone variation whose absence causes 2SLS with leniency instruments to mix responses at distinct margins.

¹More precisely, this direction holds under additional technical assumptions, including the absence of cases that are nonresponsive to the index. See [Loeser \(2024\)](#) for details.

2.2 Identifying Mean Potential Outcomes at the Margin

Our primary interest lies in margin-specific treatment effects. This is the average effect of choosing one treatment rather than another for cases indifferent between the two treatments. This section presents the identification strategy for this class of parameters and the assumptions needed for identification of these parameters.

We begin by formulating the parameter of interest within the potential-outcomes framework. Let the judge-index instrument be defined as $Z_i = (Z_i^c, Z_i^n)$, where $Z_i^c = W_i' \gamma_{J_i}^c$ and $Z_i^n = W_i' \gamma_{J_i}^n$ denote the judge indices under the assigned judge J_i for treatments c and n , respectively. Let $Y_i(t, j)$ denote the potential outcome of case i under treatment t and judge j . We impose the standard exclusion restriction that judges affect outcomes only through treatment assignment. Under the exclusion restriction, potential outcomes are indexed by treatment alone. The observed outcome can therefore be written as $Y_i = Y_i(T_i)$.

Assumption 2. (Exclusion Restriction). *For each $t \in \{d, c, n\}$ and judge j , it holds that $Y_i(t, j) = Y_i(t)$ for all cases i .*

To identify margin-specific treatment effects, we additionally require that the judge-index instruments provide exogenous variation in treatment assignment. Since $Z_i = W_i' \gamma_{J_i}$ is formed by interaction between observed case characteristics and heterogeneity in the randomly assigned judge, this exogeneity condition must be formulated conditional on W_i .

Assumption 3. (Conditional Exogeneity of Judge-index Instrument). *It holds that*

$$(T_i(z), Y_i(t))_{z \in \text{supp}(Z_i), t \in \{d, c, n\}} \perp\!\!\!\perp Z_i \mid W_i.$$

The independence in Assumption 3 is joint, not marginal: the entire array of potential treatments and potential outcomes, viewed as a single random element indexed by z and t , is independent of Z_i given W_i . Joint independence is what the identification argument uses. Under the latent-return model, $T_i(z)$ is a deterministic function of z and U_i , so Assumption 3 is equivalent to $(U_i, Y_i(d), Y_i(c), Y_i(n))$ being independent of Z_i given W_i , and the identification results below condition on events defined jointly by U_i and the potential outcomes. Under the complete randomization assumption introduced at the beginning of Section 2, Assumption 3 follows because Z_i depends only on the randomly assigned judge and observed case characteristics.

Conditional on W_i , variation in Z_i is thus induced solely by the random assignment of J_i . In applications, randomization is typically conditional rather than complete. Section 6 treats assignment as good as random within cells defined by circuit and time of disposition, and the conditioning set is then the case characteristics together with the indicators for those cells. In practice, this means that the empirical strategy must condition on W_i throughout the analysis, since exogeneity of the judge-index instrument holds only conditional on observed case characteristics.

There are three margins in our setting: the (c, d) margin, the (n, c) margin, and the (n, d) margin. For any pair $t, s \in \{d, c, n\}$ with $t \neq s$, the (t, s) margin consists of cases for which the latent returns to treatments t and s are equal and both exceed the latent return to the remaining treatment $r \in \{d, c, n\} \setminus \{t, s\}$. Formally, for each $(z, w) \in \text{supp}(Z_i, W_i)$, define the (t, s) margin event by

$$\mathcal{I}_{ts}(z, w) = \{z^t + U_i^t = z^s + U_i^s > z^r + U_i^r, W_i = w\}.$$

In words, a case is at the (t, s) margin when t and s are its two best treatments and it is indifferent between them; the remaining treatment is strictly dominated and hence irrelevant to the choice. Stated this way, the extension beyond three treatments is immediate: with any number of alternatives, the (t, s) margin ties the top two returns at treatments t and s and requires every other return to fall strictly below. The strict inequality excludes the event that all three returns tie, and whether this boundary event is included is immaterial: the margin is a one-dimensional subset of the support of U_i , within which the triple tie is a single point and carries no mass under the localization defining the conditional expectation below. The margin-specific mean potential outcome under treatment t for cases at the (t, s) margin is defined by²

$$\mu_{ts}(z, w) = \mathbb{E}[Y_i(t) | \mathcal{I}_{ts}(z, w)].$$

The treatment effect of t relative to s on this margin is then given by $\tau_{ts}(z, w) = \mu_{ts}(z, w) - \mu_{st}(z, w) = \mathbb{E}[Y_i(t) - Y_i(s) | \mathcal{I}_{ts}(z, w)]$. As a function of (z, w) , $\tau_{ts}(z, w)$ is referred to as the margin-specific effect function.

Related notions of a margin appear in the literature, and ours sits between them in resolution. Heckman et al. (2008) study the margin between one treatment, or the

²Strictly speaking, $\mathcal{I}_{ts}(z, w)$ is a null event for $(z, w) \in \text{supp}(Z_i, W_i)$. The conditional expectation is understood with respect to the sigma-algebra generated by U_i together with W_i .

best treatment within a set, and the best alternative outside it. Their marginal case is indifferent between the two sides of an exhaustive partition of the treatments, and the comparison aggregates over whichever alternative happens to be next best. Our margin instead fixes both treatments in the pair. They reach this pairwise margin as well, through an identification-at-infinity argument that drives the remaining treatments out of contention and requires large support for the indices. [Lee and Salanié \(2018\)](#) define margins at fixed values of the unobservables, a finer notion than ours. On the (c, d) margin, for instance, our event is $\{U_i^c = -z^c, U_i^n < -z^n\}$, a half-line in the space of unobserved returns, whereas their margin fixes (U_i^c, U_i^n) at a point; our margin therefore aggregates a continuum of theirs. When their conditions hold, our margin-specific effects can accordingly be recovered from theirs, by averaging along the margin with weights proportional to the density of the unobserved returns. [Mountjoy \(2022\)](#) works with the same notion as ours, defined implicitly through the two-way indifference conditions that his LIV estimands localize. The definition above formalizes it and makes the role of the dominated treatment explicit. We return to this comparison at the end of this section, where the choice of margin is paired with the derivative through which each approach identifies its parameter.

2.2.1 Local Instrumental Variables (LIV)

The identification of margin-specific mean potential outcome functions can be obtained by the multivalued LIV framework in [Mountjoy \(2022\)](#). For estimation, however, it is more advantageous to work with a representation closer in form to Theorem 1 of [Carneiro and Lee \(2009\)](#), as it rewrites the relevant LIV object as an observed conditional mean plus a propensity-based derivative correction. This avoids the ratio of two derivatives that a direct implementation of the LIV framework would require, which typically yields heavy-tailed estimates when the denominator is small. Our approach instead yields a single derivative and a simple regression-based implementation that avoids heavy tails and associated moment issues.

Let the *treatment propensity* be defined by

$$P^t(z, w) = \mathbb{P}(T_i = t \mid Z_i = z, W_i = w), \quad t \in \{d, c, n\}.$$

Under the latent-return model, treatment t is chosen under z if and only if U_i belongs

to the argmax region defined by

$$\{u \in \mathbb{R}^2 : z^t + u^t \geq z^s + u^s \text{ for all } s \neq t\},$$

with the convention $z^d = 0$ and $u^d = 0$. Since U_i is independent of (W_i, J_i) , the conditional treatment probability given $(Z_i = z, W_i = w)$ depends only on z . Thus,

$$P^t(z, w) \equiv P^t(z|f) = \int_{\mathbb{R}^2} \mathbb{1}\{z^t + u^t \geq z^s + u^s, \forall s \neq t\} f(u) du, \quad (2)$$

which does not depend on w . For notational simplicity, we write $P^t(z) = P^t(z|f)$ as the t -treatment propensity function. Let $P_i^t = P^t(Z_i)$ denote the t -treatment propensity evaluated at the observed Z_i .

When f has full support in $\mathbb{R}^2$, the propensity map $(P^c(z), P^n(z))$ is injective in $z \in \mathbb{R}^2$ (see Lemma G.4 in Appendix). Moreover, under the latent-return model, $P^c(z)$ is strictly increasing in z^c and strictly decreasing in z^n , while $P^n(z)$ is strictly increasing in z^n and strictly decreasing in z^c . Accordingly, $P^d(z)$ is strictly decreasing in both arguments. This is useful in our analysis because, conditional on one index argument, variation in the other index can be reparametrized by variation in a treatment propensity. This allows LIV estimands to be expressed as a single derivative with respect to the propensity argument rather than ratios of two derivatives with respect to the judge index.

We now introduce the second-stage regression objects used to represent the LIV estimands. Fix a margin between treatments t and s , and let $r \in \{d, c, n\} \setminus \{t, s\}$ be the remaining treatment. Let $Z_i^{t,r} = Z_i^t - Z_i^r$ and $z^{t,r} = z^t - z^r$ denote the relative index instrument and the corresponding value between treatments t and r . The variables $Z_i^{s,r}$ and $z^{s,r}$ are defined analogously. Recall the convention that $Z_i^d \equiv 0$ and $z^d \equiv 0$.

Define the *margin-specific selected-outcome regression* by

$$m_{ts}(p, a, w) = \mathbb{E}[Y_i | T_i = t, P_i^t = p, Z_i^{t,r} = a, W_i = w], \quad t \in \{d, c, n\}.$$

This function gives the mean observed outcome for cases selected into treatment t , conditional on the treatment propensity, the relevant relative index, and observed covariates. In general, however, this selected-outcome regression does not equal the average potential outcome under treatment t , because cases observed in treatment

t remain selectively sorted even after conditioning on $(P_i^t, Z_i^{t,r}, W_i)$. Consequently, differences such as $m_{ts}(p^t, z^{t,r}, w) - m_{st}(p^s, z^{s,r}, w)$ do not by themselves admit a causal interpretation. The LIV estimand enters by correcting the selection-by-outcome for the selection into treatment t . In particular, the following identity holds:

$$\partial_p m_{ts}(p, z^{t,r}, w) \Big|_{p=P^t(z)} = \frac{1}{P^t(z)} \left(\underbrace{\mu_{ts}(z, w)}_{\text{mean potential outcome}} - \underbrace{m_{ts}(P^t(z), z^{t,r}, w)}_{\text{mean observed outcome}} \right).$$

The logic is as follows. When $P^t(z)$ increases while $z^{t,r}$ is held fixed, the comparison between treatments t and r remains unchanged, while treatment s becomes strictly less attractive relative to both. This is because the increase in $P^t(z)$ is induced by a decrease in $z^{s,r}$ and hence a decrease in the return to treatment s relative to both t and r . This induces movements along the (t, s) and (s, r) margins out of treatment s . However, because the selected-outcome regression conditions on $T_i = t$, only movements into or out of treatment t affect m_{ts} . The partial derivative of m_{ts} with respect to the propensity argument therefore isolates the selection component specific to the (t, s) margin. Multiplying this derivative by $P^t(z)$ yields the gap between the selected-outcome regression and the mean potential outcome under treatment t on that margin.

Applying the preceding logic to each (t, s) pair yields a collection of LIV mean-outcome estimands. Each estimand identifies the mean potential outcome under a given treatment for cases on a margin with another treatment, with the conditioning variables chosen to hold fixed the comparison involving the remaining treatment. The resulting equations are collected in the following proposition.

Proposition 1 (Margin-Specific Mean Potential Outcomes via LIV). *Under the model in (1), Assumptions 2 and 3, for any $t, s \in \{d, c, n\}$ with $t \neq s$, and any $(z, w) \in \text{supp}(Z_i, W_i)$, it holds that*

$$\mu_{ts}(z, w) = [m_{ts}(p, a, w) + p \partial_p m_{ts}(p, a, w)]_{p=P^t(z), a=z^{t,r}}$$

with $r = r(t, s)$ denoting the remaining treatment in $\{d, c, n\} \setminus \{t, s\}$.

Once we identify the selected-outcome means relevant to the (t, s) margin via

Proposition 1, the margin-specific treatment effect can be identified as

$$\begin{aligned}\tau_{ts}(z, w) = & \left(m_{ts}(P^t(z), z^{t,r}, w) - m_{st}(P^s(z), z^{s,r}, w) \right) \\ & + P^t(z) \partial_p m_{ts}(P^t(z), z^{t,r}, w) - P^s(z) \partial_p m_{st}(P^s(z), z^{s,r}, w). \quad (3)\end{aligned}$$

This result is closely related to, and extends, Theorem 1 in [Carneiro and Lee \(2009\)](#) to settings with multiple treatments and instruments. The key difference is that identification additionally requires controlling for the relative indices $z^{t,r}$ and $z^{s,r}$ in order to isolate variation on the (t, s) margin while holding fixed the comparison with the irrelevant remaining treatment r .

We close this section by placing the identification strategy in the literature. Two choices separate the available approaches: the notion of margin, which fixes the parameter of interest, and the form of the derivative through which that parameter is constructively identified. Table 1 summarizes both. [Mountjoy \(2022\)](#) considers the same margins as we do and identifies the complier means through ratios of derivatives with respect to the indices. A marginal increase in z^t draws cases into treatment t from both remaining treatments and moves no case between them, so the mean potential outcome of the compliers leaving treatment s is identified by the ratio $\partial_{z^t} \mathbb{E}[Y_i T_i^s | Z_i = z, W_i = w] / \partial_{z^t} \mathbb{E}[T_i^s | Z_i = z, W_i = w]$. The mean outcome under the destination treatment then follows by netting the two inflows. Proposition 1 identifies the same parameter through a different derivative. Conditional on the relative index $z^{t,r}$, variation in the remaining index is reparametrized by the focal propensity P^t , and the mean potential outcome becomes an observed regression plus a single derivative in the propensity argument, in the form of [Carneiro and Lee \(2009\)](#). No ratio is formed, so the construction is not subject to the small local denominators that ratio forms divide by. The remaining two rows of the table change the parameter rather than the derivative. [Heckman et al. \(2008\)](#) identify, through the propensity of the focal treatment, the effect of choosing it over the next best alternative, a margin defined by an exhaustive partition of the treatments rather than by a pair. For the fixed pairwise margin they instead rely on identification at infinity, which removes the remaining treatments by driving their indices arbitrarily low and requires large support. Proposition 1 reaches the pairwise margin without large support conditions, since conditioning on the relative index $z^{t,r}$ holds the comparison with the remaining treatment fixed rather than pushing that treatment out of the choice set. [Lee and](#)

Salanié (2018) identify effects at fixed values of the unobservables, the finer margins discussed in Section 2.2. The finer margin calls for a finer derivative. Their constructive identification differentiates to higher order, taking cross-partial derivatives directly in the instruments, of order matching the dimension of the unobservables, whereas ours requires a first-order derivative alone. Our margin notion and derivative choice thus pair the parameter that answers the empirical question, which treatment the marginal case should receive, with the construction that survives estimation in a judge design.

Table 1: Margin Notions and Constructive Identification across Approaches

	Margin (parameter of interest)	Identifying derivative
Heckman et al. (2008)	one treatment vs the next-best alternative	propensity of the focal treatment
Lee and Salanié (2018)	at fixed unobserved returns	higher order, cross-partial in the instruments
Mountjoy (2022)	top-two indifference, pairwise	ratios of derivatives in the indices
This paper	top-two indifference, pairwise	first order, single, in the focal propensity

3 Estimation

3.1 Estimation in the First Stage

Let $D_{j,i} = \mathbb{1}\{J_i = j\}$ be the indicator for whether case i is assigned to judge j , and let $D_i = (D_{1,i}, \dots, D_{J,i})'$ be the vector of judge indicators. Let $X_i = D_i \otimes W_i$ be the $J \dim(W_i)$ -vector of regressors obtained by interacting the judge indicators with the case characteristics, where $\otimes$ denotes the Kronecker product. Let $t \in \{c, n\}$ be a non-default treatment and write $\gamma^t = (\gamma_1^t, \dots, \gamma_J^t)' \in \mathbb{R}^{\dim(X_i)}$ for the stack of judge-specific parameters for treatment t . Then the latent return for treatment t underlying the observed treatment assignment $T_i(J_i) = \arg\max_{t \in \{d, c, n\}} R_i^t(J_i)$ can be expressed as

$$R_i^t(J_i) = W_i' \gamma_{J_i}^t + U_i^t = \sum_{j=1}^J D_{j,i} W_i' \gamma_j^t + U_i^t = X_i' \gamma^t + U_i^t.$$

Let $\boldsymbol{\gamma} = (\gamma^c, \gamma^n)$ be the $\dim(X_i) \times 2$ matrix whose columns are the vectors of first-stage parameters across the two non-default treatments. This allows us to write $(X_i' \gamma^c, X_i' \gamma^n) = \boldsymbol{\gamma}' X_i$ as the vector of the two latent returns for the non-default treatments.

Let $\boldsymbol{\gamma}_o$ denote the true value of $\boldsymbol{\gamma}$ and f_o denote the true density of U_i , which we take to belong to the class $\mathcal{F}$, the collection of all continuous densities f that are everywhere positive on $\mathbb{R}^2$ and satisfy the tail condition $\sup_{u \in \mathbb{R}^2} (1 + \|u\|)^{1+\delta} |f(u)| < \infty$ for some $\delta > 0$. Accordingly, let $\gamma_{oj} = (\gamma_{oj}^c, \gamma_{oj}^n)$ denote the true parameter matrix for judge j . Let $P_o^t(z) = P^t(z | f_o)$ be the true t -treatment propensity as a function of the judge indices $z = (z^c, z^n)$. Since U_i is independent of X_i , the model implies that the observed treatment assignment follows a trinomial distribution

$$T_i | X_i \sim \text{MNL} \left(P_o^t(\boldsymbol{\gamma}'_o X_i) \right)_{t \in \{d, c, n\}}. \quad (4)$$

The model pairs a finite-dimensional parameter $\boldsymbol{\gamma}_o$, our chief object of interest, with an infinite-dimensional nuisance $P_o = (P_o^c, P_o^n)$. The model is therefore semiparametric, and the first-stage estimation problem is to estimate $\boldsymbol{\gamma}_o$ and P_o from the observed data $\{(T_i, X_i)\}_{i=1}^N$ under the trinomial model (4).

3.1.1 Sieve Maximum Likelihood Estimation

The conditional distribution (4) implies a log-likelihood function of the form

$$l_i(\theta) = l(T_i | X_i, \theta) = \sum_{t \in \{d, c, n\}} T_i^t \log P^t(\boldsymbol{\gamma}' X_i), \quad \theta = (\boldsymbol{\gamma}, P) \in \Theta = \Gamma \times \mathcal{P},$$

where Γ is the parameter space for $\boldsymbol{\gamma}$ and $\mathcal{P}$ is the class of admissible propensity functions, both of which are defined in the next two sections. We define the sieve maximum likelihood estimator (SMLE) as the solution to the following problem:

$$\hat{\theta} = (\hat{\boldsymbol{\gamma}}, \hat{P}) = \arg \max_{\theta \in \Theta_N} \mathcal{L}_N(\theta), \quad \Theta_N = \Gamma \times \mathcal{P}_N, \quad (5)$$

where $\mathcal{L}_N(\theta) = N^{-1} \sum_{i=1}^N l_i(\theta)$ is the empirical criterion, and $(\mathcal{P}_N)_{N=1}^\infty$ is a sequence of sieve spaces approximating the function class $\mathcal{P}$ with dimension increasing with the sample size N . We adopt a sieve approach to estimation because the infinite-dimensional nature of $P(\cdot)$ precludes direct maximization of the likelihood over $\mathcal{P}$.

In the SMLE problem (5), we directly approximate the propensity function $P_o(\cdot) = P(\cdot | f_o)$ rather than through its underlying density f_o . The density f_o is not of primary interest in estimation for two reasons. First, f_o is not directly related to the second-stage estimation problem, which relies on the propensity function $P_o(\cdot)$ rather than f_o . Second, f_o generally fails to be point-identified when the observed covariates have compact support, as is typically the case in empirical applications. This is intuitive because case i with a sufficiently high value of U_i^c would be assigned to treatment c regardless of W_i and J_i , and similar logic applies to cases with sufficiently high values of U_i^n as well. The upper tails of f_o therefore have no effect on the observed treatment propensities and hence cannot be identified from the data. The function $P_o(\cdot)$, by contrast, can be identified on a compact region. This is precisely the setting sieves handle best.

3.1.2 The Parameter Space

The parameter space for γ is taken to be a compact subset of $\mathbb{R}^{\dim(X_i) \times 2}$. Let $e_J \in \mathbb{R}^{\dim(X_i)}$ be the vector with ones in the J intercept coordinates and zeros elsewhere, so that $e_J' X_i \equiv 1$, and define

$$\Gamma = \{\gamma \in \mathbb{R}^{\dim(X_i) \times 2} : \|\gamma\|_F = \varrho, e_J' \gamma^c = e_J' \gamma^n = 0\} \quad \text{for some } \varrho > 0.$$

Here, $\|A\|_F := (\text{tr}(A'A))^{1/2}$ denotes the Frobenius norm of a matrix A . Some normalization is unavoidable, because the model is invariant to two transformations: translating the intercepts $(\gamma_{j,1})_{j=1}^J$ and U_i in offsetting directions, and rescaling γ and U_i by a common positive constant. Neither changes the ranking of latent returns. Neither, therefore, changes the likelihood. The restrictions $e_J' \gamma^c = e_J' \gamma^n = 0$ and $\|\gamma\|_F = \varrho$ therefore pin down the unique value of γ_o by fixing its location and scale. By doing so, the indeterminacy of $P_o(\cdot)$ is also resolved, and the true value $\theta_o = (\gamma_o, P_o)$ can be point-identified. In practice, one could instead adopt a different normalization since one normalization can be recovered from another by a suitable one-to-one correspondence. For instance, a scale normalization by $\varrho = 2$ can be recovered from a scale normalization by $\varrho = 1$ by simply multiplying the latter by 2. The location normalization sets the unweighted cross-judge mean of the intercepts to zero. This choice is likewise immaterial: any linear restriction $\ell' \gamma^c = \ell' \gamma^n = 0$ with $\ell' e_J \neq 0$ determines the intercepts uniquely, and one moves between such normalizations by a

computable shift. Henceforth, we use $\varrho = 1$ without loss of generality.

3.1.3 The Function Class

We now describe the nuisance function class $\mathcal{P}$ for the propensity function $P = (P^c, P^n)$. The function $P(\cdot)$ here includes only the two non-default treatment propensity functions since $P^d(\cdot) = 1 - P^c(\cdot) - P^n(\cdot)$ can be represented in terms of the other two by the trinomial structure of the model.

Let $\mathcal{Z} = \text{supp}(\gamma'_o X_i)$ denote the support of the index instrument $Z_i = \gamma'_o X_i$ at the true value γ_o . We assume that $\mathcal{Z}$ is a compact subset of $\mathbb{R}^2$ with dense interior. In many empirical applications, observed covariates are typically bounded and the norm of γ_o is restricted to be a constant. The support of Z_i is therefore naturally compact. The set $\mathcal{Z}$ serves as the domain of definition for $P(\cdot)$, and the compactness of $\mathcal{Z}$ is a standard condition that facilitates the sieve approximation of these functions. Let $\bar{\mathcal{Z}}$ be a compact rectangle containing $\mathcal{Z}$, which will be defined more precisely in the next section when we specify the sieve space. Let $C^1(\bar{\mathcal{Z}})$ denote the class of continuously differentiable functions $P : \bar{\mathcal{Z}} \rightarrow \mathbb{R}^2$. We define the function class $\mathcal{P} = \mathcal{P}_{M,\kappa}$ as follows: for a sufficiently small $\kappa > 0$ and a sufficiently large $M > 0$,

$$\begin{aligned} \mathcal{P} = \Big\{ P \in C^1(\bar{\mathcal{Z}}) : & \text{for all } z \in \bar{\mathcal{Z}}, P^c(z), P^n(z) \geq \kappa, 1 - P^c(z) - P^n(z) \geq \kappa, \\ & \partial_{z^n} P^c(z) = \partial_{z^c} P^n(z), \partial_{z^n} P^c(z) \leq 0, \\ & 0 \leq \partial_{z^c} P^c(z), \partial_{z^n} P^n(z) \leq M, \\ & \partial_{z^c} P^c(z) + \partial_{z^n} P^c(z) \geq 0, \quad \partial_{z^n} P^n(z) + \partial_{z^c} P^n(z) \geq 0 \Big\}, \end{aligned}$$

where $DP(z)$ is the Jacobian matrix of $P(\cdot)$ at z , with diagonal entries $\partial_{z^c} P^c(z)$ and $\partial_{z^n} P^n(z)$ and common off-diagonal entry $\partial_{z^n} P^c(z) = \partial_{z^c} P^n(z)$. The function class $\mathcal{P}$ collects four groups of regularity conditions on the propensity function P . Each is a property shared by every structural propensity $P(\cdot|f)$ induced by $f \in \mathcal{F}$. $\mathcal{P}$ therefore relaxes the structural class $\{P(\cdot|f) : f \in \mathcal{F}\}$ to all functions that share these properties, and is semi-structural in this sense. In particular the true propensity $P_o = P(\cdot|f_o)$ belongs to $\mathcal{P}$ with a sufficiently small κ and a sufficiently large M . The first set of conditions require that the propensity for each treatment is uniformly bounded away from zero by a constant $\kappa > 0$, referred to as the overlap condition. This is necessary to ensure that the likelihood and its score are well defined since the

logarithm and reciprocal of treatment propensities are involved. It is also a condition for identifying margin-specific treatment effects in the second stage, as it relies on each t -treatment group exhibiting the common support of Z_i .

The second condition requires that P is a sufficiently smooth function of the indices by imposing continuous differentiability along with the bound M on the diagonal entries of its Jacobian, $\partial_{z^c} P^c(z), \partial_{z^n} P^n(z) \leq M$, which together with the diagonal dominance below bounds the whole of $DP(z)$. Such smoothness conditions are technically required to ensure that $\mathcal{P}$ is compact so that nonparametric optimization over $\mathcal{P}$ is well-posed and that the likelihood function responds smoothly to changes in γ .

A smoothness condition on the link function $P(\cdot)$ is essential, as in any two-index regression model, where the index coefficients are identified through the gradient of the link with respect to the indices. In a generic two-index model, however, only the span $\{a\gamma^c + b\gamma^n : a, b \in \mathbb{R}\}$ is identified, because index sufficiency is invariant to invertible linear transformations of the indices. Smoothness alone therefore cannot separate γ^c from γ^n . What breaks this invariance here is the third condition, the cross-partial equality $\partial_{z^n} P^c = \partial_{z^c} P^n$. It is not an additional assumption. As shown below, it is forced by the latent-return structure through the Williams–Daly–Zachary representation. A generic two-index model does not impose it, and it is precisely this structurally implied restriction—once built into $\mathcal{P}$ —that pins down γ^c and γ^n separately rather than only their span.

Williams–Daly–Zachary Representation. The cross-partial condition ensures that the propensity function $P(\cdot)$ arises as a gradient of a scalar function, which we refer to as the *social surplus function* and denote by $G(\cdot)$. To be more precise, let the social surplus function induced by a density f be defined as

$$G(z|f) := \int_{\mathbb{R}^2} \max_{t \in \{d, c, n\}} (z^t + u^t) f(u) du = \mathbb{E}_{U_i \sim f} [\max(0, z^c + U_i^c, z^n + U_i^n)], \quad z \in \mathbb{R}^2$$

with the convention $z^d = 0$ and $u^d = 0$. According to the Williams–Daly–Zachary theorem (McFadden, 1981), the propensity function $P(\cdot) = P(\cdot|f)$ can be represented by the gradient of the social surplus function $G(\cdot|f)$ as

$$P(z|f) = \nabla G(z|f) = \left(\frac{\partial G(z|f)}{\partial z^c}, \frac{\partial G(z|f)}{\partial z^n} \right)' \quad \text{for a.e. } z \in \mathbb{R}^2.$$

If $G(\cdot|f)$ is twice continuously differentiable, by the symmetry of mixed partial derivatives, the cross-partial derivatives of the two non-default treatment propensity functions must satisfy the equality

$$\partial_{z^n} P^c(z) = \frac{\partial^2 G(z|f)}{\partial z^n \partial z^c} = \frac{\partial^2 G(z|f)}{\partial z^c \partial z^n} = \partial_{z^c} P^n(z),$$

which is the third condition in the definition of $\mathcal{P}$. This cross-partial equality is sometimes called *Slutsky symmetry*, by analogy with consumer theory. The propensity function is the gradient of the surplus function just as compensated demand is the gradient of the expenditure function. Its Jacobian therefore inherits the symmetry of a Hessian in the same way as the Slutsky matrix. We use this term in the sequel. This relationship allows us to represent the propensity function $P(\cdot)$ in terms of a single scalar function $G(\cdot)$, which is useful for deriving a suitable sieve space in the following section.

The fourth group of conditions restricts the entries of the Jacobian $DP(\cdot)$: the own-treatment effects $\partial_{z^c} P^c$ and $\partial_{z^n} P^n$ are nonnegative, the cross effect $\partial_{z^n} P^c = \partial_{z^c} P^n$ is nonpositive, and each own effect dominates the cross effect in absolute value, $\partial_{z^c} P^c \geq |\partial_{z^n} P^c|$ and $\partial_{z^n} P^n \geq |\partial_{z^c} P^n|$. Every structural propensity satisfies them. For $f \in \mathcal{F}$, Lemma G.3 shows that $DP(\cdot|f)$ exhibits this sign pattern and that the dominance gap $\partial_{z^c} P^c - |\partial_{z^n} P^c|$ equals the marginal response of the default propensity, which is strictly positive. Together with the Slutsky symmetry, these conditions make $DP(z)$ symmetric and diagonally dominant with a nonnegative diagonal, hence positive semidefinite at every z . This positive semidefiniteness distinguishes $P(\cdot)$ from its orientation-reversed image $P(-\cdot)$, whose Jacobian $-DP(-\cdot)$ is negative semidefinite. Without it, the alias $(-\gamma_o, P_o(-\cdot))$ would be observationally equivalent to (γ_o, P_o) , leaving the sign of γ_o unidentified. Imposing it removes the alias and forces the within-judge alignment to a positive scaling.

3.1.4 The Sieve Space

While there are various choices of the sieve that can be used to approximate $\mathcal{P}$, we focus on the *gradient* B-spline sieve for its theoretical and practical advantages. This section presents the construction of the gradient B-spline basis and the resulting finite-dimensional optimization problem for computing the SMLE.

Let $q \geq 4$ be the order of the B-spline basis, which ensures that the sieve defined

below is contained in the function class $\mathcal{P}$. Let $\bar{\mathcal{Z}} = [\underline{z}^c, \bar{z}^c] \times [\underline{z}^n, \bar{z}^n]$ be a compact rectangle containing the index support $\cup_{\gamma \in \Gamma} \text{supp}(\gamma' X_i)$, referred to as the knot box.³ For a chosen integer $K \geq 1$, let $\underline{z}^c = \eta_0^c < \eta_1^c < \dots < \eta_K^c = \bar{z}^c$ and $\underline{z}^n = \eta_0^n < \eta_1^n < \dots < \eta_K^n = \bar{z}^n$ be sequences of knots for the c -th and n -th dimensions, respectively. For each knot sequence, we construct the corresponding univariate B-spline basis functions of order q as $b_K^c(z^c) = \{b_{K,k}^c(z^c)\}_{k=1}^{K+q}$ and $b_K^n(z^n) = \{b_{K,k}^n(z^n)\}_{k=1}^{K+q}$. Then we construct the bivariate B-spline basis functions on $\bar{\mathcal{Z}}$ as the tensor product of the univariate bases:

$$b_K(z) = \text{vec}(\{b_{K,k_1}^c(z^c)b_{K,k_2}^n(z^n)\}_{k_1,k_2=1}^{K+q}) \in \mathbb{R}^{d_K},$$

where $d_K = (K + q)^2$ is the dimension of the bivariate basis. We denote $b_K(\cdot) = (b_{K,1}(\cdot), \dots, b_{K,d_K}(\cdot))'$ as the vector of the bivariate B-spline basis functions.

The B-spline basis functions above are not directly used to approximate the propensity function $P(\cdot)$ since the latter is a gradient of the social surplus function $G(\cdot)$. Instead, we consider the collection of gradients of the B-spline series, which is given by

$$\text{GradSpl}(K) = \left\{ \nabla b_K(\cdot)' \xi_K \in \mathcal{P} : \xi_K \in \mathbb{R}^{d_K} \right\}.$$

The gradient B-spline sieve $(\text{GradSpl}(K))_{K \geq 1}$ forms a dense subset in $\mathcal{P}$ under quasi-uniform knots. The admissible coefficient space for $\xi_K = (\xi_{K,1}, \dots, \xi_{K,d_K})'$ is the convex set

$$\begin{aligned} \Xi(K) = \left\{ \xi_K : P_{\xi_K}^c(z), P_{\xi_K}^n(z) \geq \kappa, P_{\xi_K}^c(z) + P_{\xi_K}^n(z) \leq 1 - \kappa, \right. \\ \partial_{z^n} P_{\xi_K}^c(z) \leq 0, 0 \leq \partial_{z^c} P_{\xi_K}^c(z), \partial_{z^n} P_{\xi_K}^n(z) \leq M, \\ \left. \partial_{z^c} P_{\xi_K}^c(z) + \partial_{z^n} P_{\xi_K}^c(z) \geq 0, \partial_{z^n} P_{\xi_K}^n(z) + \partial_{z^c} P_{\xi_K}^n(z) \geq 0, \forall z \in \bar{\mathcal{Z}} \right\}. \end{aligned}$$

Here, $P_{\xi_K}^t(z) = \partial_{z^t} b_K(z)' \xi_K$ is the sieve propensity for treatment t , and $\partial_{z^s} P_{\xi_K}^t(z)$ is the corresponding entry of the sieve Jacobian $DP_{\xi_K}(z) = \nabla^2 b_K(z)' \xi_K$. All of these constraints are linear in ξ_K , and hence $\Xi(K)$ is a polytope. In principle, one can enforce these constraints on a sufficiently fine grid of points in $\bar{\mathcal{Z}}$. In our iterative

³The set $\cup_{\gamma \in \Gamma} \text{supp}(\gamma' X_i)$ is compact as a continuous image of $\Gamma \times \text{supp}(X_i)$ under the mapping $(\gamma, x) \mapsto \gamma' x$, and contains the true index support $\mathcal{Z}$; unlike $\mathcal{Z}$, it does not depend on the unknown γ_o and can be computed from Γ and the data. Since every realized index $\gamma' X_i$ with $\gamma \in \Gamma$ lies in $\bar{\mathcal{Z}}$, the criterion below is well-defined over all of Γ . Computationally it is enough that the indices visited at each iteration lie in $\bar{\mathcal{Z}}$ since empty cells do not contribute to the likelihood.

algorithm, we use the Dykstra's algorithm to alternately project the unconstrained gradient B-spline coefficients onto the polytope $\Xi(K_N)$ at each iteration.

The positive constants κ and M are assumed to be such that $P_o(\cdot)$ lies in the interior of $\mathcal{P}$ and hence can be approximated by the sieve without hitting the boundary. These constants are theoretical devices rather than tuning parameters. They do not govern the convergence rate of the SMLE, intuitively because the constraints become inactive in the limit as $N \rightarrow \infty$ as long as $P_o(\cdot)$ is in the interior of $\mathcal{P}$. In practice, they are chosen to be sufficiently small and large, respectively, so that the constraints are inactive at the SMLE solution.

The cross-partial equality is automatically satisfied by the gradient structure of the sieve and need not be imposed. Since by the properties of the B-spline $\nabla b_K(\cdot)'1 \equiv 0$ where 1 is a d_K -vector of ones, we may impose $\xi_{K,d_K} = 0$ without loss of generality. Thus the effective dimension of the sieve space is at most $d_K - 1$ when all knot cells are loaded. In practice, the effective dimension is often much smaller than $d_K - 1$ because only a fraction of the knot cells are typically active. Notably, both P^c and P^n are approximated via a common coefficient vector ξ_K since they are gradients of the same social surplus function.

Let $K_N \rightarrow \infty$ be a sequence of integers increasing with the sample size N at a suitable rate, and let $\mathcal{P}_N = \text{GradSpl}(K_N)$ be the sieve space for estimation. Let $P_\xi(\cdot) = \nabla b_{K_N}(\cdot)'\xi$ denote the propensity function corresponding to a coefficient vector $\xi \in \Xi(K_N)$. Since $\bar{\mathcal{Z}} \supseteq \cup_{\gamma \in \Gamma} \text{supp}(\gamma'X_i)$ by construction, the index $\gamma'X_i$ lies in $\bar{\mathcal{Z}}$ for every $\gamma \in \Gamma$ and i . $P_\xi(\gamma'X_i)$ is therefore well-defined throughout Γ . The semiparametric MLE problem (5) is then converted into

$$(\hat{\gamma}, \hat{\xi}) = \arg \max_{\gamma \in \Gamma, \xi \in \Xi(K_N)} \mathcal{L}_N(\gamma, P_\xi), \quad (6)$$

which is a finite-dimensional optimization problem over $\Gamma \times \Xi(K_N)$.

The estimator can be placed against three semiparametric alternatives. The single- and multiple-index least-squares estimators of Ichimura (1993) and Ichimura et al. (1991) do not carry the cross-partial equality, and it is this restriction that makes each index coefficient estimable here rather than only the span of the two. The estimator is closest to the semiparametric maximum likelihood of Klein and Spady (1993) and its multinomial extension in Lee (1995). All three estimate the link rather than assume it and attain the semiparametric efficiency bound in their respective models. Ours differs

in the gradient B-spline sieve in place of kernel smoothing, the shape restrictions the sieve carries, and the joint updating of the link and the coefficients, which replaces the profiling step that is infeasible when γ collects every judge’s coefficients. Finally, the maximum score estimator of Manski (1975) dispenses with the link function entirely, but it requires the unobserved returns to be independent and identically distributed across treatments, a restriction our model avoids.

3.2 The Outcome Specification in the Second Stage

The identification results of Section 2.2 motivate a selected-outcome-regression approach to estimating margin-specific treatment effects. Implementing this approach raises three main challenges. First, the propensity function $P_o(\cdot)$ and the index instrument Z_i are not directly observed and must be estimated in the first stage, as described in Section 3.1. Second, the derivative $\partial_p m_{ts}$ in Proposition 1 is nonparametrically identified only if Z_i varies continuously conditional on W_i , whereas Z_i takes finitely many values given W_i , one per judge. This difficulty arises in designs that pursue marginal treatment effects with instruments built from judge assignment (Brinch et al., 2017; Kamat et al., 2024). Third, even where identification is secured, fully nonparametric estimation is infeasible in practice because W_i is typically moderate- to high-dimensional. This subsection introduces a semiparametric specification for the second-stage regressions that addresses the second and third challenges.

Let the (t, s) margin be of interest, and let $r \in \{d, c, n\} \setminus \{t, s\}$ be the remaining treatment. The objective is to recover the margin-specific effect function $\tau_{ts}(\cdot)$ using (3). This requires estimation of the selected-outcome regressions $m_{ts}(\cdot)$ and $m_{st}(\cdot)$, together with their partial derivatives with respect to the propensity arguments. Estimating these objects in a fully nonparametric manner is often infeasible due to the curse of dimensionality, while imposing fully parametric restrictions may be too rigid and lead to substantial misspecification bias.

As a flexible yet tractable compromise, we adopt a semiparametric varying-coefficient specification for the selected-outcome regressions. Specifically, we model each regression as linear in W_i with coefficients varying flexibly with the relevant treatment propensity and the relative index:

$$\begin{aligned} m_{ts}(P_i^t, Z_i^{t,r}, W_i) &= W_i' \beta_{ts}(P_i^t, \tilde{Z}_i^{t,r}), \\ m_{st}(P_i^s, Z_i^{s,r}, W_i) &= W_i' \beta_{st}(P_i^s, \tilde{Z}_i^{s,r}). \end{aligned} \tag{7}$$

Here, $\beta_{ts}(\cdot, \cdot)$ and $\beta_{st}(\cdot, \cdot)$ are vector-valued functions of the same dimension as W_i , and $\tilde{Z}_i^{t,r} = Z_i^{t,r} - \mathbb{E}[Z_i^{t,r} | W_i]$ and $\tilde{Z}_i^{s,r} = Z_i^{s,r} - \mathbb{E}[Z_i^{s,r} | W_i]$ denote the residualized relative-index instruments. The effect of W_i on the relative index is removed by centering $Z_i^{t,r}$ and $Z_i^{s,r}$ with respect to their conditional expectations given W_i , so that the varying coefficients closely reflect the variation from random judge assignment. Under the complete randomization assumption, the conditional expectation $\mathbb{E}[Z_i^t | W_i]$ can be expressed as

$$\mathbb{E}[Z_i^t | W_i] = \mathbb{E}[(\gamma_{J_i}^t)' W_i | W_i] = \mathbb{E}[(\gamma_{J_i}^t)]' W_i = \left(\sum_{j=1}^J \lambda_j \gamma_j^t \right)' W_i,$$

where the aggregation is through the probabilities $\lambda_j = \mathbb{P}(J_i = j)$ of assignment to each judge j . If judges are randomized within blocks, the aggregation would instead be through the block-specific probabilities of assignment to each judge. From this, we can compute $\mathbb{E}[Z_i^{t,r} | W_i] = \mathbb{E}[Z_i^t | W_i] - \mathbb{E}[Z_i^r | W_i]$, and similarly $\mathbb{E}[Z_i^{s,r} | W_i] = \mathbb{E}[Z_i^s | W_i] - \mathbb{E}[Z_i^r | W_i]$.

The varying-coefficient specification is semiparametric in the same broad spirit as the partially linear model used in [Carneiro and Lee \(2009\)](#), but its motivation for the functional form is different. The partially linear specification restricts (7) further: only the coefficient on the constant varies with the arguments, while the slopes on the remaining covariates are fixed. The partially linear specification cleanly separates the two components dependent on the covariates and the index, which is well suited to analyzing distributional treatment effects. By contrast, our goal is to estimate margin-specific average effects and their heterogeneity across observed characteristics. For this purpose, we retain a rich set of covariate interactions with nonparametric index functions compatible with the required dimension reduction. Allowing for such heterogeneity, however, comes at the cost of greater estimation complexity. In applications where average-effect heterogeneity is not of primary interest, one may instead employ more parsimonious specifications such as in [Carneiro and Lee \(2009\)](#).

The specification resolves the identification problem posed by the discreteness of Z_i given W_i . The coefficient functions are identified under the rank condition that $\mathbb{E}[W_i W_i' | T_i = t, P_i^t, \tilde{Z}_i^{t,r}]$ is positive definite. Rich variation in the case-level characteristics given the instrument substitutes for continuous variation in the instrument given the characteristics, which is empowered by the continuous variation of the judge indices across cases.

Under this specification, we have

$$\partial_p m_{ts}(P_i^t, Z_i^{t,r}, W_i) = W_i' \partial_p \beta_{ts}(P_i^t, \tilde{Z}_i^{t,r}), \quad (8)$$

and similarly for $\partial_p m_{st}$.

Semiparametric Identification of Aggregate Effects. The specification (7) need not hold exactly. In the Appendix, we read it as the first-order projection of the selected-outcome regression onto the class of covariate interactions and show that covariate-weighted averages of margin-specific effects, including the cell averages reported in Section 6.3, remain identified provided the projection error does not vary with the propensity argument (Assumption A.1 and Proposition E.1). Effects at individual covariate values require the specification to hold exactly. The empirical conclusions of Section 6 rest on the aggregates.

3.3 Estimation in the Second Stage

The second stage estimates the margin-specific treatment effect functions $\tau_{ts}(z) = \mu_{ts}(z) - \mu_{st}(z)$, where $\mu_{ts}(z, w) = \mathbb{E}[Y_i(t) | \mathcal{I}_{ts}(z, w)]$ is the mean potential outcome under treatment t on the margin of treatments t and s at (z, w) . As established in Proposition 1, these margin-specific mean potential outcome functions can be constructed from the margin-specific selected-outcome regressions that can be estimated from the data. This section describes the estimation of the selected-outcome regressions and the resulting estimation of the margin-specific treatment effect functions.

Let $t \in \{d, c, n\}$ and $s \in \{d, c, n\} \setminus \{t\}$ be given and let $r = \{d, c, n\} \setminus \{t, s\}$ be the remaining treatment. Following the specification outlined in Section 3.2, the observed outcome Y_i under $T_i = t$ can be written as

$$Y_i = W_i' \beta_{ts}(P_i^t, \tilde{Z}_i^{t,r}) + \varepsilon_i, \quad \mathbb{E}[\varepsilon_i W_i h(P_i^t, \tilde{Z}_i^{t,r}) | T_i = t] = 0 \quad \text{for all } h(\cdot), \quad (9)$$

where $\beta_{ts}(\cdot) = (\beta_{ts,1}, \dots, \beta_{ts, \dim(W)})'(\cdot)$ is the vector of varying coefficients defined by the projection in Section 3.2. The orthogonality conditions on the projection error ε_i hold by construction. The varying coefficients admit the closed form

$$\beta_{ts}(P_i^t, \tilde{Z}_i^{t,r}) = \mathbb{E}[W_i W_i' | T_i = t, P_i^t, \tilde{Z}_i^{t,r}]^{-1} \mathbb{E}[W_i Y_i | T_i = t, P_i^t, \tilde{Z}_i^{t,r}],$$

provided that $\mathbb{E}[W_i W_i' | T_i = t, P_i^t, \tilde{Z}_i^{t,r}]$ is positive definite almost surely. We will perform this regression in two steps: first, by replacing the unobserved regressors P_i^t and $\tilde{Z}_i^{t,r}$ with their estimates from the first stage, and second, by approximating the nonparametric function $\beta_{ts}(\cdot)$ with a sieve basis expansion.

Let $\hat{Z}_i = \hat{\gamma}' X_i$ be the estimated index instrument and $\hat{P}_i^t = \hat{P}^t(\hat{Z}_i)$ be the proxy for the t -treatment propensity. Let $\hat{Z}_i^{t,r} = \hat{Z}_i^t - \hat{Z}_i^r$ be the relative index with the convention $\hat{Z}_i^d = 0$. Under the assumption of complete judge randomization, we can approximate the conditional expectation of Z_i^t given W_i by $(\sum_j \hat{\lambda}_j \hat{\gamma}_j^t)' W_i$ with the empirical frequency $\hat{\lambda}_j = N^{-1} \sum_{i=1}^N \mathbb{1}\{J_i = j\}$ of judge j . We then construct the residualized indices $\hat{\hat{Z}}_i^t = \hat{Z}_i^t - (\sum_j \hat{\lambda}_j \hat{\gamma}_j^t)' W_i$ and $\hat{\hat{Z}}_i^{t,r} = \hat{Z}_i^{t,r} - \hat{\hat{Z}}_i^r$. These variables can be computed directly from the first-stage estimates.

In the second step, we approximate the functions $\beta_{ts,\iota}(\cdot)$, $1 \leq \iota \leq \dim(W)$, using the B-spline basis functions. For $\tilde{K} \geq 1$, let $\tilde{b}_{\tilde{K}}(\cdot) = (\tilde{b}_{\tilde{K},1}(\cdot), \dots, \tilde{b}_{\tilde{K},d_{\tilde{K}}}(\cdot))'$ be the vector of bivariate B-spline basis functions of order $\tilde{q}$. This basis is again obtained by the tensor product of univariate B-spline bases defined by knot sequences $\tilde{\eta}_0^t < \tilde{\eta}_1^t < \dots < \tilde{\eta}_{\tilde{K}}^t$ and $\tilde{\eta}_0^{t,r} < \tilde{\eta}_1^{t,r} < \dots < \tilde{\eta}_{\tilde{K}}^{t,r}$ covering the supports of P_i^t and $Z_i^{t,r}$, respectively. We then approximate $\beta_{ts,\iota}(\cdot)$ by $\tilde{b}_{\tilde{K}}(\cdot)' \delta_\iota$ for some coefficient vector $\delta_\iota \in \mathbb{R}^{d_{\tilde{K}}}$ with $d_{\tilde{K}} = (\tilde{K} + \tilde{q})^2$. This gives rise to the following sieve regression problem: for $\tilde{K}_N \rightarrow \infty$ as $N \rightarrow \infty$,

$$(\hat{\delta}_1, \dots, \hat{\delta}_{\dim(W)}) = \arg \min_{\delta_1, \dots, \delta_{\dim(W)} \in \mathbb{R}^{d_{\tilde{K}_N}}} \sum_{i=1}^N T_i^t \left(Y_i - \sum_{\iota=1}^{\dim(W)} W_{i,\iota} \tilde{b}_{\tilde{K}_N}(\hat{P}_i^t, \hat{\hat{Z}}_i^{t,r})' \delta_\iota \right)^2.$$

The resulting estimator of β_{ts} is given by $\hat{\beta}_{ts}(\cdot) = (\hat{\beta}_{ts,1}(\cdot), \dots, \hat{\beta}_{ts,\dim(W)}(\cdot))'(\cdot)$ with $\hat{\beta}_{ts,\iota}(\cdot) = \tilde{b}_{\tilde{K}_N}(\cdot)' \hat{\delta}_\iota$ and the partial derivative $\widehat{\partial_1 \beta_{ts,\iota}}(\cdot) = (\partial_1 \tilde{b}_{\tilde{K}_N}(\cdot))' \hat{\delta}_\iota$ with respect to the propensity argument.

Finally, the margin-specific mean potential outcome function $\mu_{ts}(z, w)$ can be estimated by

$$\hat{\mu}_{ts}(z, w) = w'(\hat{\beta}_{ts}(\hat{P}^t(z), \tilde{z}^{t,r}) + \hat{P}^t(z) \widehat{\partial_1 \beta_{ts}}(\hat{P}^t(z), \tilde{z}^{t,r})),$$

where $\tilde{z}^{t,r}$ is the value of the residualized index instrument corresponding to z and w . Repeat the same procedure for $\beta_{st}(\cdot)$ and $\mu_{st}(\cdot)$ with the roles of t and s reversed.

This results in the margin-specific treatment effect function estimated by

$$\begin{aligned}\hat{\tau}_{ts}(z, w) = & w' \left(\hat{\beta}_{ts}(\hat{P}^t(z), \tilde{z}^{t,r}) - \hat{\beta}_{st}(\hat{P}^s(z), \tilde{z}^{s,r}) \right) \\ & + w' \left(\hat{P}^t(z) \widehat{\partial_1 \beta}_{ts}(\hat{P}^t(z), \tilde{z}^{t,r}) - \hat{P}^s(z) \widehat{\partial_1 \beta}_{st}(\hat{P}^s(z), \tilde{z}^{s,r}) \right).\end{aligned}$$

Algorithm 1 summarizes the full procedure. Sections 4 and 5 establish identification and the large-sample properties.

Algorithm 1: The two-step procedure.

Input: Data $\{(Y_i, T_i, W_i, J_i) : i = 1, \dots, N\}$, first-stage sieve (q, K_N) and knot box $\tilde{Z}$, second-stage sieve $(\tilde{q}, \tilde{K}_N)$ and knot sequences.

First stage.

Form the interacted regressors $X_i = D_i \otimes W_i$ from the judge indicators and the case characteristics.;

Solve the sieve maximum likelihood problem (6) for $(\hat{\gamma}, \hat{\xi})$, jointly over the judge coefficients and the gradient B-spline coefficients.;

Second stage.

Construct the index instruments $\hat{Z}_i = \hat{\gamma}' X_i$, the treatment propensities

$\hat{P}_i^t = \hat{P}^t(\hat{Z}_i)$, and the residualized relative indices $\hat{\tilde{Z}}_i^{t,r}$.;

for each margin (t, s) **do**

 Estimate the varying-coefficient outcome regressions (9) by sieve least squares on the t - and s -treatment subsamples.;

 Form $\hat{\tau}_{ts}(\cdot)$ from the fitted coefficients and their propensity derivatives through Proposition 1.;

end

Output: The margin-specific effect functions $\hat{\tau}_{ts}(\cdot)$ for the three margins.

4 Identification of the First Stage

No regressor in the judge-IV framework serves as a numeraire, and without one the two index vectors are identified only up to their span. The latent-return model in Section 2 thus requires a novel approach to identification. We show that within-judge and cross-judge variation can be leveraged to identify the first-stage parameters $\gamma_o = (\gamma_{oj})_{j=1}^J$ and $P_o(\cdot) = P(\cdot | f_o)$ from the observed treatment propensities under a set of primitive conditions that can be verified from the data.

We say that $\theta_o = (\gamma_o, P_o) \in \Gamma \times \mathcal{P}$ is identified if no other pair $(\gamma, P) \in \Gamma \times \mathcal{P}$ satisfies

$$P_o^t(\gamma'_{oj}w) = P^t(\gamma'_jw) \quad \text{for all } t \in \{d, c, n\}, w \in \text{supp}(W_i), j = 1, \dots, J.$$

The left-hand side of the equality is determined by the observed t -treatment propensity given covariates $W_i = w$ and judge $J_i = j$ since

$$\mathbb{P}(T_i = t | W_i = w, J_i = j) = \mathbb{P}(T_i = t | Z_i = \gamma'_{oj}w) = P^t(\gamma'_{oj}w | f_o).$$

Recall that the underlying density f_o belongs to the class $\mathcal{F}$ introduced in Section 3, whose defining conditions ensure that the propensity function $P_o(\cdot)$ has an everywhere positive definite Jacobian and is therefore injective on $\mathbb{R}^2$.

We leverage two types of variation in the data to identify (γ_o, P_o) : within-judge variation across cases and cross-judge variation across judges. The within-judge variation looks at cases assigned to the same judge and exploits the variation in their covariates to identify the judge-specific coefficients up to scale and location normalizations. In contrast, the cross-judge variation compares the treatment propensities across different judges to resolve the scale and location ambiguities. For example, if two judges j_1 and j_2 exhibit the same propensity at covariate values w_1 and w_2 , respectively, then the injectivity of the propensity function $P_o(\cdot)$ in the judge index reveals that $\gamma'_{oj_1}w_1 = \gamma'_{oj_2}w_2$, an equation that ties the relative scale and location of γ_{oj_1} and γ_{oj_2} .

Such comparability between judges j_1 and j_2 presupposes that the supports of their propensities overlap. Broadly speaking, such proximity in the space of treatment propensities is necessary whenever information is pooled across discrete variation in the data. We formalize this into two notions of adjacency: *slice adjacency* and *judge adjacency*. The slice adjacency is used for within-judge identification of the coefficients corresponding to discrete covariates, while the judge adjacency is used for cross-judge comparison.

Within-judge identification treats the components of W_i asymmetrically, which calls for the partition $W_i = (1, W'_{a,i}, W'_{b,i})'$, where $W_{1,i} \equiv 1$ is the intercept, $W_{a,i}$ consists of continuous covariates, and $W_{b,i}$ contains the discrete covariates taking finitely many values, which we refer to as *slice variables*. Partition γ_j conformably as $\gamma_j = (\gamma_{j,1}, \gamma_{j,a}, \gamma_{j,b})$.

Since J_i is assumed independent of W_i , conditioning on $J_i = j$ does not alter the covariate support. The key variation for within-judge identification comes from the continuous covariates $W_{a,i}$ as in typical two-index models. The point identification of $\gamma_{j,a}$ relies on the full gradient of the link function. To that end, there must be at least two continuous covariates $W_{a,i}$ whose conditional support given some slice $W_{b,i} = w_b^{j,*}$ has a non-empty interior, and the matrix $\gamma_{j,a}$ must have full column rank. Even with rich continuous variation, the within-judge data determine the index pair only up to an invertible affine transformation; reducing this to a common scale requires that the propensity function not decouple into two univariate monotone indices along any pair of directions.

Once $\gamma_{j,a}$ is identified, pooling information across slices requires the following notion of adjacency. Slice values w_b and w'_b are adjacent in the j -judge cell if the supports of the propensity function given $(W_b = w_b, J = j)$ and given $(W_b = w'_b, J = j)$ overlap at least at one point. We then say that w_b and w'_b are in the same slice-adjacency component if they can be connected by a finite chain of adjacent points. The support of W_b is partitioned into slice-adjacency components in the j -judge cell, and we denote the unique slice-adjacency component containing a slice w_b by $C_j(w_b)$.

The cross-judge counterpart requires a slightly stronger overlap. Two judges j_1 and j_2 are adjacent if the supports of the propensity function given $J = j_1$ and given $J = j_2$ share at least one interior point. The judge-adjacency graph has the judges $\{1, \dots, J\}$ as vertices and the adjacent pairs as edges. The two notions of adjacency differ in the required amount of overlap. Within a judge, the slices share a common scale. A single common propensity value therefore already pins down the relative location of two slices. Across judges both the scale and the location are free. A single common value can therefore always be rationalized by trading one off against the other. Separating the two requires variation over an open set of common values, which is the open variation an interior overlap supplies.

We are now prepared to state the conditions under which (γ_o, P_o) is identified.

Assumption 4. *The following hold.*

- (a) (Continuous variation.) $\dim(W_a) \geq 2$, and for each judge j there exists $w_b^{j,*} \in \text{supp}(W_b)$ such that $\text{supp}(W_a | W_b = w_b^{j,*})$ has a non-empty interior in $\mathbb{R}^{\dim(W_a)}$.
- (b) (Rank.) For each judge j , $\gamma_{oj,a} \in \mathbb{R}^{\dim(W_a) \times 2}$ has full column rank.

- (c) (Nonseparability.) *For each judge j , there exist no linearly independent vectors $v_1, v_2 \in \mathbb{R}^2$ such that $v_1' DP_o(z)v_2 = 0$ for all $z \in \text{Int}(\text{supp}(\gamma_{oj}'W))$.*
- (d) (Slice connectivity.) *For each judge j , the affine hull of $C_j(w_b^{j,*})$ spans $\mathbb{R}^{\dim(W_b)}$.*
- (e) (Judge connectivity.) *The judge-adjacency graph is connected.*

Theorem 1. *Under Assumption 4, γ_o is identified in Γ and the propensity function $P_o(\cdot)$ is identified as a function on $\cup_{j=1}^J \text{supp}(\gamma_{oj}'W)$.*

The conditions in Assumption 4 map onto the two layers of variation described above. Conditions (a) and (b) supply the continuous variation that drives within-judge identification, and condition (d) the slice pooling that extends it to the discrete coefficients. In addition, the intercept in W_i leaves a two-dimensional location ambiguity within each judge—shifting each $\gamma_{j,1}$ by $\Delta \in \mathbb{R}^2$ while translating the distribution of U_i by $(-\Delta)$ leaves the latent return unchanged—that no single judge’s data can resolve.

The within-judge layer identifies each γ_{oj} only up to a judge-specific positive scale and intercept. The judge connectivity condition (e) propagates cross-judge comparisons of the kind illustrated above along the judge-adjacency graph, equalizing the judge-specific scales and locations, and the normalizations embedded in Γ pin down the remaining factors. Connectivity does not require that every pair of judges be comparable, since a chain of pairwise-adjacent judges suffices. Nor does it demand a large number of judges. Two judges satisfy it as long as their propensity supports overlap accordingly. What it rules out is a set of judges that splits into clusters with mutually disjoint behavior, in which case each cluster would be identified only on its own scale and location, and propensities would not be comparable across clusters. The condition is mild under complete random assignment because every judge faces the same covariate distribution. Adjacency fails only when two judges’ coefficients map the common support of W into essentially disjoint regions of the index space, an extreme form of behavioral heterogeneity. More judges spanning a denser range of behavior make the judge-adjacency graph easier to connect.

The nonseparability condition (c) is the substantive restriction on the propensity function. Were it violated for some judge j and directions v_1, v_2 , the propensity function would decouple on that judge’s index support. Moving z along v_2 would leave $v_1' P_o(z)$ unchanged, and moving z along v_1 would leave $v_2' P_o(z)$ unchanged. On

any convex open subset of the support, this decoupling amounts to P_o splitting into two univariate monotone indices:

$$P_o(z) = g_1(w'_1 z)w_1 + g_2(w'_2 z)w_2, \quad (10)$$

where g_1, g_2 are strictly increasing functions and (w_1, w_2) is the basis dual to (v_1, v_2) , i.e., columns of the inverse of the matrix (v_1, v_2) . Geometrically, such a propensity function maps each parallelogram spanned by v_1, v_2 onto a parallelogram spanned by w_1, w_2 . Such separability is a degenerate, knife-edge configuration rather than a robust feature. It requires the propensity surface to decouple along (v_1, v_2) throughout the support, and a generic f_o breaks it. For instance, if the index support were all of $\mathbb{R}^2$, condition (c) holds automatically. A separable propensity function maps $\mathbb{R}^2$ onto an interior of a parallelogram, whereas the image of $\mathbb{R}^2$ under P_o must be the interior of the probability simplex, which is not a parallelogram.

To see why the cross-partial equality is the engine of within-judge identification, strip the model to its core: drop the intercept and the slice variables, let $W = (W_1, W_2)'$ be continuously distributed, and fix a judge whose coefficient matrix $\gamma_{oj} \in \mathbb{R}^{2 \times 2}$ is nonsingular. The data reveal the propensity surface $\pi(w) = (\pi^c(w), \pi^n(w))'$, where $\pi^t(w) = P_o^t(\gamma'_{oj} w)$. Any candidate pair (γ, P) must reproduce it: $\pi(w) = P(\gamma' w)$ for all $w \in \text{supp}(W)$. Differentiating both sides of this representation and rearranging gives $DP(\gamma' w) = D\pi(w)\gamma^{-T}$ where $\gamma^{-T} = (\gamma^{-1})'$. The key input is the Slutsky symmetry of $DP(\cdot)$, which holds for any density in $\mathcal{F}$. Symmetry of $D\pi(w)\gamma^{-T}$ requires

$$[D\pi(w)\gamma^{-T}]_{12} = [D\pi(w)\gamma^{-T}]_{21} \quad \text{for all } w \in \text{supp}(W).$$

Each covariate value contributes one linear equation in the four entries of γ^{-T} . The support of W therefore generates a continuum of linear restrictions on γ^{-T} , and the true value γ_{oj}^{-T} always solves the system. Identification up to scale obtains exactly when the solution set of this linear system reduces to the scalar multiples of γ^{-T} , and the nonseparability condition (c) guarantees that the propensity surface varies richly enough for this reduction. Indeed, were P_o separable as in (10), rescaling $w'_1 z$ and $w'_2 z$ separately, replacing $g_1(\cdot)$ and $g_2(\cdot)$ by $g_1(d_1 \cdot)$ and $g_2(d_2 \cdot)$ for some scalars $d_1, d_2 > 0$, would trace out a two-parameter family of solutions. Condition (c) is therefore essentially necessary as well as sufficient for within-judge identification. Richer support of W enlarges this system of restrictions and therefore strengthens identification—in

the extreme case where $\text{supp}(W) = \mathbb{R}^2$, condition (c) holds automatically, as noted above.

As claimed at the beginning of this section, the conditions of Assumption 4 can be verified from the data in the sense that each is expressible through observable objects. Condition (a) restricts the observed covariate distribution. Condition (b) is equivalent to the Jacobian of the observed propensity surface with respect to w_a attaining rank two at an interior covariate value. This Jacobian equals $DP_o(\gamma'_{oj}w)\gamma'_{oj,a}$, and DP_o is everywhere nonsingular. The two ranks therefore coincide. Condition (c) can be assessed through the linear system of the preceding paragraph, whose coefficients involve only the derivatives of the observed propensity surface. A violation of (c) manifests itself as a solution space of dimension greater than one. Conditions (d) and (e) are statements about the supports of the observed propensities across slices and judges. Moreover, conditions (b) and (c) can be checked at the estimated pair $(\hat{\gamma}, \hat{P})$. Were identification to fail, the estimates would be close to an observationally equivalent pair, and these conditions hold either at every pair in an observational equivalence class or at none—they are properties of the observed propensity surface rather than of the particular pair generating it. A violation at the true pair therefore reproduces itself at the estimates, while conditions validated at the estimates certify identification of the truth. Verification in finite samples requires estimates of the propensity surface and, for conditions (b) and (c), of its derivatives. We leave the construction of formal tests for future work.

Finally, Theorem 1 identifies the propensity function P_o but not the density f_o itself. This is a feature of the model with limited support of W rather than a limitation. The observed propensities are insensitive to the behavior of f far from where the indices fall.

Existing identification results for multinomial choice models do not deliver the first-stage identification required by the judge-IV framework. One strand of the discrete-choice literature achieves nonparametric identification through alternative-specific regressors or restrictions on the error distribution: [Matzkin \(1993\)](#) requires each alternative to carry its own regressors or the unobservables to be i.i.d. across alternatives, and the special-regressor approach of [Lewbel \(2000\)](#) requires a regressor with large support that is excluded from the remaining indices. Neither device is available here. The same device recurs in multiple-index models generally, which break the rotational invariance of up-to-span identification by reserving at least one

covariate for each index (Ichimura et al., 1991). That route is closed here as well. No covariate is ex ante excluded from either index, since every case characteristic enters the returns to both treatments, and the rotation is pinned down instead by the Slutsky symmetry of $DP(\cdot)$. The result of Berry and Haile (2024) identifies the index coefficients up to span but not individually and translates to the judge design only under at least three continuous covariates with coefficients common across judges. Humphries et al. (2025) refer to this result, while their implementation takes a different route. The three-covariate requirement is intrinsic to span identification: with only two continuous covariates, the normalized 2×2 block exhausts the coefficient matrix and leaves nothing for the data to identify. Our cross-partial argument instead identifies each judge’s coefficients from two continuous covariates, under a single scale and location normalization, and allows the coefficients to vary across judges. While the perturbed utility model of Allen and Rehbeck (2019) nests our latent-return framework, their identification requires at least one alternative-specific continuous regressor per alternative and is therefore inapplicable here. A related approach, Shi et al. (2018), exploits the convexity of the social surplus through cyclic monotonicity in the mirror image of our setting, with alternative-specific regressors and a common coefficient vector in place of our common regressors and treatment-specific coefficients.

The multiple-treatment IV literature attains point identification through richer instruments. Kirkeboen et al. (2016) use knowledge of next-best alternatives together with treatment-specific instruments, and Mountjoy (2022) uses margin-specific continuous instruments. Heckman and Pinto (2018) instead discipline the response to given discrete instruments through unordered monotonicity, which their equivalence theorem (Theorem T-3) characterizes by the absence of two-way flows: no instrument shift may draw one case into a treatment while pushing another case out of it. The latent-return model with continuous covariates violates this condition in general. A change of judge moves both indices by amounts that vary with the covariates, and cases flow in both directions between c and n . In our setting, judge identity is a single discrete instrument that shifts all margins simultaneously, which is what necessitates the joint identification of the two index vectors undertaken here. Closest to this paper, Lee and Salanié (2018) identify treatment effects taking the selection thresholds as delivered by continuous instruments that enter them directly. Our identified first stage supplies exactly the threshold structure their analysis takes as given.

5 Asymptotic Theory

5.1 Asymptotic Properties of the SMLE in the First Stage

Theorem 1 shows that the first-stage model is identified under the primitive conditions in Assumption 4. We now turn to the large-sample properties of the SMLE (6) for both γ_o and $P_o(\cdot)$. The propensity function $P_o(\cdot)$ is treated as an infinite-dimensional nuisance parameter that can be estimated at a slower nonparametric rate. However, we show that the finite-dimensional component $\hat{\gamma}$ converges at the parametric rate of $N^{-1/2}$. We proceed in two steps. Under a set of mild conditions, the first-order theory establishes the consistency of $\hat{\gamma}$ and its crude rate of convergence. We then refine this result to semiparametric efficiency of $\hat{\gamma}$ under additional regularity conditions.

Throughout the paper, we consider an asymptotic regime where the sample size N grows to infinity while the number of judges J remains fixed. This parallels the small- n -large- T panel asymptotics in the sense that a limited number of judges are observed on an increasing number of cases. Two features of this regime are worth noting. First, holding J fixed sidesteps the incidental parameters problem. The judge-specific coefficients $(\gamma_{oj})_{j=1}^J$ remain finite in number, and each is estimated by a within-judge subsample whose size grows with N . Second, the link function $P_o(\cdot)$ is specified nonparametrically but, unlike the judge-specific coefficients, is estimated by pooling the within-judge subsamples. A common link is what makes the cross-judge comparisons underlying identification possible. Because the same link generates every judge's propensities, inverting it places them in a common index space.

A final convention concerns the intercept. The location ambiguity noted above is resolved by the mean-zero normalization $e_J' \gamma^c = e_J' \gamma^n = 0$ built into Γ . The parameter retains all judge-specific intercepts, subject to these two linear restrictions; the efficient information $I_{\gamma\gamma}^{\text{eff}}$ below is accordingly singular in the two intercept-mean directions and the radial direction, and its Moore–Penrose inverse acts on their orthogonal complement.

5.1.1 Consistency and the Crude Rate of Convergence of the SMLE

Assumption 5 below presents the conditions necessary for the first-order theory in addition to the identification conditions in Assumption 4.

Assumption 5. *The following hold.*

- (a) (I.I.D. sampling and finite judges.) $(T_i, W_i, J_i)_{i=1}^N$ are i.i.d. draws from a common distribution $\mathbb{P}$. Moreover, W_i has compact support and $\lambda_j = \mathbb{P}(J_i = j)$ is strictly positive for each $j = 1, \dots, J$ with J fixed.
- (b) (Smoothness and interior.) $P_o(\cdot)$ is k_o -times continuously differentiable on $\bar{\mathcal{Z}}$ with $k_o \geq 2$, and it lies in the interior of $\mathcal{P}$ in the sense that $P_o^c(z), P_o^n(z) \in (\kappa, 1 - \kappa)$ and $\|DP_o(z)\| < M$ for all $z \in \bar{\mathcal{Z}}$.
- (c) (Spline order and quasi-uniform mesh.) $q \geq \max(k_o + 1, 4)$ and there exist positive constants $c_1, c_2 > 0$ such that $c_1 K_N^{-1} \leq \eta_m^t - \eta_{m-1}^t \leq c_2 K_N^{-1}$ for all $m = 1, \dots, K_N$ and $t \in \{c, n\}$.
- (d) (Covariate and index density.) For each $w_b \in \text{supp}(W_b)$, the distribution of W_a given $W_b = w_b$ admits a continuous and positive density $p_{W_a|W_b}(\cdot | w_b)$ with respect to the Lebesgue measure on $\mathbb{R}^{\dim(W_a)}$. Moreover, the index instrument $Z = \gamma'_o X$ has a compact support $\mathcal{Z}$ with a dense interior and admits a continuous and positive density $p_Z(\cdot)$.
- (e) (Index separation.) There exist $\epsilon > 0$ and $c_3 > 0$ such that $\mathbb{E}[\text{Var}(v' X_i | \gamma' X_i)] \geq c_3 \|v\|^2$ for every γ with $\|\gamma - \gamma_o\| < \epsilon$ and every v orthogonal to the columns γ^c and γ^n of that γ and to e_J .

Recall that k_o is the degree of smoothness of the true propensity P_o and (q, q) is the order of the tensor-product B-splines used to approximate the social surplus function. The gradient B-spline basis is therefore of order $(q - 1, q)$ and $(q, q - 1)$ for each component of the propensity function. For each $t \in \{c, n\}$, the knot sequence $(\eta_m^t)_{m=0}^{K_N}$ is assumed to be quasi-uniform in the sense that the mesh size $\eta_m^t - \eta_{m-1}^t \asymp K_N^{-1}$ is comparable across all $m = 1, \dots, K_N$. This is less restrictive than the uniform knot spacing which requires that $\eta_m^t - \eta_{m-1}^t = (\bar{z}^t - \underline{z}^t)/K_N$ for all $m = 1, \dots, K_N$ and $t \in \{c, n\}$. This condition ensures that the spline approximation error decays at a rate of $K_N^{-k_o}$ as $K_N \rightarrow \infty$.

Since $W_{a,i}$ collects the continuous covariates in W_i and $W_{b,i}$ the discrete covariates, $W_{a,i}$ given $W_{b,i}$ admitting a continuous and positive density can be viewed as a mild regularity condition. The continuity and positivity of p_Z are imposed directly at the level of the index; they do not follow automatically from the slice-level densities, since

an induced density can be irregular where the null space of the index map meets the covariate support tangentially. Since Z_i has compact support, p_Z is bounded away from zero and infinity on $\mathcal{Z}$. This is a standard condition in nonparametric regression ensuring, together with Condition (c), a well-conditioned Gram matrix of B-spline basis functions. Finally, Condition (e) is stated at the level of the regressor X_i , which is the level the estimator sees: the likelihood never references the judge structure. It requires that no linear combination $v'X_i$ be approximable in mean square by a function of the two-dimensional index $\gamma'X_i$, uniformly over γ near γ_o , except in the directions for which the left-hand side vanishes identically: the index directions themselves and the constant direction e_J . The requirement is stronger than a full-rank condition on $\mathbb{E}[X_i X_i']$, since nonlinear dependence on the index is also precluded. Condition (e) plays the role of the index-separation condition embedded in the semiparametric efficiency bound for multiple-index regression models. There, the bound is finite only when the subspace spanned by the index vectors is well separated from its linear complement, and the same separation is what allows the coefficients here to be estimated with precision. Primitive sufficient conditions adapted to the judge design are given in the Appendix.

The following theorem establishes the consistency of the sieve MLE. Let $\rho_N := \sqrt{K_N^2 \log N/N + K_N^{-k_o}}$, which represents the nonparametric convergence rate of $\hat{\theta} = (\hat{\gamma}, P_{\hat{\xi}})$ in the following L^2 -metric:

$$\mathcal{L}(\hat{\theta}, \theta_o) = \|P_{\hat{\xi}}(\hat{\gamma}'\cdot) - P_o(\gamma_o'\cdot)\|_{L^2(p_X)} = \left(\mathbb{E} \sum_{t \in \{d, c, n\}} \left(P_{\hat{\xi}}^t(\hat{\gamma}'X) - P_o^t(\gamma_o'X) \right)^2 \right)^{1/2},$$

where the expectation is taken with respect to the distribution of X_i , denoted by p_X . This metric arises naturally from maximum likelihood estimation, since the Kullback–Leibler divergence between the true and fitted propensities expands quadratically in $\mathcal{L}(\hat{\theta}, \theta_o)$.

Theorem 2. (*Consistency and nonparametric rate of SMLE.*) Suppose that $K_N \rightarrow \infty$ and $K_N^2 \log N = o(N)$ as $N \rightarrow \infty$. Under Assumptions 4 and 5(a)–(c), it holds that $\hat{\gamma} \rightarrow_p \gamma_o$ and $\mathcal{L}(\hat{\theta}, \theta_o) = O_P(\rho_N) \rightarrow_p 0$ as $N \rightarrow \infty$.

The smoothness requirement $k_o \geq 2$ in Assumption 5(b) is used only to sharpen the rate of $\hat{\gamma}$ in Theorem 3 below. The consistency and the nonparametric rate of

Theorem 2 continue to hold under the weaker requirement $k_o \geq 1$.

Theorem 3. *If, in addition to the conditions of Theorem 2, Assumptions 5(d)–(e) hold, then $\|\hat{\gamma} - \gamma_o\| = O_P(\rho_N)$. Moreover, on the interior regions $\mathcal{Z}_N := \{z \in \mathcal{Z} : \text{dist}(z, \partial\mathcal{Z}) \geq (q+1)h\}$, with $h \asymp K_N^{-1}$ the mesh width, the estimated propensity function attains the uniform rate $\|P_{\hat{\xi}}^t - P_o^t\|_{L^\infty(\mathcal{Z}_N)} = O_P(K_N \rho_N)$ for $t \in \{c, n\}$.*

The rate of $\|\hat{\gamma} - \gamma_o\|$ in Theorem 3 is inherited from the nonparametric rate of Theorem 2 through a local curvature bound on the population criterion. Once Theorem 2 places $\hat{\gamma}$ in a neighborhood of γ_o , the L^2 -discrepancy admits the lower bound

$$\|\hat{\gamma} - \gamma_o\| \lesssim \mathsf{L}(\hat{\theta}, \theta_o)$$

with probability tending to one, so that $\hat{\gamma}$ cannot move without moving the implied propensities. Combined with $\mathsf{L}(\hat{\theta}, \theta_o) = O_P(\rho_N)$, this yields $\|\hat{\gamma} - \gamma_o\| = O_P(\rho_N)$. The curvature bound is where Assumptions 5(d)–(e) enter. The index-separation condition (e) induces a curvature along a direction orthogonal to γ_o , preventing $\hat{\gamma}$ from moving in that direction without changing the implied propensities, while the density condition (d) together with the nonseparability condition 4(c) create a curvature within the column space of γ_o . The rate is crude in that $\hat{\gamma}$ is tied to the slower nonparametric rate of $P_{\hat{\xi}}$. The parametric $N^{-1/2}$ rate requires the efficient-score argument of the next section.

5.1.2 Semiparametric Efficiency for the Index Coefficients

We now turn to the semiparametric efficiency of $\hat{\gamma}$ for γ_o . We first define the notion of semiparametric efficiency for the index coefficients, then identify the bound it refers to.

Definition 1. A regular estimator $\hat{\gamma} \in \Gamma$ is *semiparametrically efficient* for γ_o if its asymptotic covariance matrix is minimal, in the positive semidefinite order, among the asymptotic covariance matrices of all regular estimators of γ_o in the first-stage model.

The first-stage model is the law of (T, X) through $l(t|x, \theta)$, with the outcome law $Y|T, X$ excluded. When the outcome mean is left unrestricted, the outcome carries no information about γ and the bound of Proposition 2 is also the bound in the full model of (Y, T, X) . Remark 3 at the end of this section discusses when the outcome

can add information. The efficient information is $I_{\gamma\gamma}^{\text{eff}} = \mathbb{E}[s_{\gamma}^{\text{eff}}(s_{\gamma}^{\text{eff}})']$, formed from the efficient score

$$s_{\gamma}^{\text{eff}} = \Pi_{\gamma_o} (s_{\gamma} - \Pi_{\mathcal{T}_{\xi}} s_{\gamma}),$$

where s_{γ} and s_{ξ} are the index and nuisance scores of $l(t|x, \theta)$, $\Pi_{\mathcal{T}_{\xi}}$ is the $L^2(\mathbb{P})$ -projection onto the nuisance tangent space spanned by s_{ξ} , and Π_{γ_o} the projection onto the tangent space $\mathcal{T}_{\gamma_o} = \{\Delta\gamma : \langle \Delta\gamma, \gamma_o \rangle_F = 0, e'_J(\Delta\gamma)^c = e'_J(\Delta\gamma)^n = 0\}$ at γ_o with $\langle a, b \rangle_F = \text{vec}(a)' \text{vec}(b)$. Specifically, $\Pi_{\gamma_o} = \mathbf{I}_L - \text{vec}(\gamma_o) \text{vec}(\gamma_o)'$ is the orthogonal projection onto $\mathcal{T}_{\gamma_o}$, where $\mathbf{I}_L$ is the orthogonal projection onto $L = \{\Delta\gamma : e'_J(\Delta\gamma)^c = e'_J(\Delta\gamma)^n = 0\}$, removing from each column its cross-judge intercept mean, and $\text{vec}(\gamma_o) \text{vec}(\gamma_o)'$ removes the radial direction, using $\|\gamma_o\|_F = 1$; the two projections commute since $\gamma_o \in L$. The presence of this projection departs from the textbook setting. Because γ_o satisfies $e'_J \gamma_o^c = e'_J \gamma_o^n = 0$ and $\|\gamma_o\|_F = 1$, Π_{γ_o} annihilates the two intercept-mean directions and the radial direction $\text{vec}(\gamma_o)$. $I_{\gamma\gamma}^{\text{eff}}$ is therefore singular but positive definite on $\mathcal{T}_{\gamma_o}$:

$$v' I_{\gamma\gamma}^{\text{eff}} v > 0 \quad \text{for all } v \in \mathcal{T}_{\gamma_o} \setminus \{0\}.$$

The efficiency bound is then read through the Moore–Penrose inverse on $\mathcal{T}_{\gamma_o}$ rather than an ordinary inverse.

Proposition 2. *The asymptotic covariance matrix of any regular estimator of γ_o in the first-stage model is bounded below by $(I_{\gamma\gamma}^{\text{eff}})^+$ in the positive semidefinite order. A regular estimator is therefore semiparametrically efficient for γ_o in the sense of Definition 1 exactly when*

$$\sqrt{N} (\text{vec}(\hat{\gamma}) - \text{vec}(\gamma_o)) \rightarrow_d \mathcal{N} \left(0, (I_{\gamma\gamma}^{\text{eff}})^+ \right).$$

The bound rests on a projection argument. Any $\hat{\gamma} \in \Gamma$ satisfies $e'_J \hat{\gamma}^c = e'_J \hat{\gamma}^n = 0$ and $\langle \gamma_o, \hat{\gamma} - \gamma_o \rangle_F = -\frac{1}{2} \|\hat{\gamma} - \gamma_o\|_F^2$. $\sqrt{N} (\text{vec}(\hat{\gamma}) - \text{vec}(\gamma_o))$ therefore has vanishing intercept-mean components and radial component $o_P(1)$, the latter because regularity makes $\sqrt{N} (\text{vec}(\hat{\gamma}) - \text{vec}(\gamma_o))$ bounded in probability while the radial term is quadratic in $\hat{\gamma} - \gamma_o$. Every regular influence function therefore takes values in $\mathcal{T}_{\gamma_o}$, and the projection Π_{γ_o} is common to all estimators. Efficiency is therefore decided on $\mathcal{T}_{\gamma_o}$. A regular influence function ψ must be a gradient: $\mathbb{E}[\psi b] = 0$ for b in the nuisance tangent space, and $\mathbb{E}[\psi s'_{\gamma}] = \Pi_{\gamma_o}$. The canonical gradient, the efficient influence

function, is $\tilde{\psi} = (I_{\gamma\gamma}^{\text{eff}})^+ s_{\gamma}^{\text{eff}}$, with variance $(I_{\gamma\gamma}^{\text{eff}})^+$. Any other gradient differs from $\tilde{\psi}$ by a term orthogonal to both s_{γ} and the nuisance tangent, hence to $\tilde{\psi}$, so

$$\text{Var}(\psi) = (I_{\gamma\gamma}^{\text{eff}})^+ + \text{Var}(\psi - \tilde{\psi}) \geq (I_{\gamma\gamma}^{\text{eff}})^+.$$

The two projections in $s_{\gamma}^{\text{eff}} = \Pi_{\gamma_o}(s_{\gamma} - \Pi_{\mathcal{T}_{\xi}} s_{\gamma})$ are exactly these. The residual off the nuisance tangent removes the part of the index score explained by the nuisance, and Π_{γ_o} restricts to the identified directions.

Theorem 4 establishes that the SMLE attains this bound through an asymptotic linear representation with influence function $(I_{\gamma\gamma}^{\text{eff}})^+ s_{\gamma}^{\text{eff}}$ and variance $(I_{\gamma\gamma}^{\text{eff}})^+$. The major obstruction is to control an increasing number of bias terms while deriving the distributional approximation, and to that end we make the following additional assumptions.

Assumption 6. *The following hold.*

- (a) (Density bound and connected support.) *There are a neighborhood $\{\gamma : \|\gamma - \gamma_o\| < \epsilon\}$ of γ_o and constants $0 < \underline{p} < \bar{p} < \infty$ such that, for every γ in this neighborhood, the index $\gamma'X$ admits a density p_{γ} satisfying $\underline{p} \leq p_{\gamma}(z) \leq \bar{p}$ for all $z \in \text{supp}(\gamma'X)$. Moreover, $\mathcal{Z} = \text{supp}(\gamma'_o X)$ is connected with finite perimeter.*
- (b) (Additional smoothness.) *The smoothness index k_o of Assumption 5(b) satisfies $k_o \geq 4$.*
- (c) (Spline order growth.) *As $N \rightarrow \infty$, $N = o(K_N^{2k_o})$ and $K_N = o((N/\log N)^{1/6})$.*

Condition (a) is a mild nondegeneracy requirement. The two-sided density bound keeps the spline Gram matrix uniformly well-conditioned on the interior of $\mathcal{Z}$. The finite perimeter of $\mathcal{Z}$ bounds the number of spline cells that straddle its boundary. The partially covered cells near the edge therefore remain a vanishing fraction of the total and cannot spoil the conditioning. The condition thus excludes a support whose boundary is irregular enough to have infinite perimeter. Connectedness ensures the index domain is a single region. The argument therefore applies uniformly across it.

The smoothness and growth conditions trade bias against variance. The smoothness condition lets the order- q spline approximate P_o and its derivatives. The growth condition constrains the sieve dimension K_N from both sides: K_N must grow fast enough that the approximation bias is $o(N^{-1/2})$ and slow enough that the stochastic

terms and the increasing number of bias terms stay controlled. These two bounds are compatible only when P_o is smooth enough. The growth window is nonempty precisely because $k_o \geq 4$, which makes $2k_o > 6$ and places the lower rate $N^{1/(2k_o)}$ below the upper rate $N^{1/6}$; the lower bound makes the crude estimate $\sqrt{N}K_N^{-k_o}$ of the asymptotic bias vanish. Greater smoothness widens the admissible range for K_N .

Theorem 4. *Under Assumptions 4 and 5–6, it holds that*

$$\sqrt{N}(\text{vec}(\hat{\gamma}) - \text{vec}(\gamma_o)) = \left(I_{\gamma\gamma}^{\text{eff}}\right)^+ \frac{1}{\sqrt{N}} \sum_{i=1}^N s_{\gamma}^{\text{eff}}(T_i, X_i) + o_P(1).$$

Consequently, $\hat{\gamma}$ attains the bound of Proposition 2 and is semiparametrically efficient for γ_o in the sense of Definition 1.

Remark 1 (Per-judge precision). The parametric rate in Theorem 4 should be read judge by judge. Only the cases assigned to judge j carry information about γ_j , so the corresponding block of $I_{\gamma\gamma}^{\text{eff}}$ is proportional to the assignment probability λ_j , and the matching block of the asymptotic covariance carries the factor λ_j^{-1} , up to the dependence across judges induced by the common link. Judge j 's coefficients are therefore estimated at the effective rate $(N\lambda_j)^{-1/2} \asymp N_j^{-1/2}$, where N_j is the judge's own caseload. This concerns the judge-specific blocks alone. Components pooled across judges, such as the common time effects or cross-judge aggregates of the coefficients, draw on the full sample and attain the $N^{-1/2}$ rate with constants free of any single λ_j . The same holds when the estimator is applied outside the judge design, to cross-sectional data with coefficients common to all observations, where every component is pooled. This is the formal counterpart of the caseload threshold in Section 6.1, which excludes judges whose N_j is too small for their coefficients to be estimated with adequate precision.

Remark 2 (Estimation of asymptotic variance). As noted in Akerberg et al. (2012), the efficient information in the first-stage model can be estimated as if the sieve propensity were parametrically specified. Let $\hat{s}_{\gamma,i} = \partial_{\gamma} \log l(T_i | X_i, \hat{\theta})$ and $\hat{s}_{\xi,i} = \partial_{\xi} \log l(T_i | X_i, \hat{\theta})$ denote the estimated index and nuisance scores of observation i . Let $\Pi_{\hat{\gamma}} = \mathbf{I}_L - \text{vec}(\hat{\gamma})\text{vec}(\hat{\gamma})'$ be the projection onto the tangent space $\mathcal{T}_{\hat{\gamma}}$. Then, under

the conditions of Theorem 4, the efficient information can be estimated by

$$\hat{I}_{\gamma\gamma}^{\text{eff}} = \Pi_{\gamma} \left(\frac{1}{N} \sum_{i=1}^N \hat{s}_{\gamma,i} \hat{s}_{\gamma,i}' - \frac{1}{N} \sum_{i=1}^N \hat{s}_{\gamma,i} \hat{s}_{\xi,i}' \left(\frac{1}{N} \sum_{i=1}^N \hat{s}_{\xi,i} \hat{s}_{\xi,i}' \right)^+ \frac{1}{N} \sum_{i=1}^N \hat{s}_{\xi,i} \hat{s}_{\gamma,i}' \right) \Pi_{\gamma}.$$

To estimate the asymptotic covariance matrix $\Sigma = (I_{\gamma\gamma}^{\text{eff}})^+$, let $\hat{s}_i = (\Pi_{\gamma} \hat{s}_{\gamma,i}', \hat{s}_{\xi,i}')'$ be the grand score vector of observation i , where $\Pi_{\gamma} \hat{s}_{\gamma,i}$ is the projected index score. Compute the outer product $\hat{s}_i \hat{s}_i'$ and take its sample average $N^{-1} \sum_{i=1}^N \hat{s}_i \hat{s}_i'$. The asymptotic covariance matrix can be estimated by the upper-left block of the Moore–Penrose inverse of this average

$$\hat{\Sigma} = \left(\frac{1}{N} \sum_{i=1}^N \hat{s}_i \hat{s}_i' \right)_{1:\dim(X), 1:\dim(X)}^+.$$

Remark 3 (Two-step versus joint estimation). Estimating γ jointly with the outcome equation could in principle sharpen the first-stage bound of Theorem 4, since the varying-coefficient restriction lets the outcome carry information about γ . The possibility is examined in the Appendix. Any gain requires the specification to hold as the exact conditional mean of the outcome given (T, W, Z) , and the two-step procedure remains consistent for γ when it does not.

5.2 Consistency and Convergence Rate of the Second-Stage Estimator

The second stage uses the first-stage estimates as generated regressors, and their uniform accuracy governs the generated-regressor effect. It draws on them through the crude rates $\|\hat{\gamma} - \gamma_o\| = O_P(\rho_N)$ and, on the interior regions $\mathcal{Z}_N$ of Theorem 3, $\|P_{\hat{\xi}}^t - P_o^t\|_{L^\infty(\mathcal{Z}_N)} = O_P(K_N \rho_N)$, which Assumption 7(e) maintains. With X_i bounded, the index proxy obeys $\max_{1 \leq i \leq N} \|\hat{Z}_i - Z_i\| \leq \|\hat{\gamma} - \gamma_o\| \max_{1 \leq i \leq N} \|X_i\| = O_P(\rho_N)$. For sample points with $Z_i \in \mathcal{Z}_N$, the propensity proxy adds a Lipschitz index correction to the interior propensity error,

$$|\hat{P}_i^t - P_o^t(Z_i)| \leq M \|\hat{Z}_i - Z_i\| + \|P_{\hat{\xi}}^t - P_o^t\|_{L^\infty(\mathcal{Z}_N)} = O_P(K_N \rho_N),$$

the propensity error dominating the $O_P(\rho_N)$ index correction. The residualized index entering S_i obeys the sharper $O_P(\rho_N)$, since $\hat{Z}_i^{t,r} - \tilde{Z}_i^{t,r}$ is the within-judge

index error plus the error in the estimated residualization mean $\mathbb{E}[Z_i^{t,r} | W_i]$, both $O_P(\rho_N)$, the latter because that mean is a projection of the estimated index onto W_i and inherits its error. The generated regressor $\hat{S}_i = (\hat{P}_i^t, \hat{\tilde{Z}}_i^{t,r})$ therefore satisfies $\max_{i: S_i \in \mathcal{S}} \|\hat{S}_i - S_i\| = O_P(K_N \rho_N)$, the propensity coordinate binding. Accounting for these generated regressors is what the analysis below requires to establish the rate of the second-stage estimator $\hat{\tau}_{ts}(\cdot)$ introduced in Section 3.3.

Denote the second-stage regressor by $S_i = (P_i^t, \tilde{Z}_i^{t,r})$, with the residualized index $\tilde{Z}_i^{t,r} = Z_i^{t,r} - \mathbb{E}[Z_i^{t,r} | W_i]$, so that (9) reads $Y_i = W_i' \beta_{ts}(S_i) + \varepsilon_i$. Let $\hat{S}_i = (\hat{P}_i^t, \hat{\tilde{Z}}_i^{t,r})$ be the generated regressor. The full-support condition on f_o makes the map $z \mapsto (P_o^t(z), z^{t,r})$ injective, and residualization shifts the relative index only by $\mathbb{E}[Z_i^{t,r} | W_i]$, a function of W_i . Conditional on W_i , S_i is thus a one-to-one transformation of Z_i and conditioning on (S_i, W_i) is the same as conditioning on (Z_i, W_i) . Let $\mathcal{S}$ be a compact subset of the interior of $\text{supp}(S_i)$. The uniform rates below are stated on $\mathcal{S}$, whereas estimation uses the full t -subsample without trimming.

Assumption 7. *The following hold.*

- (a) (Error moment and design.) $\mathbb{E}[\varepsilon_i^2 | T_i, Z_i, W_i] \leq \bar{\sigma}^2$ and $\mathbb{E}[W_i W_i' | T_i, S_i] \geq \underline{c} \mathbf{I}_{\dim(W)}$ almost surely on $\{S_i \in \mathcal{S}\}$ for some constants $\bar{\sigma}^2, \underline{c} > 0$.
- (b) (Smoothness.) $\beta_{ts}(\cdot) \in C^\chi(\mathcal{S})$ componentwise for some $\chi \geq 2$, and the second-stage spline order satisfies $\tilde{q} \geq \chi + 1$.
- (c) (Regressor density and mesh.) S_i admits a density on $\mathcal{S}$ bounded above and below by positive constants, and the second-stage knot sequences are quasi-uniform.
- (d) (Sieve growth and coupling.) $\tilde{K}_N \rightarrow \infty$, $\tilde{K}_N^2 \log N = o(N)$, and $\tilde{K}_N^3 K_N \rho_N \rightarrow 0$.
- (e) (First-stage passthrough.) The first-stage estimators satisfy $\|\hat{\gamma} - \gamma_o\| = O_P(\rho_N)$ and, on the interior regions $\mathcal{Z}_N$ of Theorem 3, $\|P_\xi^t - P_o^t\|_{L^\infty(\mathcal{Z}_N)} = O_P(K_N \rho_N)$ for $t \in \{c, n\}$, with $K_N \rho_N \rightarrow 0$.

These conditions carry the first-stage regularity into the varying-coefficient regression (9), and they refer throughout to the regressor S_i and the interior region $\mathcal{S}$. Condition (a) fixes the noise level and the design. The error bound conditions on the index Z_i to match the mean-zero restriction in (9), which is the same as conditioning on the regressor S_i . The design block is the conditional full-rank condition for the regression. It identifies $\beta_{ts}(\cdot)$ at each point of $\mathcal{S}$ and keeps the least-squares problem

uniformly well-posed there. Condition (b) controls the spline bias. With $\tilde{q} \geq \chi + 1$, the same fit approximates $\beta_{ts}(\cdot)$ to $\tilde{K}_N^{-\chi}$ and its propensity-derivative $\partial_1 \beta_{ts}(\cdot)$ to $\tilde{K}_N^{-(\chi-1)}$ in the sup norm. The derivative is the term that enters the correction in $\tau_{ts}(\cdot)$. Condition (c) is why the uniform rate is stated on $\mathcal{S}$ rather than on all of $\text{supp}(S_i)$. The two-sided density bound and the quasi-uniform mesh keep the spline Gram well-conditioned, which the sup norm requires but the boundary would violate. Condition (d) caps the sieve dimension so the variance vanishes, and the entropy bound of the spline class then controls the empirical process induced by the generated regressor $\hat{S}_i$ without cross-fitting. The condition $\tilde{K}_N^3 K_N \rho_N \rightarrow 0$ couples the two sieves so that the generated regressor leaves the second-stage Gram asymptotically unperturbed. It is required for well-posedness and ties $\tilde{K}_N$ to the first-stage tuning K_N . Condition (e) records the first-stage input the second stage requires, the interior uniform rate $O_P(K_N \rho_N)$ of Theorem 3; the requirement $K_N \rho_N \rightarrow 0$ holds under $K_N^4 \log N = o(N)$. The rates below therefore hold whether or not $\hat{\gamma}$ is efficient, and the conditions of Assumption 6 need not be met.

The following theorem gives the rates of the second-stage estimators. The first-stage estimates enter only through the generated regressor $\hat{S}_i$, whose uniform level error is $O_P(K_N \rho_N)$ by Theorem 3.

Theorem 5. *(Convergence rates of the second-stage estimator.) Suppose Assumptions 5 and 7 hold. Then, for each margin (t, s) and each component $\iota = 1, \dots, \dim(W)$,*

$$\begin{aligned} \|\hat{\beta}_{ts,\iota} - \beta_{ts,\iota}\|_{L^\infty(\mathcal{S})} &= O_P \left(\tilde{K}_N \sqrt{\log N/N} + \tilde{K}_N^{-\chi} + K_N \rho_N \right), \\ \|\widehat{\partial_1 \beta}_{ts,\iota} - \partial_1 \beta_{ts,\iota}\|_{L^\infty(\mathcal{S})} &= O_P \left(\tilde{K}_N^2 \sqrt{\log N/N} + \tilde{K}_N^{-(\chi-1)} + \tilde{K}_N K_N \rho_N \right). \end{aligned}$$

Consequently, the margin-specific mean potential outcome $\hat{\mu}_{ts}$ and treatment effect $\hat{\tau}_{ts}$ converge uniformly at the derivative rate,

$$\begin{aligned} \sup_{(z,w)} |\hat{\mu}_{ts}(z, w) - \mu_{ts}(z, w)| &= O_P \left(\tilde{K}_N^2 \sqrt{\log N/N} + \tilde{K}_N^{-(\chi-1)} + \tilde{K}_N K_N \rho_N \right), \\ \sup_{(z,w)} |\hat{\tau}_{ts}(z, w) - \tau_{ts}(z, w)| &= O_P \left(\tilde{K}_N^2 \sqrt{\log N/N} + \tilde{K}_N^{-(\chi-1)} + \tilde{K}_N K_N \rho_N \right), \end{aligned}$$

with the suprema running over (z, w) whose arguments $(P_o^t(z), \tilde{z}^{t,r})$ and $(P_o^s(z), \tilde{z}^{s,r})$ lie in $\mathcal{S}$.

Remark 4 (Efficiency is not needed for the second-stage rates). Theorem 5 draws on the first stage only through the generated regressor $\hat{S}_i$ at its crude uniform rate $\|\hat{S}_i - S_i\|_\infty = O_P(K_N \rho_N)$ on the interior region, which is Assumption 7(e). Theorem 3 supplies this rate under Assumptions 4 and 5. Notably, the first-stage efficiency of Theorem 4 is not strictly needed for the second-stage rates. Any first-stage estimator attaining the crude rate is therefore enough. The second-stage guarantees thus stand apart from the first-stage efficiency theory.

Each rate splits into a sieve variance term, a sieve bias term, and a first-stage passthrough term. For the level these are $\tilde{K}_N \sqrt{\log N/N}$ and $\tilde{K}_N^{-\chi}$, the usual two-dimensional series-regression terms. The derivative carries one extra factor $\tilde{K}_N$ in all three terms, which is the cost of differentiating the spline fit. The passthrough enters through the generated regressor $\hat{S}_i$, with $\|\hat{S}_i - S_i\|_\infty = O_P(K_N \rho_N)$ on the interior region. By Lipschitz continuity of $\beta_{ts}(\cdot)$ it contributes ρ_N to the level and $\tilde{K}_N \rho_N$ to the derivative. Both vanish under Condition (d). The estimated treatment effect function $\hat{\tau}_{ts}(\cdot)$ depends on $\widehat{\partial_1 \beta_{ts}}(\cdot)$ through the correction term. It therefore converges at the slower derivative rate.

6 Empirical Application

6.1 Data and Background

We study felony cases in the Virginia Circuit Courts, the same setting and data source as [Humphries et al. \(2025\)](#) (henceforth HOSSvD). Each case is assigned to one of the three dispositions d , c , n and to a judge whose assignment is as good as random within cells differentiated by circuit and time of disposition. Table 6 in the Appendix reports the correlations of the estimated judge-specific coefficients with case characteristics. Our extract spans 2000–2019 across 13 circuits and keeps 137 judges with more than 500 cases. We further drop judges with fewer than 50 cases involving female defendants, black defendants, or defendants convicted within the previous five years, so that each judge’s slopes on these covariates are estimated with adequate precision. This results in a full sample of 256,475 cases disposed by 129 judges. HOSSvD instead keep judges with at least 100 cases within each court-by-rolling-three-year-window cell. As a robustness check, we double the minimum caseload to 1,000 cases and find that the results are largely unchanged, so we report

the larger sample to maximize precision.

For our varying-coefficient second stage, we restrict to cases disposed through 2012 as in HOSSvD. The cutoff here serves the same purpose as for them. It allows each case the full seven-year follow-up that the longest recidivism window requires. We then trim the 0.1% of cases in the upper and lower tails of the recovered index. Specifically, we retain cases whose fitted indices $\hat{Z}_i^c$ and $\hat{Z}_i^n$ are both within the 0.025% and 99.975% percentiles of their respective sample distributions. This removes 148 outliers and leaves 169,013 cases for the second stage.

The covariate vector W_i collects female and black indicators, three offense-type indicators for violent, property, and drug charges, which are on if any charge of the corresponding type is present, and the indicator for prior conviction within the last five years of disposition. Three further covariates are the time since the last prior conviction, the number of charges, and the number of misdemeanor charges, which enter as continuous regressors. The time since the last prior conviction is measured in days and is genuinely continuous. The number of charges and the number of misdemeanor charges are counts, but they provide fine-grained variation sufficient for our purposes, as the spread of the recovered index in Figure 2 illustrates. Including the intercept, W_i has dimension 10 and there are 20 parameters per judge in the first-stage model. Our variable construction closely follows that in HOSSvD.

We also follow HOSSvD in the outcomes we study. The outcome of interest is recidivism, which is measured three ways: a new felony charge, a new felony conviction, and a new carceral sentence. We track each over three follow-up windows: the first year, years two to four, and years five to seven after disposition. HOSSvD extend the binary judge-IV design to this multiple-treatment setting with 2SLS. They conclude that noncarceral conviction relative to dismissal is a substantial and long-lasting driver of recidivism, while incarceration relative to noncarceral conviction produces a short-run reduction consistent with incapacitation. We take these findings as the benchmark and revisit them with the estimators in Sections 3.1 and 3.3.

Table 2 reports summary statistics by disposition, with rows and variable definitions matching Table I of HOSSvD. The entries are virtually identical to theirs. The two samples differ only in the judge screens, which cost about eight percent of their observations and leave the composition of the three disposition groups unchanged. The divergence documented in Section 6.2 is therefore not driven by sample composition.

Table 2: Summary Statistics by Disposition

	Dismissed (1)	Convicted (2)	Incarcerated (3)
<i>Offenses</i>			
Drugs	0.36	0.33	0.29
Larceny	0.17	0.29	0.25
Assault	0.19	0.08	0.18
Fraud	0.09	0.16	0.10
Traffic	0.04	0.05	0.13
Burglary	0.06	0.07	0.08
Robbery	0.04	0.02	0.06
Sexual assault	0.03	0.02	0.03
Kidnapping	0.03	0.01	0.02
Murder	0.01	0.00	0.01
<i>Defendant characteristics</i>			
Black	0.57	0.51	0.60
Female	0.22	0.32	0.16
% of people in ZIP earning < \$25K	0.46	0.45	0.46
Has misdemeanor	0.06	0.09	0.07
Prior conviction within five years	0.14	0.09	0.20
<i>Sentencing</i>			
Incarceration length	0.00	0.00	26.97
Probation length	0.00	32.08	39.86
Median incar. length	0	0	12
Median prob. length	0	12	12
Percent of sample	16	30	55
Observations	26,313	50,447	92,401

Notes. This table reports means, and medians where indicated, of case characteristics by disposition for the outcome sample of cases disposed through 2012, before the index trim described in the text. The columns correspond to cases ending in dismissal, noncarceral conviction, and incarceration. The rows and variable definitions follow Table I of HOSSvD. Incarceration and probation lengths are means in months, with probation top-coded at 20 years.

6.2 Tracing the Divergence from HOSSvD

The latent-return model of Section 2.1 is the basis for our first stage. We take it to the data by augmenting with time effects, and we set it against two reference specifications: the judge-fixed-effects model that HOSSvD estimate as an alternative to their 2SLS design, and an intermediate logit that shares the judge-specific slopes of our model but imposes a parametric distribution. The three differ in two features: whether the slopes on W_i vary across judges, and whether the distribution of the unobserved returns is restricted. Comparing them shows what each feature contributes over their benchmark.

We estimate the latent-return model of Section 2.1, augmented with common time effects:

$$R_i^t(j) = \underbrace{W_i' \gamma_j^t}_{Z_i^t} + W_i' \gamma_{\text{year}(i)}^t + \alpha_{\text{month}(i)}^t + \alpha_{\text{dow}(i)}^t + U_i^t, \quad t \in \{c, n\},$$

where $\gamma_{\text{year}(i)}^t$ includes both year fixed effects and year-specific slopes on the non-intercept covariates, and $\alpha_{\text{month}(i)}^t$ and $\alpha_{\text{dow}(i)}^t$ are month and day-of-week fixed effects for treatment t . These year slopes and time fixed effects are common to all judges, and absorb temporal variation in the disposition process. Each judge serves in a single circuit, so circuit-level differences are spanned by the judge intercepts. The specification includes no circuit-by-window interaction unlike HOSSvD: in a companion fit that adds these cells, a Wald test does not reject additivity of the circuit and window effects on either axis, with details relegated to the Appendix. Because the time effects are judge-invariant, they carry no cross-judge variation and do not enter the index residual: for a case with $J_i = j$, the circuit-partialled index residual $\tilde{Z}_i^t$ is

$$\tilde{Z}_i^t = \left(\gamma_j^t - \underbrace{\frac{\sum_{j' \in \text{circ}(j)} \lambda_{j'} \gamma_{j'}^t}{\sum_{j' \in \text{circ}(j)} \lambda_{j'}}}_{\text{within-circuit mean}} \right)' W_i, \quad t \in \{c, n\}, \quad (11)$$

where $\text{circ}(j)$ denotes the circuit containing judge j , so that $\tilde{Z}_i^t$ measures judge j 's leniency relative to its circuit. We remove the circuit effect because it is not a source of random variation. This within-circuit centering generalizes the all-judge centering of the residualized instrument in Section 2.2, which is needed because assignment is

random only within a circuit.

The time effects nonetheless aid identification rather than hinder it. Since the support of W_i varies little with time, the time effects supply a separate source of variation that spreads each judge’s index support across periods, enlarging the overlap between judges and helping to satisfy the connectivity condition of Assumption 4. The identification analysis of Section 2.1 carries over to this specification unchanged. Because the common time terms enter additively, each condition of Assumption 4 applies with judge j ’s index support taken as that of the full systematic index above rather than of $W_i' \gamma_j$ alone, which the time terms only enlarge. The recovered indices corroborate these conditions visually. The index cloud in Figure 2 below varies continuously, reflecting the fine-grained variation the count covariates supply, and is nearly connected across judges, in line with the connectivity condition.

As an alternative to their main 2SLS specification, HOSSvD estimate a restricted, judge-fixed-effects version of the same model, fitted separately for each circuit and year: for judge j in a given cell, the latent returns are

$$R_i^t(j) = W_i' \gamma_{\text{circ} \times \text{year}}^t + \alpha_{j \times \text{circ} \times \text{year}}^t + U_i^t, \quad t \in \{c, n\}, \quad (12)$$

where (U_i^c, U_i^n) is specified as a logistic error. Each cell is indexed by a circuit and a year, but is estimated on the three years centered at that year, so the windows of consecutive years overlap in two of their three years. The intercept in $\gamma_{\text{circ} \times \text{year}}^t$ and the level of $\alpha_{j \times \text{circ} \times \text{year}}^t$ are indistinguishable within a cell, so only their sum is identified. They use this sum as the instrument for cases handled by judge j in that $\text{circ} \times \text{year}$ cell. The specification carries no year, month, or day-of-week term, so these effects are pooled within each circuit-by-year cell. HOSSvD use a mixed logit as the main specification. We impose the plain logit here, which is computationally simpler, and their Appendix shows that the second-stage estimates change little between the two.⁴ HOSSvD construct the thresholds within each judge-by-circuit-by-window cell as the log of the c and n shares of the judge’s cases relative to the d share.⁵ We instead estimate (12) by maximum likelihood, so the instrument in Table 3 may differ in levels from their reported estimates. The slope structure of (12) and that of our

⁴See HOSSvD’s Online Appendix F and Tables F.1 and F.2, which report the alternative approach under a plain logit and under a diagonal random-effects specification and find results broadly similar to the correlated mixed logit.

⁵See their Online Appendix F.2 and their replication code.

augmented specification are not nested: we decompose the slope on W_i additively into a time-invariant judge component γ_j^t and a common year component γ_{year}^t shared across judges, whereas HOSSvD let it vary by circuit and window but not by judge. For the instrument, what matters is that HOSSvD’s judge enters only as a level shift, uniform across his cases, so he contributes only a discrete, one-dimensional source of propensity variation. Re-estimating the same judge separately in each circuit-by-year cell mitigates this: the judge is refitted cell by cell, so a judge carries a distinct value in every year he serves, which enriches the variation. This comes at the cost of the sample available per cell compared to its pooled alternative. Our semiparametric specification lifts both restrictions of (12): the cell-common slope $\gamma_{\text{circ} \times \text{year}}^t$ now varies with judge and the logit error gives way to an unrestricted distribution of U_i^t . To separate the two, we introduce an intermediate specification, the *interactive-effects logit*, which adopts the judge-specific slopes but retains a multinomial logit distribution. Reading across HOSSvD, the intermediate logit, and our semiparametric model isolates the added slope flexibility from the relaxed distribution.

We estimate the semiparametric model by spline sieve maximum likelihood of order $q = 5$ (quartic splines). Other tuning parameters—the knot box, spacing, and their resolution—and their robustness are relegated to the Appendix. The interactive-effects logit profiles the distribution at the logistic and is estimated by maximum likelihood. The judge-fixed-effects model is also estimated via maximum likelihood cell by cell with rolling three-year windows.

HOSSvD’s specification does not deliver a single likelihood comparable to the others, since it is fitted cell by cell on separate subsamples, so we compare it only through the second-stage estimates below. The interactive-effects logit and the semiparametric model, by contrast, are both estimated on the full sample and admit a single log-likelihood, which we return to after the slopes.

Judge-specific coefficients. Figure 1 compares the recovered coefficients of the two fits. Each panel plots the kernel density, across the 129 judges, of the judge-specific slope on one covariate in the conviction index z^c or the incarceration index z^n , overlaying the semiparametric and interactive-effects logit estimates. Since the two fits are identified up to their own normalizations, both are placed on a common gauge before overlaying: judge intercepts are centered across judges on each axis, and the slope block of each fit is rescaled to a common norm.

The figure carries two messages. First, the slopes disperse substantially across judges under either fit, and this is precisely the variation that the judge-fixed-effects specification (12) excludes. For most covariates the cross-judge spread is comparable to the mean loading, and the sign pattern is the same under either fit. For covariates such as the female indicator, judges agree in sign even as the magnitude varies: the slope is positive in the conviction index and negative in the incarceration index for nearly every judge. For others, such as the black indicator in the incarceration index, the judges split, with a large majority positive and a sizable minority negative. That this pattern of agreement and disagreement is the same whether or not the distribution of unobserved returns is restricted is itself informative: the qualitative structure of judge heterogeneity does not depend on which of the two, quite different, estimation approaches recovers it.

Second, the central tendency of the two fits nearly coincides for almost every covariate and both margins. The semiparametric and logit densities are centered at close to the same point. The pooled, judge-averaged loading is therefore not sensitive to whether the distribution of unobserved returns is restricted. The two fits diverge instead in the shape of the cross-judge distribution, and this divergence is far more pronounced on the incarceration index than on the conviction index. On the conviction axis the two densities nearly overlap for most covariates. On the incarceration axis the logit density is often markedly flatter and more diffuse than the semiparametric one. For the intercept it even splits into two separate modes that the semiparametric fit does not show. The loading on prior conviction in the incarceration index, the strongest in the model, is a case in point: the semiparametric fit places a tall, narrow peak with a long right tail, while the logit spreads the same mass over a wider and shorter range. Since the semiparametric estimator is consistent under either specification, this shape divergence reads as distortion from the distributional restriction that falls disproportionately on the incarceration margin, in light of the rejection below. These patterns accord with the comparison below, where the two fits agree most on the conviction index, less on the incarceration index, and least on the joint structure of the two.

The unrestricted distribution is favored over the interactive-effects logit. The two fit the choice data comparably in aggregate: log-likelihoods of $-232,550$ for the logit and $-232,025$ for the semiparametric fit. This comparison is, however, descriptive rather than a formal test. We assess the two instead through the Hausman test below.

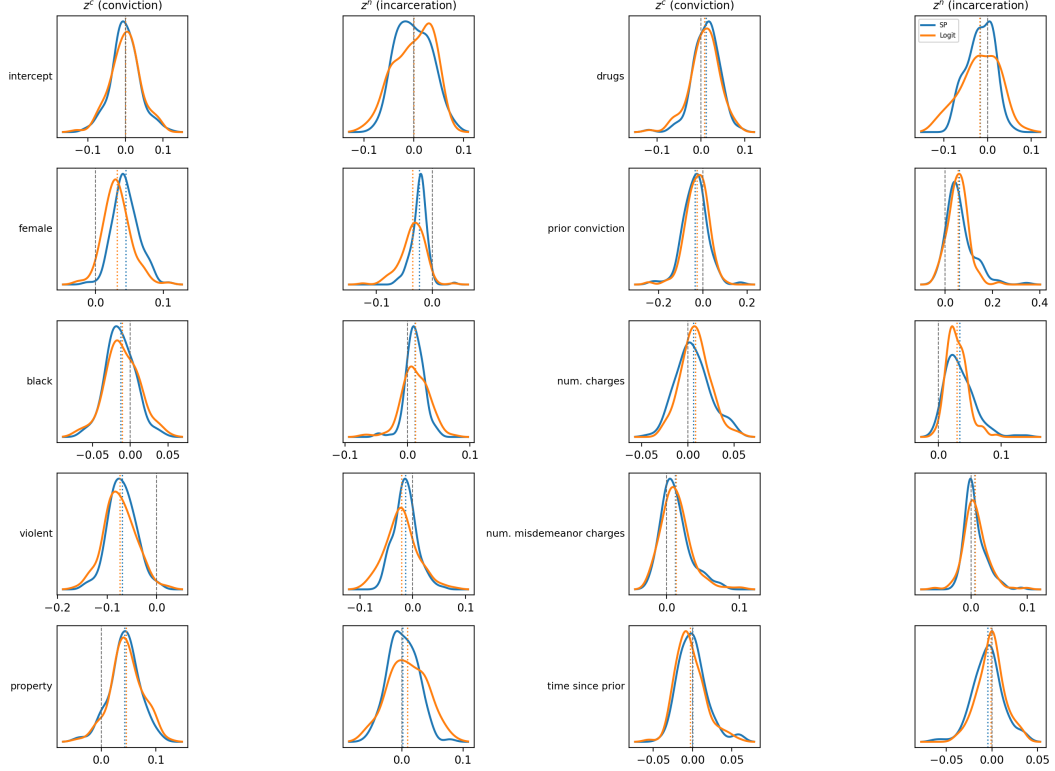
The Hausman test can be developed from the asymptotic normality result of Theorem 4 and the fact that the interactive-effects logit, which fixes the distribution of unobserved returns at the logistic distribution, is a parametric submodel of the semiparametric specification up to the sieve approximation of the propensity function. This error decays faster than stochastic order $N^{-1/2}$, so the bias is asymptotically negligible. Under the null that the logit is correctly specified, its maximum likelihood estimator is efficient, so the difference between the two estimators is asymptotically normal with mean zero and covariance given by the difference of their asymptotic variances. We apply the test judge by judge rather than as a single joint test, for two reasons. First, the judge-level heterogeneity is of direct interest, and the second stage consumes the coefficients judge by judge, so the judge level is where the restriction needs to hold. Second, the calibration of the test rests on the difference of two estimated covariance matrices, which is better behaved for a single judge’s 20 coefficients than for the 2,580 judge-specific coefficients at once. For judge j , the test statistic is given by

$$N(\text{vec}(\hat{\gamma}_{\text{semi},j}) - \text{vec}(\hat{\gamma}_{\text{logit},j}))' \left([\hat{\Sigma}_{\text{semi},j} - \hat{\Sigma}_{\text{logit},j}]_+ \right)^+ (\text{vec}(\hat{\gamma}_{\text{semi},j}) - \text{vec}(\hat{\gamma}_{\text{logit},j})) \approx \chi_{r_j}^2,$$

where $\hat{\gamma}_{\bullet,j}$ collects the estimated coefficients of judge j , $\hat{\Sigma}_{\bullet,j}$ denotes the corresponding block of the estimated asymptotic covariance matrix of $\sqrt{N}(\text{vec}(\hat{\gamma}_{\bullet}) - \text{vec}(\gamma_o))$ for each estimator ($\bullet = \text{semi}, \text{logit}$), and $[\cdot]_+$ denotes the matrix positive part. We restrict the statistic to the judge-specific coefficients and profile out the time effects and the common year slopes as nuisance parameters. The result is unchanged when the time effects are included. The degrees of freedom r_j equal the rank of $\text{plim}[\hat{\Sigma}_{\text{semi},j} - \hat{\Sigma}_{\text{logit},j}]_+$, which we estimate by the rank of its sample counterpart. Of the 129 judges, 56 reject the logit after a Benjamini–Hochberg correction for multiple testing at the 5% false discovery rate. The distributional restriction thus binds for roughly two-fifths of the judges individually.

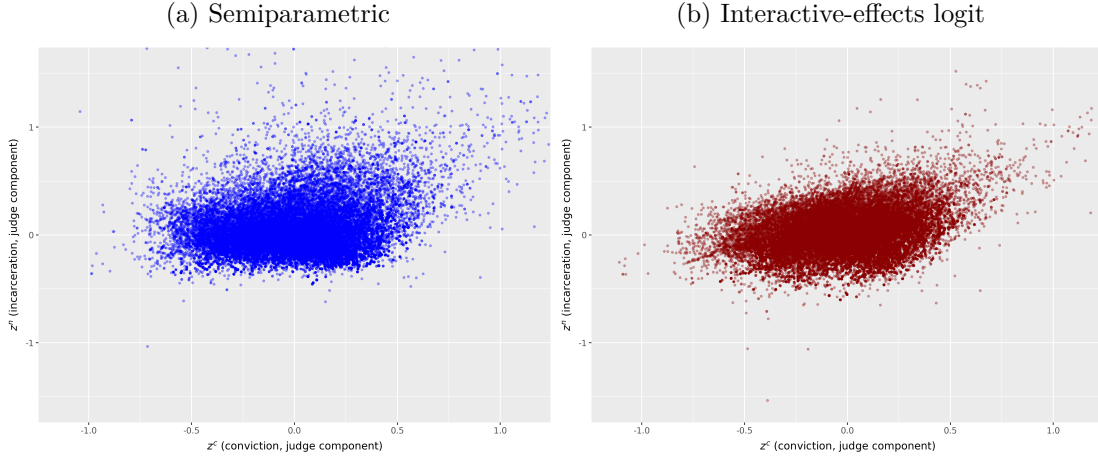
The two fits also differ in the index cloud of Figure 2, and this difference bears directly on the second stage. The index under the two fits correlate at 0.97 on the conviction axis and 0.88 on the incarceration axis. These correlations may look reassuring, but a correlation of 0.88 leaves a discrepancy whose variance is about 22% of that of the index itself. The conviction and incarceration indices have a correlation of 0.33 under the logit but their correlation is milder (0.17) under the semiparametric

Figure 1: Estimated Marginal Coefficient Distributions by First-Stage Specification



Notes: Each panel shows the kernel-smoothed density of the judge-specific coefficient across the judges on the labeled covariate in the conviction index z^c or the incarceration index z^n , for the semiparametric and interactive-effects logit fits. The Gaussian kernel is used and the bandwidth is set to $\sigma J^{-1/5}$ with σ the sample standard deviation of each coefficient across the judges under each fit and $J = 129$. Continuous covariates (number of charges, number of misdemeanor charges, and time since last conviction) are standardized to have unit variance. The two fits are placed on a common gauge: intercepts are centered across judges within each axis, and each fit is rescaled so that the covariance-weighted Frobenius norm is the same. Dotted vertical lines mark the cross-judge means and the dashed line marks zero. The cross-judge spread reflects per-judge estimation noise in addition to judge heterogeneity.

Figure 2: Recovered Index by First-Stage Specification



Notes: Each panel scatters the recovered conviction index Z_i^c against the incarceration index Z_i^n across the sample. The standard deviation of the conviction index is 0.241 in panel (a) and 0.232 in panel (b), and the standard deviation of the incarceration index is 0.201 in panel (a) and 0.181 in panel (b).

fit. Whether this difference in the joint structure moves the estimated effects is a separate, quantitative question, and Tables 4 and 5 below speak to it at HOSSvD's evaluation point. Identification of the (t, s) margin varies the propensity P^t while holding the relative index $z^{t,r} = z^t - z^r$ fixed, isolating the (t, s) -margin comparison as in Section 2.2. This conditioning is as reliable as the recovered indices. If the fitted joint structure misstates the true one, conditioning on the fitted $z^{t,r}$ no longer holds the true relative index fixed, and variation in P^t mixes flows across margins. The semiparametric fit remains consistent whether or not the logistic restriction holds.

How the first stage revises HOSSvD. To compare the first stages on common ground, we hold the second stage fixed at HOSSvD's, the estimator of Mountjoy (2022) described in the notes to Table 3, and vary only the instrument. The three tables use, in order, HOSSvD's judge thresholds, estimated by maximum likelihood of (12), our semiparametric index, and the interactive-effects logit index.

On the (c, d) margin, HOSSvD's instrument delivers sizable noncarceral-conviction effects, significant in the first four years and over the cumulative window, with weaker significance in years five to seven. Our semiparametric index attenuates them at every horizon. The first-year estimates fall by about half, and only the effect on

new incarceration remains significant at the five percent level. In years two to four the estimates for new charges and new convictions stay within about a quarter of HOSSvD's and remain significant at the one percent level, whereas the estimate for new incarceration halves and loses significance. Beyond four years the revision is sharper. The years-five-to-seven estimates fall well below HOSSvD's and are insignificant for all three measures, and the cumulative estimates retain between a quarter and a third of HOSSvD's magnitude and are no longer significant. The interactive-effects logit tracks the semiparametric index closely. Outside the cumulative window the two differ by no more than about a percentage point in every cell, and in the cumulative window the logit estimates lie nearer zero. Because the two index-based first stages share the judge-specific slopes and revise this margin in nearly the same way, the attenuation traces to the slopes rather than to the distribution of the unobserved returns.

On the (n, c) margin, HOSSvD's instrument shows a first-year reduction that is significant for new charges and new convictions, the incapacitation effect. Both index-based first stages sharpen it. The first-year reduction is roughly half again as large for those two measures and is significant for new incarceration as well. The two also read the later windows differently from HOSSvD. HOSSvD's instrument turns mildly positive in years two to four, marginally significant for new incarceration, whereas under either index the years-two-to-four estimates sit at or below zero and are significantly negative for new convictions. Neither index delivers a significant estimate in years five to seven or over the cumulative window, where all three measures lie close to zero. Under our first stage the early reduction therefore does not give way to an offsetting increase within the seven-year horizon, though the cumulative effects cannot be distinguished from zero.

The comparison across the three first stages serves an illustrative purpose. The two index-based fits agree in sign in nearly every cell, their confidence intervals overlap throughout, and the cells in which their significance differs are ones where both estimates sit near a threshold. On these tables the logistic restriction leaves little visible trace once the judge-specific slopes are in place. Where it registers, it does so in the cumulative window of the (c, d) margin, in which the logit estimates lie closer to zero than ours.

These tables attribute the observed differences to the first stage. They hold HOSSvD's second stage fixed and evaluate it at the mean of the instruments, and the same second stage is imposed in all three, which is what leaves the differences

across them attributable to the instrument. Their levels are not comparable to the estimates of Section 6.3.⁶ These tables also carry no panel for the (n, d) margin, on which HOSSvD report nothing.

6.3 Effect Heterogeneity from the Varying-Coefficient Second Stage

Our main results use the second stage of Section 3.3, augmented with the circuit and time fixed effects to absorb cross-circuit and temporal variation not mediated by the index. The specification parallels the augmented first stage. For each margin (t, s) , we estimate the outcome regression on the subsample with $T_i = t$,

$$Y_i = W_i' \beta_{ts}(S_i) + W_i' \beta_{\text{year}(i)} + \alpha_{\text{circ}(J_i)} + \alpha_{\text{month}(i)} + \alpha_{\text{dow}(i)} + \varepsilon_i, \quad \hat{S}_i = (\hat{P}_i^t, \hat{Z}_i^{t,r}),$$

where $\beta_{\text{year}(i)}$ includes year fixed effects and year-specific slopes on the non-intercept covariates, $\alpha_{\text{circ}(J_i)}$ is a circuit fixed effect corresponding to the assigned judge, and $\alpha_{\text{month}(i)}$ and $\alpha_{\text{dow}(i)}$ are the time fixed effects same as in the first stage. The index residual is constructed following (11) to eliminate the circuit effect: first, regress $\hat{\gamma}_{J_i}^t$ on the circuit indicator and take the residuals; then interact the residuals with W_i . The augmented specification does not undermine consistency of the second stage established in Section 5.2, with only finite-dimensional parameters added and not generated regressors. The margin-specific effect function $\tau_{ts}(\cdot)$ is then estimated from $\hat{\beta}_{ts}(\cdot)$ as described in Section 3.3. The coefficient functions are formed by tensor B-splines of order three with four basis functions per axis (16 basis functions in total). We compare this choice with two richer bases, six basis functions per axis at order three and five per axis at order four, and the estimates are largely unchanged. This choice of tuning parameters and its robustness are examined in the Appendix.

For presentation, we summarize the estimates through covariate-cell averages. For a cell $\mathcal{W}$, formed by fixing one of the six binary case characteristics at zero or one,

⁶Under the varying-coefficient model, this estimator does not identify the same object as our second stage does. It enters the slopes on the residualized covariates as locally constant, whereas the specification of Section 3.2 lets them vary with the propensity.

Table 3: Margin-Specific Treatment Effects: HOSSvD’s Specification

	Year 1	Years 2-4	Years 5-7	Years 1-7
Panel A: Noncarceral conviction vs dismissal (C vs D)				
Felony charge:	0.071** [0.006,0.153] {0.075}	0.128** [0.008,0.262] {0.179}	0.117* [-0.004,0.251] {0.144}	0.184** [0.028,0.406] {0.351}
Felony conviction:	0.091*** [0.021,0.176] {0.052}	0.155*** [0.053,0.268] {0.137}	0.096* [-0.000,0.238] {0.136}	0.247*** [0.100,0.435] {0.277}
Felony incarceration:	0.073** [0.019,0.145] {0.046}	0.113** [0.024,0.231] {0.117}	0.060 [-0.042,0.176] {0.101}	0.176*** [0.020,0.342] {0.255}
Panel B: Incarceration vs noncarceral conviction (N vs C)				
Felony charge:	-0.037** [-0.067,-0.003] {0.075}	0.048 [-0.013,0.124] {0.162}	0.023 [-0.045,0.103] {0.132}	-0.010 [-0.107,0.099] {0.320}
Felony conviction:	-0.029* [-0.061,0.003] {0.065}	0.036 [-0.022,0.111] {0.149}	0.023 [-0.041,0.090] {0.118}	-0.004 [-0.100,0.098] {0.292}
Felony incarceration:	-0.006 [-0.032,0.020] {0.045}	0.053* [-0.005,0.109] {0.092}	0.016 [-0.046,0.077] {0.089}	-0.013 [-0.102,0.077] {0.215}
Controls	Yes	Yes	Yes	Yes

Notes. The three tables estimate margin-specific treatment effects by the method of Mountjoy (2022), held fixed across tables so that only the first-stage instrument differs. Each margin’s effect is recovered from local linear regressions of the treatment indicators and of their products with the outcome on the two recovered indices, evaluated at the mean of the indices; ratios of the local slopes give the complier mean potential outcomes, whose differences are the reported effects. This table uses HOSSvD’s judge thresholds, estimated by maximum likelihood of the judge-fixed-effects model described in the text; HOSSvD construct theirs as the log of treatment shares relative to the dismissal share within each judge-by-circuit-by-window cell, so levels may differ from their reported estimates. Panel A is noncarceral conviction versus dismissal, Panel B incarceration versus noncarceral conviction. All specifications include the case-characteristic controls of Section 6.1 along with circuit and year fixed effects. Following HOSSvD, each outcome is estimated on the longest disposition window that its follow-up horizon permits: through 2018 for the first year, through 2015 for years two to four, and through 2012 for years five to seven and for the cumulative seven-year window. 95% confidence intervals from a cluster bootstrap with 500 replications that resamples judge-by-year-by-court blocks are in brackets; control-group complier means are in braces (the dismissal-complier mean in Panel A, the noncarceral-conviction-complier mean in Panel B). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Margin-Specific Treatment Effects: Semiparametric

	Year 1	Years 2-4	Years 5-7	Years 1-7
Panel A: Noncarceral conviction vs dismissal (C vs D)				
Felony charge:	0.055 [-0.011,0.122] {0.093}	0.160*** [0.051,0.264] {0.174}	0.006 [-0.085,0.109] {0.144}	0.066 [-0.062,0.229] {0.363}
Felony conviction:	0.060* [-0.011,0.130] {0.072}	0.164*** [0.067,0.273] {0.137}	0.041 [-0.062,0.137] {0.116}	0.099 [-0.054,0.229] {0.304}
Felony incarceration:	0.055** [0.001,0.113] {0.040}	0.078 [-0.015,0.170] {0.116}	0.003 [-0.076,0.083] {0.093}	0.060 [-0.056,0.194] {0.247}
Panel B: Incarceration vs noncarceral conviction (N vs C)				
Felony charge:	-0.046*** [-0.074,-0.020] {0.089}	-0.010 [-0.053,0.035] {0.158}	0.023 [-0.020,0.074] {0.135}	0.012 [-0.050,0.077] {0.300}
Felony conviction:	-0.039*** [-0.065,-0.014] {0.077}	-0.025 [-0.065,0.019] {0.150}	0.028 [-0.021,0.075] {0.118}	-0.007 [-0.072,0.057] {0.283}
Felony incarceration:	-0.019* [-0.042,0.003] {0.054}	0.012 [-0.026,0.049] {0.110}	0.032 [-0.008,0.076] {0.078}	0.041 [-0.015,0.109] {0.202}
Controls	Yes	Yes	Yes	Yes

Notes. As in Table 3, with our semiparametric index as the first-stage instrument.

Table 5: Margin-Specific Treatment Effects: Interactive-Effects Logit

	Year 1	Years 2-4	Years 5-7	Years 1-7
Panel A: Noncarceral conviction vs dismissal (C vs D)				
Felony charge:	0.042 [-0.016,0.093] {0.088}	0.098** [0.010,0.193] {0.172}	0.018 [-0.063,0.110] {0.143}	0.012 [-0.095,0.141] {0.363}
Felony conviction:	0.047* [-0.011,0.108] {0.070}	0.109** [0.019,0.194] {0.136}	0.043 [-0.047,0.125] {0.113}	0.035 [-0.096,0.148] {0.305}
Felony incarceration:	0.041** [0.000,0.089] {0.036}	0.052 [-0.036,0.128] {0.114}	0.017 [-0.053,0.082] {0.085}	0.006 [-0.108,0.125] {0.242}
Panel B: Incarceration vs noncarceral conviction (N vs C)				
Felony charge:	-0.055*** [-0.080,-0.028] {0.095}	-0.036* [-0.076,0.011] {0.181}	0.020 [-0.023,0.067] {0.137}	-0.018 [-0.079,0.040] {0.327}
Felony conviction:	-0.047*** [-0.070,-0.024] {0.083}	-0.044** [-0.083,-0.003] {0.171}	0.022 [-0.026,0.065] {0.124}	-0.033 [-0.090,0.025] {0.312}
Felony incarceration:	-0.024** [-0.044,-0.004] {0.059}	-0.002 [-0.038,0.030] {0.121}	0.030 [-0.009,0.071] {0.081}	0.013 [-0.039,0.075] {0.227}
Controls	Yes	Yes	Yes	Yes

Notes. As in Table 3, with the interactive-effects logit index as the first-stage instrument.

the average effect on the (t, s) margin is

$$\bar{\tau}_{ts}(\mathcal{W}) = \frac{\sum_{i=1}^{\tilde{N}} \mathbb{1}\{W_i \in \mathcal{W}\} \mathbb{1}\{\hat{S}_i \in \mathcal{S}\} \hat{\tau}_{ts}(W_i, \hat{Z}_i)}{\sum_{i=1}^{\tilde{N}} \mathbb{1}\{W_i \in \mathcal{W}\} \mathbb{1}\{\hat{S}_i \in \mathcal{S}\}},$$

where $\tilde{N} = 169,013$ denotes the outcome sample size, and the interior region $\mathcal{S}$ collects the cases with squared Mahalanobis distance $(\hat{Z}_i - \tilde{N}^{-1} \sum_{i=1}^{\tilde{N}} \hat{Z}_i)' \hat{V}_{\hat{Z}}^{-1} (\hat{Z}_i - \tilde{N}^{-1} \sum_{i=1}^{\tilde{N}} \hat{Z}_i)$ below its 0.99 sample quantile, a criterion that appears reasonable in light of the index support in Figure 2. Here, $\hat{V}_{\hat{Z}}$ is the covariance of $\hat{Z}_i$ within the outcome sample. Individual effects are clipped to $[-1, 1]$, the logical range of an effect on a binary outcome, before averaging. On average, 0.31% of cases are affected by this winsorization across all outcomes and margins, with the worst case being new convictions over years one to seven on the (n, c) margin in which 0.85% of the cases are affected. Overall, the winsorization does not materially shift the results we report. Because each cell fixes one characteristic at a time, the cells overlap and do not partition the sample. The complement of an offense-type cell, for instance, pools all cases without that charge type. We report each measure cumulated over years one to four and years one to seven. The measures are the new felony charge and the new incarceration, together with a third, the new noncarceral conviction, defined as the difference between the new-conviction and new-incarceration indicators. Separating the noncarceral convictions from the carceral ones reveals into which type of reoffense each margin pushes the marginal case. The 95% confidence intervals are percentile intervals from a cluster bootstrap with 200 replications, which resamples judge-by-year-by-court blocks and re-estimates the second stage while holding the first-stage estimates fixed. This convention parallels HOSSvD, whose bootstrap also holds the first-stage fixed, though it likely understates the uncertainty from the generated regressors.

Figures 3 and 4 present the cell averages, and reading across the measures within each margin shifts the emphasis of HOSSvD from noncarceral conviction to incarceration. Noncarceral conviction relative to dismissal feeds the bottom of the severity ladder. The (c, d) effect on new charges is positive in most cells. Its effect on new noncarceral convictions is positive in nearly every cell, typically smaller than the charge effect. Its effect on new incarceration is comparable in magnitude to the noncarceral-conviction effect, sometimes larger, though less consistently separated from zero. Incarceration relative to noncarceral conviction mostly shifts the composition of recidivism rather than its level. The (n, c) effect on new noncarceral

Figure 3: Cell Average Effects by Margin: Cumulative Years 1 to 4

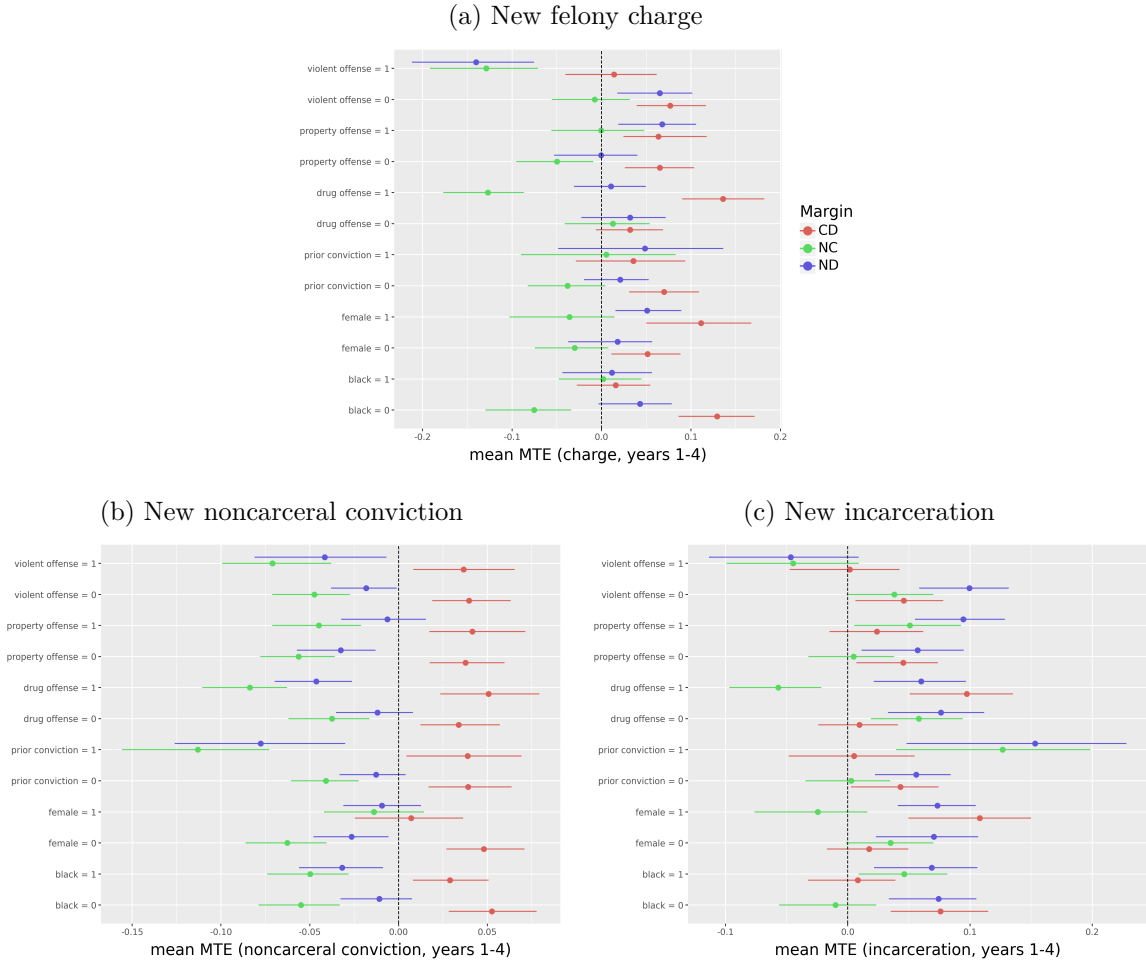
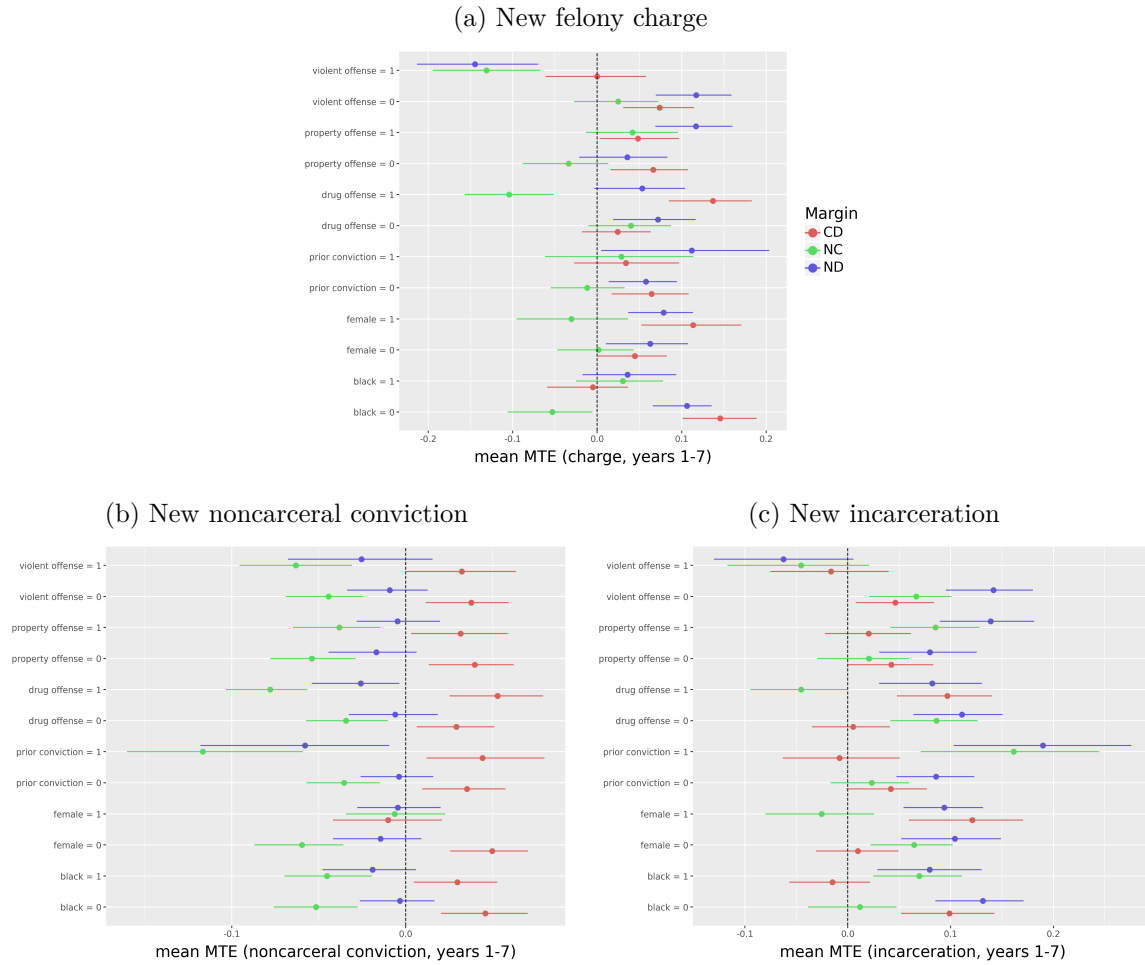


Figure 4: Cell Average Effects by Margin: Cumulative Years 1 to 7



Notes: As in Figure 3 but with each measure cumulated over years one to seven.

convictions is negative in nearly every cell, and the effect on new incarceration is positive in most cells, strengthening from the four-year to the seven-year horizon. For most cells the two effects roughly offset, so the small net effect on new convictions reflects a real shift in type rather than no change: the noncarceral-conviction effect is negative and the incarceration effect is positive, and reconvictions that would have been noncarceral instead return as carceral ones. This offsetting is not uniform across covariate cells, as the exceptions below make clear. On the (n, d) margin, for which HOSSvD report no estimate, the excess recidivism is almost entirely carceral. The effect on new incarceration over years one to seven is negative for violent offenses and positive in every other cell, ranging from 0.08 to 0.19, while the effect on new noncarceral convictions is flat or negative. This incarceration effect also grows from the four-year to the seven-year horizon in nearly every cell, though the two windows are not statistically distinguishable, suggesting incarceration leaves a longer-lasting mark on this margin as well. It also exceeds its (n, c) counterpart cell by cell. Incarceration is therefore more consequential for cases that would otherwise have been dismissed than for those that would have been convicted. Read together, the margins describe a revolving door. Once the marginal case crosses into incarceration, its subsequent convictions arrive with carceral sentences.

This anatomy is largely uniform across the covariate cells, with a few informative exceptions. Violent offenses break the pattern: incarceration reduces recidivism outright rather than changing its type, pointing to a decline specific to violent crime rather than a composition shift. Drug offenses depart differently: the conviction itself, rather than the subsequent incarceration decision, appears to be the consequential step, with incarceration adding little beyond it. Property offenses go the other way, and in the direction the pooled near-offsetting pattern obscures: the incarceration effect exceeds the decline in noncarceral convictions, so this margin encourages new incarceration for property cases more than the aggregate picture suggests, rather than merely recomposing recidivism. Defendants with a prior conviction form a smaller cell with visibly wider intervals, but the revolving-door pattern is strongest there, consistent with incarceration most directly redirecting reoffending toward re-incarceration for defendants already in the system. The confidence intervals are pointwise rather than adjusted for multiple testing and, as noted above, hold the first-stage estimates fixed. We therefore read the cell patterns through the signs and magnitudes of the estimates rather than through cell-by-cell significance.

As a check on the role of the distribution, we re-estimate the second stage with the interactive-effects logit first stage in place of the semiparametric one. The qualitative findings are unchanged, and the revision falls mainly on the magnitudes of the effects. The corresponding exhibits are collected in the Appendix.

7 Conclusion

Margin-specific effects are the objects of policy interest in judge IV designs with multiple treatments, yet the variation these designs supply is not margin-specific. This paper shows that the required instruments can be constructed rather than assumed. From case characteristics and random judge assignment, the two-step procedure recovers a treatment-specific index for every judge and converts the indices into margin-specific effects that vary with case characteristics. Judge designs thereby extend to settings where no margin-specific instrument is observed and treatment effects would otherwise be blended across margins. The application illustrates how the first stage matters: allowing judges to differ in slopes rather than intercepts alone revises the estimated effects of conviction and incarceration on recidivism, and the heterogeneity the second stage recovers is where the revolving-door pattern of incarceration emerges.

Two tasks are left to future work. The first is inference on the margin-specific effects. The index coefficients admit standard inference, but for the effect functions we establish uniform consistency without a limiting distribution. The empirical conclusions rest on covariate-cell averages of the effect function, and such smooth functionals are typically estimable at the parametric rate even where the pointwise derivative is not. Confidence intervals that account for both stages of estimation are therefore a concrete next step.

The second is formal specification and identification testing. Tests of the maintained index specification come first, since the identification conditions of Assumption 4 are stated within that structure. The identification conditions can meanwhile be corroborated visually, as Section 6.2 illustrates through the recovered index cloud, and the rank and nonseparability conditions are expressible through derivatives of the observed propensity surface and can be checked at the estimated first stage. Formal counterparts of these checks would let researchers carry out the same verification in their own settings.

References

- Daniel Akerberg, Xiaohong Chen, and Jinyong Hahn. A Practical Asymptotic Variance Estimator for Two-Step Semiparametric Estimators. *The Review of Economics and Statistics*, 94(2), 2012.
- Roy Allen and John Rehbeck. Identification with Additively Separable Heterogeneity. *Econometrica*, 87(3), 2019.
- Carolina Arteaga. Parental incarceration and children’s educational attainment. *Review of Economics and Statistics*, 105(6), 2023.
- Steven T Berry and Philip A Haile. Nonparametric identification of differentiated products demand using micro data. *Econometrica*, 92(4):1135–1162, 2024.
- Manudeep Bhuller and Henrik Sigstad. 2SLS with multiple treatments. *Journal of Econometrics*, 242(1), 2024.
- Christian N. Brinch, Magne Mogstad, and Matthew Wiswall. Beyond LATE with a Discrete Instrument. *Journal of Political Economy*, 125(4), 2017.
- Pedro Carneiro and Sokbae Lee. Estimating distributions of potential outcomes using local instrumental variables with an application to changes in college enrollment and wage inequality. *Journal of Econometrics*, 149(2), 2009.
- Victor Chernozhukov, Denis Chetverikov, and Kengo Kato. Gaussian approximation of suprema of empirical processes. *Annals of Statistics*, 42(4), 2014.
- Carl De Boor and Carl De Boor. *A practical guide to splines*, volume 27. springer New York, 1978.
- Ronald A DeVore and George G Lorentz. *Constructive approximation*, volume 303. Springer Science & Business Media, 1993.
- Brigham Frandsen, Lars Lefgren, and Emily Leslie. Judging judge fixed effects. *American Economic Review*, 113(1), 2023.
- James J. Heckman and Rodrigo Pinto. Unordered Monotonicity. *Econometrica*, 86(1), 2018.

- James J. Heckman, Sergio Urzua, and Edward Vytlacil. Instrumental Variables in Models with Multiple Outcomes: The General Unordered Case. *Annales d'Économie et de Statistique*, (91/92), 2008.
- John Eric Humphries, Aurelie Ouss, Kamelia Stavreva, Megan T Stevenson, and Winnie van Dijk. Conviction, Incarceration, and Recidivism: Understanding the Revolving Door*. *The Quarterly Journal of Economics*, 2025.
- Hidehiko Ichimura. Semiparametric least squares (SLS) and weighted SLS estimation of single-index models. *Journal of econometrics*, 58(1-2), 1993.
- Hidehiko Ichimura, Lung-Fei Lee, W. A. Barnett, J. L. Powell, and G. Tauchen. Semiparametric least squares estimation of multiple index models: Single equation estimation. In *Nonparametric and Semiparametric Methods in Econometrics and Statistics: Proceedings of the Fifth International Symposium in Economic Theory and Econometrics*. Cambridge, 1991.
- Guido W. Imbens and Joshua D. Angrist. Identification and Estimation of Local Average Treatment Effects. *Econometrica*, 62(2), 1994.
- Vishal Kamat, Samuel Norris, and Matthew Pecenco. Conviction, incarceration, and policy effects in the criminal justice system. *Available at SSRN*, 2024.
- Lars J. Kirkeboen, Edwin Leuven, and Magne Mogstad. Field of study, earnings, and self-selection. *The quarterly journal of economics*, 131(3), 2016.
- Roger W. Klein and Richard H. Spady. An efficient semiparametric estimator for binary response models. *Econometrica*, 1993.
- Lung-Fei Lee. Semiparametric maximum likelihood estimation of polychotomous and sequential choice models. *Journal of Econometrics*, 65(2), 1995.
- Sokbae Lee and Bernard Salanié. Identifying Effects of Multivalued Treatments. *Econometrica*, 86(6), 2018.
- Arthur Lewbel. Semiparametric qualitative response model estimation with unknown heteroscedasticity or instrumental variables. *Journal of econometrics*, 97(1), 2000.
- John Loeser. Local average treatment effects for multiple treatments with multiple instruments. 2024.

- Charles F. Manski. Maximum score estimation of the stochastic utility model of choice. *Journal of Econometrics*, 3(3), 1975.
- Rosa L. Matzkin. Nonparametric identification and estimation of polychotomous choice models. *Journal of Econometrics*, 58(1), 1993.
- Daniel McFadden. Econometric models of probabilistic choice. *Structural analysis of discrete data with econometric applications*, 1981.
- Jack Mountjoy. Community colleges and upward mobility. *American Economic Review*, 112(8), 2022.
- Michael Mueller-Smith. The criminal and labor market impacts of incarceration. *Unpublished working paper*, 18, 2015.
- Roman Rivera. Release, detain, or surveil? The effect of electronic monitoring on defendant outcomes. *American Economic Journal: Applied Economics*, 2023.
- Larry Schumaker. *Spline functions: basic theory*. Cambridge university press, 2007.
- Xiaoxia Shi, Matthew Shum, and Wei Song. Estimating Semi-Parametric Panel Multinomial Choice Models Using Cyclic Monotonicity. *Econometrica*, 86(2), 2018.
- Joel A. Tropp. An introduction to matrix concentration inequalities. *Foundations and trends® in machine learning*, 8(1-2), 2015.
- Aad W Van Der Vaart and Jon A Wellner. Weak convergence. In *Weak convergence and empirical processes: with applications to statistics*. Springer, 1996.

Appendix

This appendix collects the proofs of the results in the main text, together with the supporting lemmas and additional results.

Notation

We begin by fixing notation used throughout the proofs.

Treatment propensities and the surplus. The signed propensity extension $P^t(z|f)$ and the social surplus function $G(z|f)$ are defined in the main text ((2) and Section 3.1). The Williams–Daly–Zachary representation $\partial G(z|f)/\partial z^t = P^t(z|f)$ is Lemma G.3, and explicit formulas for the derivatives $D_l P^t$ and $D_{ll'} P^t$ are in Lemma G.2.

Covering numbers. For a class $\mathcal{G}$ of functions and a semi-metric $d(\cdot, \cdot)$, $N(\epsilon, \mathcal{G}, d)$ is the smallest cardinality of an ϵ -net covering $\mathcal{G}$.

Differential operators. D is the Jacobian and ∇ is the gradient. The index is $z = (z^c, z^n) \in \mathbb{R}^2$. For a coordinate $t \in \{c, n\}$, ∂_{z^t} is the partial derivative in that coordinate. We also write D_l for ∂_{z^l} and $D_{ll'}$ for $\partial_{z^l} \partial_{z^{l'}}$, and at times $D_1 = \partial_{z^c}$ and $D_2 = \partial_{z^n}$. The Hessian is D^2 . For a propensity P^t , its gradient is $DP^t = \nabla P^t$ and its Hessian is $D^2 P^t$. The 2×2 Jacobian of the propensity map $P(z) = (P^c(z), P^n(z))$ is $DP = \nabla^2 G$, with entries $(\partial_{z^l} P^t(z))_{l,t \in \{c,n\}}$. This Jacobian is symmetric and positive definite.

Probability measures and stochastic order. $\mathbb{P}$ and $\mathbb{E}$ are the population probability and expectation under the law of (Y_i, T_i, W_i, J_i) . $\mathbb{P}_N$ and $\mathbb{E}_N$ are the empirical measure and expectation. We use them interchangeably, with $\mathbb{P}_N g = \mathbb{E}_N g = N^{-1} \sum_i g(Y_i, T_i, W_i, J_i)$. The empirical process is $\mathbb{G}_N = \sqrt{N}(\mathbb{P}_N - \mathbb{P})$. For a positive sequence a_N and random variables X_N , $X_N = O_P(a_N)$ means X_N/a_N is tight, and $X_N = o_P(a_N)$ means $X_N/a_N \rightarrow_p 0$.

B-spline surplus sieve. The tensor B-spline basis $b_K(z) \in \mathbb{R}^{d_K}$, the spline space $\text{Spl}(K)$, the propensity sieve $\text{GradSpl}(K) = \{\nabla b_K(\cdot)' \xi : \xi \in \Xi(K)\}$, and the constraint set $\Xi(K)$ enforcing the cap $P_\xi^t := \partial_{z^t} b_K' \xi \in [\kappa, 1 - \kappa]$ are defined in Section 3.1, with $d_K = (K + q)^2 \asymp K^2$ and mesh width $h \asymp K^{-1}$. The proofs use the following further objects. The knot box $\tilde{\mathcal{Z}}$ contains $\cup_{\gamma \in \Gamma} \text{supp}(\gamma' X)$, and the cells Q are its mesh rectangles. The surplus potential with coefficient a is $s_a := b_K' a$, and $b^t := \partial_{z^t} b_K$, $b^d := -(b^c + b^n)$, so $b^t(z)' a = \partial_{z^t} s_a(z)$. The set of charged cells is $\mathcal{A}$, and the charged

region is $\Omega_{\mathcal{A}} := \cup_{Q \in \mathcal{A}} Q$. The charged seminorm is $\|a\|_{\mathcal{A}}^2 := \|\nabla s_a\|_{L^2(\Omega_{\mathcal{A}})}^2 = a' L_{\mathcal{A}} a$, and $L_{\mathcal{A}}$ is its Gram matrix.

Three pseudo-metrics. Fix parameters $\theta = (\gamma, P)$ and $\tilde{\theta} = (\tilde{\gamma}, \tilde{P})$. The Kullback–Leibler divergence $\text{KL}(\theta \| \tilde{\theta})$, the Hellinger distance $\text{H}(\theta, \tilde{\theta})$, and the L^2 (propensity) distance $\text{L}(\theta, \tilde{\theta})$ are

$$\begin{aligned} \text{KL}(\theta \| \tilde{\theta}) &= \sum_{t \in \{d, c, n\}} \mathbb{E} \left[P^t(\gamma' X) \log \frac{P^t(\gamma' X)}{\tilde{P}^t(\tilde{\gamma}' X)} \right], \\ \text{H}(\theta, \tilde{\theta}) &= \left(\mathbb{E} \sum_{t \in \{d, c, n\}} \left(\sqrt{P^t(\gamma' X)} - \sqrt{\tilde{P}^t(\tilde{\gamma}' X)} \right)^2 \right)^{1/2}, \\ \text{L}(\theta, \tilde{\theta}) &= \left(\mathbb{E} \sum_{t \in \{d, c, n\}} \left(P^t(\gamma' X) - \tilde{P}^t(\tilde{\gamma}' X) \right)^2 \right)^{1/2}. \end{aligned}$$

H and L are symmetric and satisfy the triangle inequality. In general KL does neither.

Treatment-propensity classes. For $t \in \{c, n\}$, the class of t -treatment propensities over the sieve is $\mathcal{G}_K^t := \{z \mapsto P_{\xi}^t(z) = \partial_{z^t} b_K(z)' \xi : \xi \in \Xi(K)\}$, and we collect the two as $\mathcal{G}_K := \mathcal{G}_K^c \cup \mathcal{G}_K^n$. Each $\mathcal{G}_K^t$ lies in the d_K -dimensional linear space $\{\partial_{z^t} b_K(\cdot)' h : h \in \mathbb{R}^{d_K}\}$ and has constant envelope 1. It is therefore a VC-subgraph class of index at most $d_K + 2$.

Uniform entropy and the uniform entropy integral. For a class $\mathcal{F}$ with envelope F , $N(\epsilon, \mathcal{F}, \|\cdot\|)$ is the ϵ -covering number in $\|\cdot\|$. With the supremum taken over finitely supported probability measures Q , the uniform entropy and the uniform entropy integral are

$$\begin{aligned} &\sup_Q \log N \left(\epsilon \|F\|_{L^2(Q)}, \mathcal{F}, L^2(Q) \right), \\ \tilde{J}(\delta, \mathcal{F}) &:= \int_0^{\delta} \sqrt{1 + \sup_Q \log N \left(\epsilon \|F\|_{L^2(Q)}, \mathcal{F}, L^2(Q) \right)} d\epsilon. \end{aligned}$$

These quantities enter the standard maximal inequality $\mathbb{E} \|\mathbb{G}_N\|_{\mathcal{F}} \lesssim \tilde{J}(1, \mathcal{F}) \|F\|_{L^2(\mathbb{P})}$. It turns an entropy bound into a fluctuation bound. Here $\|\mathbb{G}_N\|_{\mathcal{F}} = \sup_{f \in \mathcal{F}} |\mathbb{G}_N(f)|$. This supremum is measurable when $\mathcal{F}$ is pointwise separable. The classes we use have this property.

A Identification

Proof of Theorem 1. Suppose $\theta = (\boldsymbol{\gamma}, P) \in \Gamma \times \mathcal{P}$ is observationally equivalent to $\theta_o = (\boldsymbol{\gamma}_o, P_o)$, i.e.,

$$P^t(\gamma'_j w) = P_o^t(\gamma'_{oj} w) \quad t \in \{d, c, n\}, \quad w \in \text{supp}(W), \quad j \in \{1, \dots, J\}. \quad (\text{A.1})$$

We show (A.1) forces $\boldsymbol{\gamma} = \boldsymbol{\gamma}_o$.

Step 1 (Functional dependence via injectivity): Lemma G.4 applies to $f_o \in \mathcal{F}$, so $P_o(z) = P(z|f_o)$ is an injective function of $z \in \mathbb{R}^2$. Letting $h = P_o^{-1} \circ P$, which depends only on P and P_o and not on j , we have

$$h(\gamma'_j w) = \gamma'_{oj} w \quad \forall w \in \text{supp}(W), \quad j = 1, \dots, J. \quad (\text{A.2})$$

The function h above is well-defined since the range of $P(\cdot)$ on $\cup_{j=1}^J \text{supp}(\gamma'_j W)$ is contained in the range of $P_o(\cdot)$ by (A.1).

Step 2 (Slope alignment on W_a): Fix j and choose $w_b^{j,*} \in \text{supp}(W_b)$ as in Assumption 4(a). Let $B \subset \mathbb{R}^{\dim(W_a)}$ be an open ball with $B \subseteq \text{supp}(W_a | W_b = w_b^{j,*})$. On the slice $\{(1, W_a, w_b^{j,*}) : W_a \in B\}$, (A.2) reads off as

$$h(\gamma_{j,1} + \gamma'_{j,a} w_a + \gamma'_{j,b} w_b^{j,*}) = \gamma_{oj,1} + \gamma'_{oj,a} w_a + \gamma'_{oj,b} w_b^{j,*}, \quad \forall w_a \in B.$$

By the inverse mapping theorem, $h = P_o^{-1} \circ P$ is continuously differentiable. Differentiating both sides with respect to w_a gives

$$\nabla h(\gamma_{j,1} + \gamma'_{j,a} w_a + \gamma'_{j,b} w_b^{j,*}) \cdot \gamma'_{j,a} = \gamma'_{oj,a}, \quad \forall w_a \in B.$$

Since $\text{rank}(\gamma_{oj,a}) = 2$ by Assumption 4(b), the above equality implies $\text{rank}(\gamma_{j,a}) = 2$ and $\text{rank}(\nabla h(\gamma_{j,1} + \gamma'_{j,a} w_a + \gamma'_{j,b} w_b^{j,*})) = 2$, and hence

$$\gamma_{oj,a} = \gamma_{j,a} M_j$$

where $M_j = \nabla h(\gamma_{j,1} + \gamma'_{j,a} w_a + \gamma'_{j,b} w_b^{j,*})'$ is non-singular for any $w_a \in B$. Letting $\tilde{z} = \gamma_{j,1} + \gamma'_{j,a} w_a + \gamma'_{j,b} w_b^{j,*}$, as w_a ranges over B , $\tilde{z}$ ranges over an open subset of $\mathbb{R}^2$ since $\text{rank}(\gamma_{j,a}) = 2$ and $\dim(W_a) \geq 2$ from Assumption 4(a) and (b). By substituting $\gamma_{j,1} + \gamma'_{j,a} W_a + \gamma'_{j,b} w_b^{j,*}$ with $\tilde{z}$, on the image of the slice $I_j(w_b^{j,*}) := \text{supp}(\gamma'_j W | W_b =$

$w_b^{j,*} \subseteq \mathbb{R}^2$, h takes the affine form $h(\tilde{z}) = M_j' \tilde{z} + c_j^{(w_b^{j,*})}$ for all $\tilde{z} \in I_j(w_b^{j,*})$, where $c_j^{(w_b^{j,*})}$ is a constant given by

$$c_j^{(w_b^{j,*})} = (\gamma_{oj,1} - M_j' \gamma_{j,1}) + (\gamma_{oj,b} - \gamma_{j,b} M_j)' w_b^{j,*}. \quad (\text{A.3})$$

Step 3 (Slope alignment on W_b): For any $w_b \in \text{supp}(W_b)$, we have $h(\tilde{z}) = M_j' \tilde{z} + c_j^{(w_b)}$ on $I_j(w_b)$ where $c_j^{(w_b)}$ is defined as (A.3) with $w_b^{j,*}$ replaced by w_b . Suppose that $I_j(w_b^{j,*}) \cap I_j(w_b) \neq \emptyset$. Then by (A.1) and injectivity of P_o , we have

$$\text{supp}(\gamma_{oj}' W \mid W_b = w_b^{j,*}) \cap \text{supp}(\gamma_{oj}' W \mid W_b = w_b) \neq \emptyset,$$

i.e., $w_b^{j,*}$ and w_b are adjacent. On the intersection $I_j(w_b^{j,*}) \cap I_j(w_b)$, the two affine forms must agree, so we have $c_j^{(w_b^{j,*})} = c_j^{(w_b)}$. Iterating this over a path joining $w_b^{j,*}$ to any $w_b \in C_j(w_b^{j,*})$, we have

$$c_j^{(w_b^{j,*})} = c_j^{(w_b)} \iff (\gamma_{oj,b} - \gamma_{j,b} M_j)' (w_b^{j,*} - w_b) = 0$$

for all $w_b \in C_j(w_b^{j,*})$. Invoking Assumption 4(d), we have that $w_b^{j,*} - w_b$ spans $\mathbb{R}^{\dim(W_b)}$ as w_b ranges over $C_j(w_b^{j,*})$, so the above condition implies $\gamma_{oj,b} = \gamma_{j,b} M_j$. Combining with Step 2, we conclude that

$$\gamma_{oj,-1} = \gamma_{j,-1} M_j, \quad \gamma_{oj,1} = M_j' \gamma_{j,1} + c_j, \quad (\text{A.4})$$

where c_j is the residual judge-specific intercept after aligning the slopes and equals the common value of $c_j^{(w_b)}$ for all $w_b \in C_j(w_b^{j,*})$. The function h takes the global affine form $h(\tilde{z}) = M_j' \tilde{z} + c_j$ on the union $\bigcup_{w_b \in \text{supp}(W_b)} I_j(w_b) = \text{supp}(\gamma_j' W) \subseteq \mathbb{R}^2$ and has the range $\text{supp}(\gamma_{oj}' W)$.

Step 4 (M_j is diagonalizable with real eigenvalues): We have shown that $P_o^t(M_j' \tilde{z} + c_j) = P^t(\tilde{z})$ for all $\tilde{z} \in \text{supp}(\gamma_j' W)$ and $t \in \{c, n\}$. Equivalently, $P_o^t(z) = P^t(M_j^{-T}(z - c_j))$ for all $z \in \text{supp}(\gamma_{oj}' W)$ by substituting $M_j' \tilde{z} + c_j$ with z . Here, $M_j^{-T} = (M_j^{-1})'$ is the inverse transpose of M_j . Differentiating this identity in z and stacking over $t \in \{c, n\}$, we have

$$DP_o(z) = DP(M_j^{-T}(z - c_j)) M_j^{-T}, \quad z \in \text{Int}(\text{supp}(\gamma_{oj}' W)),$$

where $DP_o(\cdot)$ and $DP(\cdot)$ are Jacobians of $P_o(\cdot)$ and $P(\cdot)$, respectively.

Taking the transpose of $DP(M_j^{-T}(z - c_j)) = DP_o(z)M'_j$ and using the symmetry of $DP(\cdot)$ and $DP_o(\cdot)$, we obtain $DP(M_j^{-T}(z - c_j)) = M_jDP_o(z)$. Subtracting the two identities, we obtain

$$DP_o(z)M'_j = M_jDP_o(z).$$

Fix any $z_0 \in \text{Int}(\text{supp}(\gamma'_{oj}W))$ and let $H = DP_o(z_0)$. The displayed identity at z_0 reads $M_jH = HM'_j$, with H symmetric positive definite. We show that M_j is diagonalizable with real eigenvalues. Multiplying $M_jH = HM'_j$ on the left by $H^{-1/2}$ and on the right by $H^{-1/2}$ gives,

$$H^{-1/2}M_jH^{1/2} = H^{1/2}M'_jH^{-1/2}.$$

The right-hand side is the transpose of the left-hand side, so $H^{1/2}M'_jH^{-1/2}$ equals its own transpose, i.e., is symmetric. A symmetric real matrix is diagonalizable with real eigenvalues. Since M_j is similar to $H^{1/2}M'_jH^{-1/2}$ via conjugation by $H^{1/2}$, M_j inherits diagonalizability and real spectrum. Write

$$M_j = U_jD_jU_j^{-1},$$

with U_j and $D_j = \text{diag}(d_{j,1}, d_{j,2}) \in \mathbb{R}^{2 \times 2}$ both invertible.

Step 5 (Separability obstruction): Substituting $M_j = U_jD_jU_j^{-1}$ into the condition $M_jDP_o(z) = DP_o(z)M'_j$ from Step 4 gives, after multiplying on the left by U_j^{-1} and on the right by $U_j^{-T} = (U'_j)^{-1}$,

$$D_jU_j^{-1}DP_o(z)U_j^{-T} = U_j^{-1}DP_o(z)U_j^{-T}D_j \quad \text{for all } z \in \text{Int}(\text{supp}(\gamma'_{oj}W)).$$

Suppose, for contradiction, that $d_{j,1} \neq d_{j,2}$. Then any symmetric matrix commuting with D_j must be diagonal. In particular, $U_j^{-1}DP_o(z)U_j^{-T}$ is diagonal at every $z \in \text{Int}(\text{supp}(\gamma'_{oj}W))$. Now let v_1, v_2 denote the columns of U_j^{-T} , and reparametrize $z = U_j^{-T}x$, so that $x = U'_jz$. Define $\tilde{G}_o(x) = G_o(U_j^{-T}x)$ where $G_o(\cdot) = G(\cdot | f_o)$. The Hessian transforms as

$$\nabla_x^2 \tilde{G}_o(x) = U_j^{-1}DP_o(U_j^{-T}x)U_j^{-T},$$

which is diagonal on the corresponding open subset by the previous display. Pulling

back to z ,

$$0 \equiv \frac{\partial^2}{\partial x^c \partial x^n} \tilde{G}_o(x) = v_1' \nabla^2 G_o(U_j^{-T} x) v_2 = v_1' DP_o(z) v_2, \quad z \in \text{Int}(\text{supp}(\gamma_{oj}' W)).$$

This is exactly the separability ruled out by Assumption 4(c)—a contradiction. Hence $d_{j,1} = d_{j,2} =: d_j$ and $M_j = d_j I_2$.

It remains to fix the sign of $d_j \neq 0$. With $M_j = d_j I_2$, the level identity of Step 4 reads $P_o(d_j \tilde{z} + c_j) = P(\tilde{z})$, and differentiating in $\tilde{z}$ gives

$$DP(\tilde{z}) = d_j DP_o(d_j \tilde{z} + c_j), \quad \tilde{z} \in \text{Int}(\text{supp}(\gamma_j' W)).$$

Here $DP_o(\cdot)$ is positive definite by Lemma G.3, and $DP(\tilde{z})$ is positive semidefinite by Lemma G.6 since $P \in \mathcal{P}$. If $d_j < 0$, the right-hand side would be negative definite, contradicting the positive semidefiniteness of the left-hand side. Since M_j is nonsingular we have $d_j \neq 0$, and therefore $d_j > 0$. Consequently $DP(\cdot) = d_j DP_o(d_j \cdot + c_j)$ is positive definite on $\text{Int}(\text{supp}(\gamma_j' W))$ for every j , so P is injective on $\cup_{j=1}^J \text{Int}(\text{supp}(\gamma_j' W))$ by Lemma G.6. Returning to the within-judge identities,

$$\gamma_{oj,-1} = d_j \gamma_{j,-1}, \quad \gamma_{oj,1} = d_j \gamma_{j,1} + c_j,$$

for some judge-specific scalar $d_j > 0$ and intercept $c_j \in \mathbb{R}^2$.

Step 6 (Cross-judge comparison and location determination): Take any pair of adjacent judges j_1, j_2 , and set $\mathcal{O} := \text{Int}(\text{supp}(\gamma_{o,j_1}' W)) \cap \text{Int}(\text{supp}(\gamma_{o,j_2}' W))$. By adjacency the propensity supports share an interior point; these supports equal $P_o(\text{supp}(\gamma_{o,j_i}' W))$ by compactness of the index supports, so the open-map identity of Lemma G.5 places that shared point in $P_o(\mathcal{O})$. Hence $\mathcal{O}$ is nonempty, and as an intersection of open sets it is open. Fix $z_0 \in \mathcal{O}$. By Step 5 applied to each judge, the level identity at z_0 reads

$$P^t(d_{j_1}^{-1}(z_0 - c_{j_1})) = P_o^t(z_0) = P^t(d_{j_2}^{-1}(z_0 - c_{j_2})) \quad \forall t \in \{c, n\}.$$

The affine map $\xi \mapsto d_{j_i}^{-1}(\xi - c_{j_i})$ carries $\text{Int}(\text{supp}(\gamma_{o,j_i}' W))$ onto $\text{Int}(\text{supp}(\gamma_{j_i}' W))$, so the two arguments above lie in $\cup_{j=1}^J \text{Int}(\text{supp}(\gamma_j' W))$, where DP is positive definite. By Lemma G.6, P is injective there, so

$$d_{j_1}^{-1}(z_0 - c_{j_1}) = d_{j_2}^{-1}(z_0 - c_{j_2}).$$

Rearranging,

$$(d_{j_1}^{-1} - d_{j_2}^{-1})z_0 = c_{j_2} - c_{j_1} \in \mathbb{R}^2.$$

The same equation evaluated at any other point $z'_0 \in \mathcal{O}$ gives $(d_{j_1}^{-1} - d_{j_2}^{-1})z'_0 = c_{j_2} - c_{j_1}$. Subtracting,

$$(d_{j_1}^{-1} - d_{j_2}^{-1})(z_0 - z'_0) = 0.$$

Since $\mathcal{O}$ is open and nonempty, it contains two distinct points, so this forces $d_{j_1} = d_{j_2}$; by the connectedness of the judge-adjacency graph, iterating this argument over the path joining j_1 to j_2 in the judge-adjacency graph, we have $d_{j_1} = d_{j_2}$ for every pair of connected judges, and hence $d_j \equiv d$ for some scalar d across all judges. Substituting this back into the level identity at z_0 , we have $c_{j_1} = c_{j_2}$ for every pair of connected judges by the same argument, so $c_j \equiv c$ for a common $c \in \mathbb{R}^2$. Summing the intercept identities $\gamma_{oj,1} = d\gamma_{j,1} + c$ over judges gives $\sum_j \gamma_{oj,1} = d\sum_j \gamma_{j,1} + Jc$; both sums vanish by the location normalization $e'_J \gamma^t = e'_J \gamma_o^t = 0$, whence $c = 0$.

Step 7 (Scale determination): From Step 6, $\gamma_{oj} = d\gamma_j$ for every j . The constraint $\|\gamma\|_F = 1 = \|\gamma_o\|_F$ gives

$$1 = \|\gamma_o\|_F^2 = \sum_{j=1}^J \|\gamma_{oj}\|^2 = \sum_{j=1}^J \|d\gamma_j\|^2 = d^2 \sum_{j=1}^J \|\gamma_j\|^2 = d^2 \|\gamma\|_F^2 = d^2,$$

and hence $|d| = 1$, i.e., $d = \pm 1$. Using $d > 0$ from Step 6, we have $d = 1$, so $\gamma_{oj} = \gamma_j$ for every j . This establishes $\gamma_o = \gamma$, i.e., the first-stage coefficient matrix is identified.

Step 8 (Identification of P_o): With $\gamma = \gamma_o$, (A.1) becomes

$$P^t(z) = P_o^t(z), \quad z \in \text{supp}(\gamma'_o X), \quad t \in \{d, c, n\}.$$

Thus $P_o(\cdot)$ is identified as a function of $z \in \text{supp}(\gamma'_o X)$. This completes the proof. $\square$

B Consistency and the Crude Rate of Convergence

This subsection proves Theorems 2 and 3. The argument runs as follows. Lemmas B.1–B.5 collect the preliminary inputs: the compact closure of $\mathcal{P}$, an entropy bound, a stability bound, the simultaneous spline approximation of (P_o, G_o) , and metric comparisons. Proposition B.1 derives the nonparametric rate $\mathbf{H}(\hat{\theta}, \theta_o) = O_P(\rho_N)$ from these inputs via the rate theorem of Van Der Vaart and Wellner (1996, The-

orem 3.4.1). Lemma B.6 converts Theorem 1 into a global separation statement in the $\mathbf{L}$ metric, which combined with the rate proves Theorem 2. Lemmas B.7–B.10 and Propositions B.2 and B.3 then establish the local curvature bound $\mathbf{L} \gtrsim \|\gamma - \gamma_o\|$, which proves the first claim of Theorem 3; Proposition B.4 addresses the uniform rate of P_{ξ} .

Conventions. Throughout this subsection, $\Theta_N := \Gamma \times \{P_{\xi} : \xi \in \Xi(K_N)\}$ denotes the feasible set of the SMLE problem (6), and $\theta_K := (\gamma_o, P_{\xi_K^o})$ with ξ_K^o the approximant of Lemma B.4. We write $d_K = (K + q)^2$ and $R_X := \sup_{x \in \text{supp}(X_i)} \|x\|$, which is finite because W_i has compact support. Since $\bar{\mathcal{Z}} \supseteq \cup_{\gamma \in \Gamma} \text{supp}(\gamma' X_i)$, every realized index $\gamma' X_i$ with $\gamma \in \Gamma$ lies in $\bar{\mathcal{Z}}$, so every criterion value below is well defined. Under Assumption 5(c), $q \geq 4$, a sieve propensity P_{ξ} with $\xi \in \Xi(K)$ is a derivative of an order- (q, q) tensor spline with simple knots and is therefore $C^{q-3} \subseteq C^1$; since the constraints in $\Xi(K)$ reproduce the remaining defining conditions of $\mathcal{P}$, it follows that $\text{GradSpl}(K) = \{P_{\xi} : \xi \in \Xi(K)\} \subseteq \mathcal{P}$. Throughout, $k_o \geq 2$ denotes the smoothness of P_o (Assumption 5(b)), so that $\rho_N = \sqrt{K_N^2 \log N/N + K_N^{-k_o}}$; the approximation theory (Lemma B.4) uses $q \geq k_o + 1$, also granted by Assumption 5(c).

Lemma B.1 (closure of the propensity class). *Let $\bar{\mathcal{P}}$ denote the closure of $\mathcal{P}$ in the sup norm on the continuous maps $\bar{\mathcal{Z}} \rightarrow \mathbb{R}^2$. Then $\bar{\mathcal{P}}$ is compact, and every $P \in \bar{\mathcal{P}}$ satisfies the cap, is $2M$ -Lipschitz, is the gradient of a C^1 surplus function on $\bar{\mathcal{Z}}$, and is monotone in the sense of $(z - z')'(P(z) - P(z')) \geq 0$.*

Proof. For $P \in \mathcal{P}$: the cap holds by definition; the diagonal entries of DP lie in $[0, M]$ and dominate the common off-diagonal entry in absolute value, so $\|DP\| \leq 2M$ pointwise and P is $2M$ -Lipschitz on the convex $\bar{\mathcal{Z}}$; DP is symmetric, so line integrals of P do not depend on the path and P is the gradient of a C^1 surplus function on the rectangle $\bar{\mathcal{Z}}$; and DP is positive semidefinite by Lemma G.6, so the line-integral identity in the proof of Lemma G.6 gives monotonicity. Each of the four properties passes to uniform limits (line integrals along fixed paths and the monotonicity inequality converge directly, so path independence and the surplus representation are preserved), so they hold on $\bar{\mathcal{P}}$. Compactness follows from the Arzelà–Ascoli theorem: $\bar{\mathcal{P}}$ is uniformly bounded, uniformly $2M$ -Lipschitz, and closed. $\square$

Lemma B.2 (entropy of the sieve criterion class). *For $t \in \{c, n\}$ let $\mathcal{G}_K^t = \{z \mapsto \partial_{z^t} b_K(z)' \xi : \xi \in \Xi(K)\}$, and let $\mathcal{L}_K := \{(t, x) \mapsto \sum_t t^t \log P_{\xi}^t(\gamma' x) : \gamma \in \Gamma, \xi \in \Xi(K)\}$ be the criterion class. Then, for $0 < \epsilon \leq 1$:*

- (i) $\sup_Q \log N(\epsilon, \mathcal{G}_K^t, L^2(Q)) \lesssim d_K \log(1/\epsilon)$, the supremum over finitely supported probability measures;
- (ii) $\mathcal{L}_K$ has the constant envelope $\log(1/\kappa)$ and $\sup_Q \log N(\epsilon, \mathcal{L}_K, L^2(Q)) \lesssim d_K \log(C/\epsilon)$ for a constant $C = C(\kappa, M, R)$;
- (iii) $\mathbb{E} \sup_{\theta \in \Theta_N} |(\mathbb{P}_N - \mathbb{P})l_i(\theta)| \lesssim \sqrt{d_{K_N}/N}$.

Proof. (i) $\mathcal{G}_K^t$ lies in the d_K -dimensional vector space $\{\partial_{z^t} b_K(\cdot)'h : h \in \mathbb{R}^{d_K}\}$ and has envelope 1 by the cap. A finite-dimensional vector space of functions is VC-subgraph of index at most $d_K + 2$ (Van Der Vaart and Wellner, 1996, Lemma 2.6.15), and the uniform entropy bound for VC-subgraph classes with bounded envelope (Van Der Vaart and Wellner, 1996, Theorem 2.6.7) gives the claim; restricting to the convex set $\Xi(K)$ only shrinks the class.

(ii) On $[\kappa, 1 - \kappa]$ the logarithm is κ^{-1} -Lipschitz. Fix γ and ξ, ξ' ; then $|l(t, x; \gamma, \xi) - l(t, x; \gamma, \xi')| \leq \kappa^{-1} \sum_{t \in \{c, n, d\}} |P_\xi^t(\gamma'x) - P_{\xi'}^t(\gamma'x)|$, and the d -component is the sum of the other two, so covering each $\mathcal{G}_K^t$ to accuracy $\epsilon\kappa/6$ under the pushforward of Q by $x \mapsto \gamma'x$ covers the ξ -direction; by (i) this costs $\lesssim d_K \log(1/\epsilon)$ uniformly in γ and Q . Fix ξ and γ, γ' ; then by Lemma B.3 and boundedness of X , $|l(t, x; \gamma, \xi) - l(t, x; \gamma', \xi)| \leq \kappa^{-1} \cdot 3 \cdot 2MR_X \|\gamma - \gamma'\|$. Since Γ is a compact subset of a fixed-dimensional Euclidean space, an $\epsilon\kappa/(12MR_X)$ -net of Γ has log-cardinality $\lesssim \log(C/\epsilon)$. Combining the two nets gives (ii); the envelope is $|\log \kappa|$ since $P^t \geq \kappa$.

(iii) The uniform entropy integral of $\mathcal{L}_{K_N}$ satisfies

$$\tilde{J}(1, \mathcal{L}_{K_N}) \lesssim \int_0^1 \sqrt{1 + d_{K_N} \log(C/\epsilon)} d\epsilon \lesssim \sqrt{d_{K_N}},$$

so the maximal inequality $\mathbb{E} \|\mathbb{G}_N\|_{\mathcal{L}_{K_N}} \lesssim \tilde{J}(1, \mathcal{L}_{K_N}) \|F\|_{L^2(\mathbb{P})}$ with the constant envelope $F = |\log \kappa|$ gives $\mathbb{E} \sup_{\theta \in \Theta_N} |(\mathbb{P}_N - \mathbb{P})l_i| \lesssim \sqrt{d_{K_N}/N}$. $\square$

Lemma B.3 (stability of the sieve propensities). *For every K , every $\xi \in \Xi(K)$, and $t \in \{c, n\}$:*

- (i) $\sup_z \|DP_\xi^t(z)\| \leq 2M$ and P_ξ^t is $2M$ -Lipschitz on $\bar{\mathcal{Z}}$;
- (ii) if $q \geq 4$, as Assumption 5(c) implies, then $\sup_z \|D^2 P_\xi^t(z)\| \leq C(q, c_1, c_2)MK$.

The same bounds in (i) hold for every $P \in \mathcal{P}$.

Proof. (i) The constraints in $\Xi(K)$ bound the diagonal entries of DP_ξ by M and, through diagonal dominance, the off-diagonal entry by the diagonal ones, so every entry of DP_ξ^t is bounded by M in absolute value; the Lipschitz bound follows by integrating along segments of the convex $\bar{\mathcal{Z}}$. The same argument applies to $P \in \mathcal{P}$.

(ii) Each entry of DP_ξ^t is a tensor spline of coordinate orders at least $q - 2 \geq 2$ on the quasi-uniform mesh, so the Markov inequality for splines (DeVore and Lorentz, 1993, Ch. 5, Theorem 1.2) applied at $\ell = 1$, $p = \infty$ gives $\|D(DP_\xi^t)\|_{L^\infty} \leq C(q)\delta_0^{-1}\|DP_\xi^t\|_{L^\infty}$ with $\delta_0 \asymp K^{-1}$ by Assumption 5(c); combined with (i) this yields the claim. $\square$

Lemma B.4 (simultaneous approximation and feasibility). *Let $G_o = G(\cdot | f_o)$, so that $P_o = \nabla G_o$ and $G_o \in C^{k_o+1}(\bar{\mathcal{Z}})$, and let $q \geq k_o + 1$. There exists $\xi^\circ = \xi_K^\circ \in \mathbb{R}^{d_K}$ such that*

$$\begin{aligned} \|D^i(G_o - b'_K \xi^\circ)\|_{L^\infty(\bar{\mathcal{Z}})} &\lesssim K^{-(k_o+1-i)}, \quad 0 \leq i \leq k_o, \\ \|D^j(P_o^t - P_{\xi^\circ}^t)\|_{L^\infty(\bar{\mathcal{Z}})} &\lesssim K^{-(k_o-j)}, \quad 0 \leq j \leq k_o - 1, \end{aligned}$$

for $t \in \{c, n\}$. Moreover, $\xi^\circ \in \Xi(K)$ for all K large enough.

Proof. Approximation. Set $b'_K \xi^\circ := Q_K G_o$, where $Q_K = Q^{(1)}Q^{(2)}$ is the tensor product of the univariate de Boor–Fix quasi-interpolant Q of order q , acting coordinatewise on the box $\bar{\mathcal{Z}}$. On a quasi-uniform mesh Q is local, reproduces polynomials of degree $\leq q - 1$, and approximates a function together with its derivatives: for $g \in C^m(I)$ with $0 \leq a \leq m \leq q$,

$$\|D^a(g - Qg)\|_{L^\infty(I)} \lesssim h^{m-a}\|D^m g\|_{L^\infty(I)}, \quad (\text{B.1})$$

with $h \asymp K^{-1}$ (De Boor and De Boor, 1978, Ch. XII); see also Schumaker (2007, Theorem 6.27). Since $\bar{\mathcal{Z}}$ is a product and each $Q^{(\nu)}$ acts on one coordinate, the tensor quasi-interpolant inherits (B.1) on $\bar{\mathcal{Z}}$ (De Boor and De Boor, 1978, Ch. XVII). With $G_o \in C^{k_o+1}(\bar{\mathcal{Z}})$ and $q \geq k_o + 1$, this gives the first display; the second follows since $D^j(P_o^t - P_{\xi^\circ}^t) = D^j \partial_{z^t}(G_o - b'_K \xi^\circ)$ is the surplus bound at $i = j + 1$.

Feasibility. Write $\epsilon_K := \max_t \|P_o^t - P_{\xi^\circ}^t\|_{L^\infty(\bar{\mathcal{Z}})} + \max_t \|D(P_o^t - P_{\xi^\circ}^t)\|_{L^\infty(\bar{\mathcal{Z}})} \lesssim K^{-(k_o-1)} \rightarrow 0$. Each constraint in $\Xi(K)$ holds for P_o with a uniform margin on the compact $\bar{\mathcal{Z}}$: the cap with margin by Assumption 5(b) and continuity; $\|DP_o\| < M$ with margin likewise; the diagonal entries of DP_o and the dominance gaps are continuous and strictly positive by Lemma G.3, hence bounded away from zero; and

$\partial_{z^n} P_o^c(z) = -\int_{-z^c}^{\infty} f_o(u^c, z^c - z^n + u^c) du^c$ is continuous and strictly negative by Lemma G.2 and the positivity of densities in $\mathcal{F}$, hence bounded away from zero. Since every constraint is a level or first-derivative functional of the propensity, P_{ξ° violates none of them once ϵ_K is smaller than the smallest margin, so $\xi^\circ \in \Xi(K)$ for K large. $\square$

Lemma B.5 (reverse metric comparison under the cap). *Let $\theta = (\gamma, P)$ and $\tilde{\theta} = (\tilde{\gamma}, \tilde{P})$ have propensities bounded below by κ along the relevant indices. Then $\text{KL}(\theta \|\tilde{\theta}) \leq \kappa^{-1} \mathbb{L}^2(\theta, \tilde{\theta})$. In particular, $\text{KL}(\theta_o \|\theta_K) \leq 3\kappa^{-1} \max_t \|P_o^t - P_{\xi^\circ}^t\|_{L^\infty(\tilde{\mathcal{Z}})}^2 \lesssim K^{-2k_o}$ and $\text{H}(\theta_o, \theta_K) \lesssim K^{-k_o}$.*

Proof. Pointwise in x , with $p^t := P^t(\gamma'x)$ and $q^t := \tilde{P}^t(\tilde{\gamma}'x)$, the inequality $\log(1+t) \leq t$ gives $\sum_t p^t \log(p^t/q^t) \leq \sum_t p^t(p^t - q^t)/q^t = \sum_t (p^t - q^t)^2/q^t \leq \kappa^{-1} \sum_t (p^t - q^t)^2$, where the middle equality uses $\sum_t (p^t - q^t) = 0$. Taking expectations proves the first claim; the second follows since θ_K shares the index γ_o , so $\mathbb{L}^2(\theta_o, \theta_K) \leq 3 \max_t \|P_o^t - P_{\xi^\circ}^t\|_\infty^2$, and $\text{H}^2 \leq \text{KL}$ by Lemma G.1. $\square$

Proposition B.1 (nonparametric rate). *Suppose Assumptions 5(a)–(c) hold, $K_N \rightarrow \infty$ with $K_N^2 \log N = o(N)$, and $\xi_{K_N}^\circ \in \Xi(K_N)$ for all large N (Lemma B.4). Then $\text{H}(\hat{\theta}, \theta_o) = O_P(\rho_N)$ and $\mathbb{L}(\hat{\theta}, \theta_o) = O_P(\rho_N)$.*

Proof. We verify the conditions of Van Der Vaart and Wellner (1996, Theorem 3.4.1) for the criterion $\mathbb{M}_N(\theta) := \mathcal{L}_N(\theta) = \mathbb{P}_N l_i(\theta)$, its population version $\mathbb{M}(\theta) := \mathbb{E} l_i(\theta)$, the semimetric $d = \text{H}$ centered at θ_o , and the feasible set Θ_N .

Step 0: approximate maximization and consistency. Since $\hat{\theta}$ maximizes $\mathcal{L}_N$ over $\Theta_N \ni \theta_{K_N}$,

$$\mathbb{M}_N(\hat{\theta}) \geq \mathbb{M}_N(\theta_{K_N}) = \mathbb{M}_N(\theta_o) - \mathbb{P}_N(l_i(\theta_o) - l_i(\theta_{K_N})). \quad (\text{B.2})$$

The term $h_N := l_i(\theta_o) - l_i(\theta_{K_N})$ is a fixed function for each N with mean $\text{KL}(\theta_o \|\theta_{K_N})$ and, by the κ^{-1} -Lipschitz property of the logarithm and Lemma G.1, variance $\lesssim \text{H}^2(\theta_o, \theta_{K_N})$. Chebyshev's inequality and Lemma B.5 give

$$\mathbb{P}_N h_N = \text{KL}(\theta_o \|\theta_{K_N}) + O_P(N^{-1/2} \text{H}(\theta_o, \theta_{K_N})) = O(K_N^{-2k_o}) + O_P(N^{-1/2} K_N^{-k_o}) = O_P(\rho_N^2),$$

using $N^{-1/2} K_N^{-k_o} \leq \frac{1}{2}(N^{-1} + K_N^{-2k_o}) \leq \rho_N^2$. For consistency, $\mathbb{M}(\theta) - \mathbb{M}(\theta_o) =$

$-\text{KL}(\theta_o \parallel \theta) \leq -\text{H}^2(\theta, \theta_o)$ for every $\theta \in \Theta_N$ by Lemma G.1, so

$$\begin{aligned} \text{H}^2(\hat{\theta}, \theta_o) &\leq \mathbb{M}(\theta_o) - \mathbb{M}(\hat{\theta}) \leq \mathbb{M}_N(\theta_o) - \mathbb{M}_N(\hat{\theta}) + 2 \sup_{\Theta_N} |(\mathbb{P}_N - \mathbb{P})l_i| \\ &\leq \mathbb{P}_N h_N + 2 \sup_{\Theta_N} |(\mathbb{P}_N - \mathbb{P})l_i|, \end{aligned}$$

and the right side is $o_P(1)$ by Lemma B.2(iii) and $d_{K_N} = o(N)$. Hence $\text{H}(\hat{\theta}, \theta_o) = o_P(1)$.

Step 1: curvature. As displayed above, $\mathbb{M}(\theta) - \mathbb{M}(\theta_o) \leq -\text{H}^2(\theta, \theta_o)$ on Θ_N .

Step 2: modulus of continuity. Let $\mathcal{M}_\delta := \{l_i(\theta) - l_i(\theta_o) : \theta \in \Theta_N, \text{H}(\theta, \theta_o) \leq \delta\}$.

By the Lipschitz property of the logarithm and Lemma G.1, $\text{Var}(l_i(\theta) - l_i(\theta_o)) \leq 3\kappa^{-2}\text{L}^2(\theta, \theta_o) \leq 12\kappa^{-2}\text{H}^2(\theta, \theta_o)$, so the class $\mathcal{M}_\delta$ has second moments bounded by $\sigma^2 \asymp \delta^2$, the constant envelope $2|\log \kappa|$, and the uniform entropy of Lemma B.2(ii). The maximal inequality of Chernozhukov et al. (2014, Theorem 5.2) yields

$$\mathbb{E} \sup_{\mathcal{M}_\delta} |\mathbb{G}_N(l_i(\theta) - l_i(\theta_o))| \lesssim \phi_N(\delta) := \delta \sqrt{d_{K_N} \log(A/\delta)} + \frac{d_{K_N} \log(A/\delta)}{\sqrt{N}}$$

for a constant A , and $\delta \mapsto \phi_N(\delta)/\delta^{3/2}$ is decreasing on $(0, A/e)$.

Step 3: rate. Since $\log(A/\rho_N) \lesssim \log N$, so

$$\rho_N^{-2} \phi_N(\rho_N) \lesssim \rho_N^{-1} \sqrt{d_{K_N} \log N} + \rho_N^{-2} \frac{d_{K_N} \log N}{\sqrt{N}} \lesssim \sqrt{N},$$

where both bounds use $\rho_N^2 \geq K_N^2 \log N/N \gtrsim d_{K_N} \log N/N$. By Step 0, $\mathbb{M}_N(\hat{\theta}) \geq \mathbb{M}_N(\theta_o) - O_P(\rho_N^2)$ and $\text{H}(\hat{\theta}, \theta_o) = o_P(1)$. Theorem 3.4.1 of Van Der Vaart and Wellner (1996) then gives $\text{H}(\hat{\theta}, \theta_o) = O_P(\rho_N)$, and $\text{L} \leq 2\text{H}$ (Lemma G.1) completes the proof. $\square$

Lemma B.6 (global identification in the L metric). *Suppose Assumptions 4 and 5(a) hold. Then, for every $\epsilon > 0$,*

$$\delta(\epsilon) := \inf \left\{ \text{L}((\gamma, P), \theta_o) : \gamma \in \Gamma, \|\gamma - \gamma_o\| \geq \epsilon, P \in \bar{\mathcal{P}} \right\} > 0.$$

Proof. Suppose not. Then there are $(\gamma_n, P_n) \in \Gamma \times \bar{\mathcal{P}}$ with $\|\gamma_n - \gamma_o\| \geq \epsilon$ and $\text{L}((\gamma_n, P_n), \theta_o) \rightarrow 0$. By compactness of Γ (Euclidean) and of $\bar{\mathcal{P}}$ (Lemma B.1), pass to a subsequence with $\gamma_n \rightarrow \gamma^* \in \Gamma$, $\|\gamma^* - \gamma_o\| \geq \epsilon$, and $P_n \rightarrow \bar{P} \in \bar{\mathcal{P}}$ uniformly. Since $|P_n^t(\gamma'_n x) - \bar{P}^t(\gamma^{*'} x)| \leq \|P_n - \bar{P}\|_\infty + 2MR_X \|\gamma_n - \gamma^*\| \rightarrow 0$ uniformly over $\text{supp}(X)$,

we get $L((\gamma^*, \bar{P}), \theta_o) = 0$, i.e.

$$\bar{P}^t(\gamma'^* x) = P_o^t(\gamma'_o x) \quad \text{for } \mathbb{P}\text{-almost every } x, t \in \{d, c, n\}. \quad (\text{B.3})$$

Both sides of (B.3) are continuous in x , and every point of $\text{supp}(X)$ is a limit of points where (B.3) holds (the equality set has full measure, hence is dense in the support), so (B.3) holds for *every* $x \in \text{supp}(X)$. By Assumption 5(a), $\lambda_j > 0$ for every j , so this is the observational equivalence (A.1) between $(\gamma^*, \bar{P})$ and (γ_o, P_o) .

It remains to show that observational equivalence forces $\gamma^* = \gamma_o$, contradicting $\|\gamma^* - \gamma_o\| \geq \epsilon$. Theorem 1 does not apply verbatim, because $\bar{P} \in \bar{\mathcal{P}}$ is only Lipschitz. We rerun the proof of Theorem 1, replacing its differentiation step by a linear-algebra argument and recovering smoothness from the resulting affine form; the remaining steps carry over.

Linear algebra in place of differentiation. $h := P_o^{-1} \circ \bar{P}$ is well defined and continuous on $\cup_j \text{supp}(\gamma_j'^* W)$, and $h(\gamma_j'^* w) = \gamma_{oj}' w$ on $\text{supp}(W)$ for every j , where γ_j^* denotes judge j 's block of γ^* . Fix j and the slice ball $B \subseteq \text{supp}(W_a | W_b = w_b^{j,*})$ of Assumption 4(a). On the slice, $h(\gamma_{j,1}^* + \gamma_{j,a}^* w_a + \gamma_{j,b}^* w_b^{j,*}) = \gamma_{oj,1} + \gamma_{oj,a}' w_a + \gamma_{oj,b}' w_b^{j,*}$ for all $w_a \in B$. If $\gamma_{j,a}'(w_a - \tilde{w}_a) = 0$ for $w_a, \tilde{w}_a \in B$, the left sides coincide, hence so do the right sides: $\ker(\gamma_{j,a}') \cap (B - B) \subseteq \ker(\gamma_{oj,a}')$. Since $B - B$ contains an open neighborhood of the origin and $\text{rank}(\gamma_{oj,a}') = 2$ by Assumption 4(b), this forces $\text{rank}(\gamma_{j,a}') = 2$. Choose a linear right inverse S with $\gamma_{j,a}' S = I_2$. For $\tilde{z}$ in the open image $I_j(w_b^{j,*})$ of the slice, write $w_a = S(\tilde{z} - \gamma_{j,1}^* - \gamma_{j,b}^* w_b^{j,*}) + \nu$ with $\nu \in \ker(\gamma_{j,a}')$; the kernel inclusion shows $\gamma_{oj,a}' \nu = 0$, so

$$h(\tilde{z}) = M_j' \tilde{z} + c_j^{(w_b^{j,*})}, \quad M_j' := \gamma_{oj,a}' S,$$

on $I_j(w_b^{j,*})$, with $c_j^{(w_b^{j,*})}$ as in (A.3) and $\gamma_{oj,a} = \gamma_{j,a}^* M_j$, M_j nonsingular.

Only level identities are used, so the slice-pooling argument goes through and yields $\gamma_{oj,-1} = \gamma_{j,-1}^* M_j$ and the global affine form $h(\tilde{z}) = M_j' \tilde{z} + c_j$ on $\text{supp}(\gamma_j'^* W)$.

Smoothness from the affine form. The level identity now reads $\bar{P}(\tilde{z}) = P_o(M_j' \tilde{z} + c_j)$ on $\text{supp}(\gamma_j'^* W)$, so on the interior of that set $\bar{P}$ is C^1 with $D\bar{P}(\tilde{z}) = M_j DP_o(M_j' \tilde{z} + c_j)$ after using the symmetry of DP_o . Moreover $\bar{P}$ is the gradient of a C^1 surplus function (Lemma B.1), which is C^2 wherever $\bar{P}$ is C^1 ; hence $D\bar{P}$ is symmetric there, and transposing gives $M_j DP_o = DP_o M_j'$ on $\text{Int}(\text{supp}(\gamma_o' W))$. Steps 4 and 5 of Theorem 1 are algebra from this identity and Assumption 4(c), so $M_j = d_j I_2$ with d_j real and

nonzero. For the sign, monotonicity of $\bar{P}$ (Lemma B.1) makes $D\bar{P}$ positive semidefinite where it exists; $D\bar{P} = d_j DP_o(d_j \cdot + c_j)$ with DP_o positive definite rules out $d_j < 0$.

The cross-judge comparison uses only level identities, the injectivity of $\bar{P}$ on the sets where its Jacobian is positive definite (Lemma G.6, whose proof applies to $\bar{P}$ restricted to those open sets), Lemma G.5, and the normalizations; it yields $d_j \equiv d = 1$ and $c_j \equiv 0$, hence $\gamma^* = \gamma_o$. $\square$

Proof of Theorem 2. The rate claim is Proposition B.1. For consistency of $\hat{\gamma}$, fix $\epsilon > 0$. The estimated propensity satisfies $P_{\hat{\xi}} \in \mathcal{P} \subseteq \bar{\mathcal{P}}$ (standing conventions), so on the event $\{\|\hat{\gamma} - \gamma_o\| \geq \epsilon\}$ Lemma B.6 gives $L(\hat{\theta}, \theta_o) \geq \delta(\epsilon) > 0$. Since $L(\hat{\theta}, \theta_o) = o_P(1)$, the probability of this event vanishes, i.e. $\hat{\gamma} \rightarrow_p \gamma_o$. $\square$

Cells and notation for the curvature bound. A cell is a pair $c = (j, w_b)$ with $j \in \{1, \dots, J\}$ and $w_b \in \text{supp}(W_b)$; there are finitely many cells, each with probability $\pi_c = \mathbb{P}(J = j, W_b = w_b) \geq \underline{\pi} > 0$ by Assumption 5(a) and the definition of the support. On cell c , the index is affine in the continuous covariates, $\gamma'X = \gamma_{j,1} + \gamma'_{j,a}W_a + \gamma'_{j,b}w_b$. Write $f_c(\cdot | \gamma)$ for the density of $\gamma'X$ given the cell c (Lemma B.7), $f_\gamma := \sum_c \pi_c f_c(\cdot | \gamma)$ for the index density, and e_J for the intercept stack of Section 3.1, so that $e'_J X_i \equiv 1$. By Assumption 4(b) and continuity, there are a convex neighborhood $\mathcal{N}$ of γ_o in $\{\gamma : e'_J \gamma^c = e'_J \gamma^n = 0\}$ and $\underline{\sigma} > 0$ with $\sigma_2(\gamma_{j,a}) \geq \underline{\sigma}$ for every j and $\gamma \in \mathcal{N}$.

Lemma B.7 (cell index densities: existence, bound, L^1 continuity). *Suppose Assumptions 4(b) and 5(a),(d) hold. Then, for every cell c and $\gamma \in \mathcal{N}$, $\gamma'X$ given c admits a density $f_c(\cdot | \gamma)$ on $\mathbb{R}^2$ with $\sup_{\gamma \in \mathcal{N}} \|f_c(\cdot | \gamma)\|_\infty \leq \bar{f} < \infty$, and as $\gamma \rightarrow \gamma_o$,*

$$\int_{\mathbb{R}^2} |f_c(z | \gamma) - f_c(z | \gamma_o)| dz \rightarrow 0.$$

Proof. On cell c , $\gamma'X = \gamma'_{j,a}W_a + b_\gamma$, where $\gamma'_{j,a} \in \mathbb{R}^{2 \times \dim(W_a)}$ has rank 2 and $b_\gamma = \gamma_{j,1} + \gamma'_{j,b}w_b$. By the coarea formula, the pushforward of $p_{W_a|W_b}(\cdot | w_b)dw_a$ under $w_a \mapsto \gamma'_{j,a}w_a + b_\gamma$ has the density

$$f_c(z | \gamma) = \mathcal{J}(\gamma)^{-1} \int_{\ker \gamma'_{j,a}} p_{W_a|W_b}((\gamma'_{j,a})^+(z - b_\gamma) + \nu | w_b) d\nu,$$

where $\mathcal{J}(\gamma) = \sqrt{\det(\gamma'_{j,a}\gamma_{j,a})}$. The fiber is contained in a ball of radius R_X , the conditional density is bounded (continuous with compact support), and $\mathcal{J}(\gamma) \geq \underline{\sigma}^2$ on

$\mathcal{N}$, which gives the uniform bound $\bar{f}$. For the L^1 continuity, parametrize $\ker \gamma'_{j,a}$ by an orthonormal basis chosen continuously in γ on the (shrunk) neighborhood $\mathcal{N}$; the map $(z, \nu) \mapsto (\gamma'_{j,a})^+(z - b_\gamma) + O_\gamma \nu$ is affine with coefficients converging to those at γ_o , and $\mathcal{J}(\gamma) \rightarrow \mathcal{J}(\gamma_o)$. Since composition of a fixed L^1 function with converging invertible affine maps is continuous in L^1 (continuity of translation and dilation in L^1 , applied to the bounded, compactly supported $p_{W_a|W_b}$), integrating out ν over the bounded fiber and z over the bounded range yields the displayed convergence by Fubini and dominated convergence. $\square$

Lemma B.8 (local regularity). *Suppose Assumptions 5(a),(d) hold, and let $\{B_e\}_{e \in \mathcal{E}}$ be any finite collection of open balls with closures in $\text{Int}(\mathcal{Z})$. There are $\eta > 0$, $\pi_0 > 0$, and a neighborhood $\mathcal{N}$ of γ_o such that, with the eroded support $\mathcal{Z}_\eta := \{z \in \mathcal{Z} : \text{dist}(z, \partial \mathcal{Z}) \geq \eta\}$: (i) $\bar{\mathcal{Z}}_\eta \subset \text{Int}(\mathcal{Z})$ and $B_e \subset \mathcal{Z}_\eta$ for every $e \in \mathcal{E}$; (ii) $\mathbb{P}(\gamma'X \in \mathcal{Z}_\eta | c) \geq \pi_0$ for every cell c and every $\gamma \in \mathcal{N}$; (iii) $\inf_{\mathcal{Z}_\eta} p_Z > 0$.*

Proof. For each cell c , $\mathbb{P}(\gamma'_o X \in \text{Int}(\mathcal{Z}) | c) = 1$ since the index has a density and $\mathcal{Z} \setminus \text{Int}(\mathcal{Z})$ has measure zero (compact support with dense interior), so $\mathbb{P}(\gamma'_o X \in \mathcal{Z}_{2\eta} | c) \uparrow 1$ as $\eta \downarrow 0$; choose η small enough that this probability exceeds $1/2$ for every cell and that (i) holds. With $R_X \|\gamma - \gamma_o\| < \eta$ on $\mathcal{N}$, any realization with $\gamma'_o X \in \mathcal{Z}_{2\eta}$ has $\gamma'X \in \mathcal{Z}_\eta$, giving (ii) with $\pi_0 = 1/2$. Claim (iii) holds because p_Z is continuous and positive on the compact $\mathcal{Z} \supseteq \bar{\mathcal{Z}}_\eta$. $\square$

Lemma B.9 (non-degeneracy of the index field). *Suppose Assumption 4(c) and Assumption 5(b),(d) hold, and let $\mathcal{Z}_\eta$ be as in Lemma B.8 with η small. For $M \in \mathbb{R}^{2 \times 2}$ define $\Phi_M(z) := DP_o(z)M'z$ and let $\mathcal{G}$ be the class of gradient fields on $\mathcal{Z}_\eta$. There is $\underline{\lambda} > 0$ such that, for every M with $\text{vec}(M) \perp \text{vec}(I_2)$,*

$$\inf_{g \in \mathcal{G}} \int_{\mathcal{Z}_\eta} \|\Phi_M(z) - g(z)\|^2 dz \geq \underline{\lambda} \|M\|_F^2.$$

Proof. Let $H := DP_o = \nabla^2 G_o$, symmetric, positive definite (Lemma G.3), and C^1 on $\bar{\mathcal{Z}}$ ($k_o \geq 2$). The infimum equals the squared $L^2(\mathcal{Z}_\eta)$ norm of the component of Φ_M orthogonal to all gradient fields, which is positive if and only if Φ_M violates the cross-partial equality somewhere on $\mathcal{Z}_\eta$. Because third derivatives of G_o commute, the cross-partial asymmetry of the field is

$$\partial_{z^c} \Phi_M^n - \partial_{z^n} \Phi_M^c = \langle S_M, H(z) \rangle, \quad S_M = \begin{pmatrix} -M_{21} & \frac{1}{2}(M_{11} - M_{22}) \\ \frac{1}{2}(M_{11} - M_{22}) & M_{12} \end{pmatrix},$$

with $\langle S, H \rangle := S_{11}H_{11} + 2S_{12}H_{12} + S_{22}H_{22}$. The map $M \mapsto S_M$ is linear with kernel $\text{span}(I_2)$ and $\|S_M\| \asymp \|M\|_F$ on $\text{vec}(I_2)^\perp$. For nonzero symmetric S , $\langle S, H \rangle \neq 0$ on $\text{Int}(\mathcal{Z})$: if S is indefinite, $S = \frac{1}{2}(v_1v_2' + v_2v_1')$ with v_1, v_2 linearly independent and $\langle S, H \rangle = v_1'Hv_2$, which is not identically zero by Assumption 4(c) applied on $\text{Int}(\text{supp}(\gamma'_{oj}W)) \subseteq \text{Int}(\mathcal{Z})$; if S is definite, $\langle S, H \rangle = \text{tr}(SH) \neq 0$; if $S = \pm uu'$, $\langle S, H \rangle = \pm u'Hu \neq 0$, the last two by $H \succ 0$. Hence $c_0 := \inf_{\|S\|=1} \int_{\text{Int}(\mathcal{Z})} \langle S, H \rangle^2 dz > 0$ by compactness of the unit sphere and continuity. Eroding to $\mathcal{Z}_\eta$ removes the interior strip $\text{Int}(\mathcal{Z}) \setminus \mathcal{Z}_\eta$, whose Lebesgue measure vanishes as $\eta \downarrow 0$; since $|\langle S, H \rangle| \leq \|H\|_\infty$, $\int_{\mathcal{Z}_\eta} \langle S, H \rangle^2 \geq c_0 - \|H\|_\infty^2 \text{meas}(\text{Int}(\mathcal{Z}) \setminus \mathcal{Z}_\eta) \geq c_0/2$ for η small, uniformly in $\|S\| = 1$. Finally $\underline{\lambda} := \min\{\inf_g \int_{\mathcal{Z}_\eta} \|\Phi_M - g\|^2 : \|\text{vec}(M)\| = 1, \text{vec}(M) \perp \text{vec}(I_2)\} > 0$ by compactness, since each value is positive by the cross-partial computation above. $\square$

Lemma B.10 (overlap chaining). *Suppose Assumptions 4(e) and 5(a),(d) hold, and suppose each judge's index $\gamma'_{oj}W_i$ admits a continuous density f_j . There exist a connected graph $\mathcal{G}^*$ on $\{1, \dots, J\}$ and, for each of its edges (j, j') , a closed ball $B_{jj'} \subset \text{Int}(\mathcal{Z})$ on which $\min\{f_j, f_{j'}\} \geq \underline{f} > 0$.*

Proof. The sets $O_j := \{f_j > 0\}$ are open by the assumed continuity, and $\text{supp}(\gamma'_{oj}W) = \bar{O}_j$. Fix an edge (j, j') of the judge-adjacency graph and let C be a connected component of the open overlap $\text{Int}(\text{supp}(\gamma'_{oj}W)) \cap \text{Int}(\text{supp}(\gamma'_{oj'}W))$, which is nonempty by Assumption 4(e). First, C meets both O_j and $O_{j'}$: C is an open subset of $\bar{O}_j$, and an open set disjoint from O_j cannot lie in $\bar{O}_j \setminus O_j$, the boundary of an open set having empty interior. Second, $C \subseteq \cup_l O_l$: at every $z \in C \subseteq \mathcal{Z}$ the mixture $p_Z(z) = \sum_l \lambda_l f_l(z)$ is positive, so some $f_l(z) > 0$. A connected set covered by open sets is chained: any two of its points are joined by a finite sequence of the covering sets with consecutive pairwise intersections nonempty in C . Applying this between a point of $C \cap O_j$ and a point of $C \cap O_{j'}$ yields judges $j = l_0, l_1, \dots, l_m = j'$ with $O_{l_{s-1}} \cap O_{l_s} \cap C \neq \emptyset$ for every s ; each such intersection is open and nonempty, and continuity supplies a closed ball inside it on which both densities exceed a positive constant. Collect these pairs (l_{s-1}, l_s) , over all edges of the judge-adjacency graph, as the edges of $\mathcal{G}^*$, with the corresponding balls, taking $\underline{f}$ as the minimum over the finitely many balls. Every edge of the judge-adjacency graph connects its endpoints within $\mathcal{G}^*$, so connectedness of $\mathcal{G}^*$ follows from Assumption 4(e). $\square$

Proposition B.2 (primitive sufficient conditions for Assumption 5(e)). *Suppose Assumptions 4 and 5(a),(d) hold, and in addition: (i) each judge's index $\gamma'_{oj}W_i$ admits*

a continuous density; (ii) writing $W_{-1,i} = (W'_{a,i}, W'_{b,i})'$ for the covariates excluding the intercept and $\gamma_{j,-1}$ conformably, there exist $\epsilon, \underline{v} > 0$ such that $\text{Var}(v'W_{-1,i} | \gamma'_j W_i) \geq \underline{v} \|v\|^2$ almost surely, for every judge j and every (v, γ_j) with $\gamma'_{j,-1}v = 0$ and $\|\gamma_j - \gamma_{oj}\| < \epsilon$. Then Assumption 5(e) holds: there are $\delta' > 0$ and a neighborhood $\mathcal{N}$ of γ_o such that, for every $\gamma \in \mathcal{N}$ and every $u \in \mathbb{R}^{\dim(X_i)}$ with $u \perp \text{span}(\gamma^c, \gamma^n, e_J)$,

$$\mathbb{E}[\text{Var}(u'X | \gamma'X)] \geq \delta' \|u\|^2.$$

Proof. Let $\mathcal{G}^*$ and the balls $\{B_{jj'}\}$ be as in Lemma B.10, whose hypotheses hold under (i). Fix $\gamma \in \mathcal{N}$ and $u \perp \text{span}(\gamma^c, \gamma^n, e_J)$; all constants below are uniform over both. Decompose each judge block as $u_j = (u_{j,1}, u_{j,-1})$ and $u_{j,-1} = \gamma_{j,-1}\alpha_j + w_j$ with $\gamma'_{j,-1}w_j = 0$; the maps $u \mapsto \alpha_j$ and $u \mapsto w_j$ are linear with coefficients continuous in γ on $\mathcal{N}$, by the rank condition and continuity. Since $\gamma'_j W = \gamma_{j,1} + \gamma'_{j,-1}W_{-1}$, on the event $\{J = j, \gamma'X = z\}$,

$$u'X = u'_j W = \alpha'_j z + c_j + w'_j W_{-1}, \quad c_j := u_{j,1} - \alpha'_j \gamma_{j,1},$$

and, because W is independent of J , the conditional variance decomposes over judges as

$$\mathbb{E}[\text{Var}(u'X | Z)] = \underbrace{\sum_j \mathbb{E}[\mathbb{P}(J = j | Z) \text{Var}(w'_j W_{-1} | \gamma'_j W = Z)]}_{\text{within}} + \underbrace{\mathbb{E}[\text{Var}_J(a_J(Z) | Z)]}_{\text{across}}, \quad (\text{B.4})$$

with $Z := \gamma'X$, $a_j(z) := \alpha'_j z + c_j + \mu_j(z)$, and $\mu_j(z) := \mathbb{E}[w'_j W_{-1} | \gamma'_j W = z]$, so that $|\mu_j| \leq R_X \|w_j\|$.

Within. Hypothesis (ii) floors the conditional variance by $\underline{v} \|w_j\|^2$ for almost every z , uniformly over $\gamma \in \mathcal{N}$, and $\mathbb{E}[\mathbb{P}(J = j | Z)] = \lambda_j$ by independence. Hence within $\geq \underline{v} \lambda \sum_j \|w_j\|^2 =: c_1 \|w\|^2$.

Across. Let $f_j(\cdot | \gamma)$ denote the index density of judge j , the slice mixture of the cell densities of Lemma B.7, which inherits the bound $\bar{f}$ and the L^1 continuity in γ . By the pairwise variance identity, for each edge (j, j') of $\mathcal{G}^*$,

$$\text{across} \geq \int_{B_{jj'}} \mathbb{P}(j | z) \mathbb{P}(j' | z) (a_j(z) - a_{j'}(z))^2 f_\gamma(z) dz = \lambda_j \lambda_{j'} \int_{B_{jj'}} \frac{f_j(z | \gamma) f_{j'}(z | \gamma)}{f_\gamma(z)} (a_j - a_{j'})^2 dz.$$

Lemma B.10 floors $f_j(\cdot|\gamma_o)$ and $f_{j'}(\cdot|\gamma_o)$ on $B_{jj'}$, and the L^1 continuity gives, after shrinking $\mathcal{N}$, a set $S_\gamma \subseteq B_{jj'}$ with $\text{meas}(S_\gamma) \geq \frac{3}{4} \text{meas}(B_{jj'})$ on which both densities exceed half their floors, while $f_\gamma \leq \bar{f}$. The difference $a_j - a_{j'} = g_{jj'} + (\mu_j - \mu_{j'})$ splits into the affine function $g_{jj'}(z) := (\alpha_j - \alpha_{j'})'z + (c_j - c_{j'})$ and a remainder bounded by $2R_X\|w\|$, so $(a_j - a_{j'})^2 \geq \frac{1}{2}g_{jj'}^2 - 4R_X^2\|w\|^2$, and the sublevel-strip bound for affine functions on S_γ gives $\int_{S_\gamma} g_{jj'}^2 \geq c_B(\|\alpha_j - \alpha_{j'}\|^2 + |c_j - c_{j'}|^2)$. Combining over the edges,

$$\text{across} \geq c_2 \sum_{(j,j')} \left(\|\alpha_j - \alpha_{j'}\|^2 + |c_j - c_{j'}|^2 \right) - C_3\|w\|^2.$$

Combining. Averaging the two bounds with weights adapted to C_3 , (B.4) gives $\mathbb{E}[\text{Var}(u'X|Z)] \geq c_4\|T_\gamma u\|^2$ with the linear map $T_\gamma u := ((w_j)_j, (\alpha_j - \alpha_{j'}, c_j - c_{j'})_{(j,j')})$, whose coefficients are continuous in γ . Its kernel is exactly $\text{span}(\gamma^c, \gamma^n, e_J)$: $w_j \equiv 0$ forces $u_{j,-1} = \gamma_{j,-1}\alpha_j$, and connectedness of $\mathcal{G}^*$ (Lemma B.10) forces $\alpha_j \equiv \alpha$ and $c_j \equiv c$, whence $u_{j,-1} = \gamma_{j,-1}\alpha$ and $u_{j,1} = \alpha'\gamma_{j,1} + c$, i.e. $u = \gamma\alpha + ce_J$. Positivity and uniformity follow from continuity of $(\gamma, u) \mapsto \|T_\gamma u\|^2$ on the compact set $\{\gamma \in \bar{\mathcal{N}}, \|u\| = 1, u \perp \text{span}(\gamma^c, \gamma^n, e_J)\}$. $\square$

Proposition B.3 (variation control). *Suppose Assumptions 4 and 5(a),(d),(e) hold, Assumption 6(a) holds, and $k_o \geq 2$. There are $\epsilon_0 > 0$ and $c > 0$ such that, for every $\gamma \in \Gamma$ with $\|\gamma - \gamma_o\| < \epsilon_0$,*

$$\inf_{P \in \bar{\mathcal{P}}} \mathcal{L}((\gamma, P), \theta_o) \geq c\|\gamma - \gamma_o\|.$$

Proof. Fix $\gamma \in \Gamma \cap \mathcal{N}$, $P \in \bar{\mathcal{P}}$, and write $z := \gamma'X$, $z_o := \gamma_o'X$, $E := \gamma - \gamma_o$, $P_\gamma := \gamma(\gamma'\gamma)^{-1}\gamma'$, and $H := DP_o$.

Step 1: exact decomposition of the index shift. Regressing z_o on z ,

$$z - z_o = E'X = Mz - r, M := I_2 - \gamma_o'\gamma(\gamma'\gamma)^{-1} = E'\gamma(\gamma'\gamma)^{-1}, r := \gamma_o'(I - P_\gamma)X = -E'_\perp X,$$

with $E_\perp := (I - P_\gamma)E$, using $\gamma'(I - P_\gamma) = 0$. With $B := (\gamma'\gamma)^{-1}\gamma'E$, so that $E = \gamma B + E_\perp$, one has $M = B'$ exactly.

Step 2: the residual block is orthogonal to e_J . Because $\gamma, \gamma_o \in \Gamma$, the columns of E satisfy $e_J'E^t = 0$, and $e_J \perp \text{col}(\gamma)$ since $e_J'\gamma = 0$, so $e_J'E_\perp^t = e_J'E^t = 0$. Each column of $E_\perp$ therefore lies in $\text{span}(\gamma^c, \gamma^n, e_J)^\perp$. Writing $m_\perp(z) := \mathbb{E}[E'_\perp X|z]$ and $\tilde{E} := E'_\perp X - m_\perp(z)$, we have $r = -E'_\perp X$, $r - \mathbb{E}[r|z] = -\tilde{E}$.

Step 3: two orthogonal blocks in $\mathbf{L}^2$. By a first-order expansion ($P_o \in C^2$), $P_o(z) - P_o(z_o) = H(z)(Mz - r) + R_2$ with $\|R_2\| \leq C\|E\|^2$. Collect the z -measurable terms. Splitting $M = M_\perp + \tau I_2$ into traceless and trace parts, the trace part is absorbed as a gradient, $\Phi_{\tau I_2}(z) = \tau H(z)z = \tau \nabla(z'P_o(z) - G_o(z))$. Hence

$$P(z) - P_o(z_o) = \underbrace{\left[\tilde{Q}(z) + \Phi_{M_\perp}(z) + H(z)m_\perp(z) \right]}_{=:g(z)} + H(z)\tilde{E} + R_2,$$

where $\tilde{Q} := (P - P_o) + (\text{absorbed gradient fields})$ is a gradient field bounded by a constant C_b (the cap bounds $P - P_o$; the absorbed fields are bounded on the compact $\tilde{\mathcal{Z}}$ by $C\|E\|$ -multiples of fixed functions). Since g is z -measurable and $\mathbb{E}[\tilde{E}|z] = 0$, the cross term vanishes and

$$\mathbf{L}((\boldsymbol{\gamma}, P), \theta_o) \geq \left(\|g\|_{L^2}^2 + \|H\tilde{E}\|_{L^2}^2 \right)^{1/2} - C\|E\|^2. \quad (\text{B.5})$$

Step 4: orthogonal block. On all of $\tilde{\mathcal{Z}}$, $\lambda_{\min}(H) \geq c_H > 0$: the Jacobian DP_o is positive definite at every point by Lemma G.3, using that densities in $\mathcal{F}$ are everywhere positive, and it is continuous on the compact $\tilde{\mathcal{Z}}$. Applying Assumption 5(e) to each unit column $E_\perp^t / \|E_\perp^t\|$, which is orthogonal to the columns of the running $\boldsymbol{\gamma}$ and to e_J ,

$$\|H\tilde{E}\|_{L^2}^2 \geq c_H^2 \mathbb{E}\|\tilde{E}\|^2 = c_H^2 \sum_t \mathbb{E} \left[\text{Var}(E_\perp^{t'} X | z) \right] \geq c_H^2 c_3 \|E_\perp\|^2 =: c_5 \|E_\perp\|^2.$$

Step 5: index block. By Assumption 6(a) there are $0 < \underline{p} \leq \bar{p}$ and a neighborhood $\mathcal{N}$ of $\boldsymbol{\gamma}_o$ with $\underline{p} \leq p_\boldsymbol{\gamma} \leq \bar{p}$ on $\text{supp}(\boldsymbol{\gamma}'X)$ for every $\boldsymbol{\gamma} \in \mathcal{N}$. Since $\tilde{\mathcal{Z}}_\eta \subset \text{Int}(\mathcal{Z})$ with $\text{dist}(\tilde{\mathcal{Z}}_\eta, \partial\mathcal{Z}) \geq \eta > 0$, and $\text{supp}(\boldsymbol{\gamma}'X)$, the image of the compact $\text{supp}(X)$ under $x \mapsto \boldsymbol{\gamma}'x$, varies continuously in $\boldsymbol{\gamma}$, a further shrinking of $\mathcal{N}$ gives $\mathcal{Z}_\eta \subseteq \text{supp}(\boldsymbol{\gamma}'X)$ for every $\boldsymbol{\gamma} \in \mathcal{N}$. Hence $p_\boldsymbol{\gamma} \geq \underline{p}$ on $\mathcal{Z}_\eta$ throughout $\mathcal{N}$. The field $\tilde{Q}$ is a gradient field in $L^2(\mathcal{Z}_\eta)$, so by the triangle inequality in L^2 and $\|Hm_\perp\|_{L^2} \leq C\|E_\perp\|$,

$$\begin{aligned} \|g\|_{L^2} &\geq \left(\int_{\mathcal{Z}_\eta} \|\Phi_{M_\perp}(z) + \tilde{Q}(z)\|^2 p_\boldsymbol{\gamma}(z) dz \right)^{1/2} - C\|E_\perp\| \\ &\geq \left(\underline{p} \int_{\mathcal{Z}_\eta} \|\Phi_{M_\perp} + \tilde{Q}\|^2 dz \right)^{1/2} - C\|E_\perp\|, \end{aligned}$$

and Lemma B.9, whose infimum runs over all gradient fields and is homogeneous of degree two, bounds the integral below by $\lambda \|M_\perp\|_F^2$. Hence $\|g\|_{L^2} \geq c_6 \|M_\perp\|_F - C \|E_\perp\|$ with $c_6 := (p\lambda)^{1/2}$, uniformly over $\gamma \in \mathcal{N}$.

Step 6: norm accounting and combination. The normalization $\|\gamma\|_F = \|\gamma_o\|_F = 1$ gives $\text{tr}(\gamma' \gamma B) = \text{tr}(\gamma' E) = \frac{1}{2} \|E\|_F^2$. Writing $B = B_0 + \tau I_2$ with $\text{tr}(\gamma' \gamma B_0) = 0$ yields $|\tau| = \frac{1}{2} \|E\|^2 / \text{tr}(\gamma' \gamma) = \frac{1}{2} \|E\|^2$. On the subspace $\{\text{tr}(\gamma' \gamma B_0) = 0\}$ the linear map $B_0 \mapsto (B'_0)_{\text{traceless}}$ is injective (its kernel $B_0 \propto I_2$ meets the subspace only at 0, since $\text{tr}(\gamma' \gamma) = 1 \neq 0$), so $\|(B'_0)_{\text{traceless}}\| \geq c_7 \|B_0\|$ with c_7 uniform over $\mathcal{N}$. Hence, with $M = B'$,

$$\|M_\perp\| \geq c_7 \|B\| - C \|E\|^2.$$

Also $\|E\|^2 = \|\gamma B\|^2 + \|E_\perp\|^2 \leq C(\|B\|^2 + \|E_\perp\|^2)$ by the uniform positive definiteness of $\gamma' \gamma$ on $\mathcal{N}$. Fix Λ large. If $\|B\| \leq \Lambda \|E_\perp\|$, then (B.5) and Step 4 give $\mathbf{L} \geq c_5^{1/2} \|E_\perp\| - C \|E\|^2 \gtrsim \|E\| - C \|E\|^2$. If $\|B\| > \Lambda \|E_\perp\|$, then Steps 5–6 give $\mathbf{L} \geq c_6(c_7 - C/\Lambda) \|B\| - C \|E\|^2 \gtrsim \|E\| - C \|E\|^2$ for Λ large. Choosing ϵ_0 small absorbs the quadratic terms and proves the claim. $\square$

Proof of Theorem 3, first claim. By Theorem 2, $\|\hat{\gamma} - \gamma_o\| < \epsilon_0$ with probability tending to one. On that event, since $P_\xi \in \mathcal{P} \subseteq \bar{\mathcal{P}}$, Proposition B.3 gives $\|\hat{\gamma} - \gamma_o\| \leq c^{-1} \mathbf{L}(\hat{\theta}, \theta_o)$, and the right side is $O_P(\rho_N)$ by Proposition B.1. $\square$

Proposition B.4 (uniform rate of the estimated propensity, interior version). *Suppose the conditions of Theorem 3 hold, and let $\mathcal{Z}_{(q+1)h} := \{z \in \mathcal{Z} : \text{dist}(z, \partial \mathcal{Z}) \geq (q+1)h\}$ with $h \asymp K_N^{-1}$ the mesh width. Then, for $t \in \{c, n\}$,*

$$\|P_\xi^t - P_o^t\|_{L^\infty(\mathcal{Z}_{(q+1)h})} = O_P(K_N \rho_N).$$

Proof. First remove the index error: by Lemma B.3(i) and Theorem 3, $(\mathbb{E} \sum_t (P_\xi^t(\gamma'_o X) - P_o^t(\gamma'_o X))^2)^{1/2} \leq \mathbf{L}(\hat{\theta}, \theta_o) + 2MR_X \|\hat{\gamma} - \gamma_o\| = O_P(\rho_N)$. Since $p_Z \geq \underline{p}_Z := \min_{\mathcal{Z}} p_Z > 0$ (Assumption 5(d)), it follows that $\|P_\xi^t - P_o^t\|_{L^2(\mathcal{Z}, dz)} = O_P(\rho_N)$. Interpose the approximant of Lemma B.4: $s := P_\xi^t - P_{\xi_o}^t$ is a spline of order $(q-1, q)$ on the mesh with $\|s\|_{L^2(\mathcal{Z}, dz)} = O_P(\rho_N + K_N^{-k_o}) = O_P(\rho_N)$. For $z \in \mathcal{Z}_{(q+1)h}$, every basis function active at z has support contained in $\mathcal{Z}$, so the local stability of the B-spline basis (De Boor and De Boor, 1978, Ch. XI) gives $|s(z)| \leq C \|(\text{active coefficients})\|_{\ell^\infty} \leq C \|(\text{active coefficients})\|_{\ell^2} \leq Ch^{-1} \|s\|_{L^2(\mathcal{Z}, dz)}$, whence $\|s\|_{L^\infty(\mathcal{Z}_{(q+1)h})} = O_P(K_N \rho_N)$. The triangle inequality with $\|P_{\xi_o}^t - P_o^t\|_\infty \lesssim K_N^{-k_o}$ completes the proof. $\square$

C Asymptotic Normality and Efficiency

This subsection proves Theorem 4. The argument has four parts. Part I controls the empirical information matrices uniformly over a shrinking neighborhood (Lemmas C.3–C.9): a restricted eigenvalue in the gradient seminorm on the core, relative convergence of the design Gram and of the gradient information, and a boundary layer lemma that disposes of the boundary cells. Part II sets up the profile expansion (Lemmas C.10–C.12): a uniform rate for the inner sieve MLE over the neighborhood, interiority of the cap, and the sphere-constrained expansion in integral form. Part III computes the two ingredients of the expansion: the profile Hessian limit (Lemmas C.14–C.17) and the central limit theorem for the profiled score with the corrected Neyman centering (Lemmas C.18–C.22). Part IV identifies the limit variance with the semiparametric efficiency bound (Lemma C.23) and assembles the proof.

Notation. Write $\omega_i = (T_i, X_i)$, $Z = \gamma'_o X$, and, for $\xi \in \Xi(K_N)$, $b^t := b_{K_N}^t = \partial_{z^t} b_{K_N}$ as in the conventions above, with $b^d = -(b^c + b^n)$. The scores and Hessian blocks of $l(\omega, \gamma, P)$ are

$$\begin{aligned} s_\gamma(\omega, \gamma, P) &= \sum_t \frac{T^t}{P^t(\gamma'X)} DP^t(\gamma'X) \otimes X, & s_\xi(\omega, \gamma, P) &= \sum_t \frac{T^t}{P^t(\gamma'X)} b^t(\gamma'X), \\ s_{\gamma\gamma}(\omega, \gamma, P) &= \sum_t \frac{T^t}{P^t} D^2 P^t \otimes (XX') - \sum_t \frac{T^t}{(P^t)^2} DP^t DP^{t'} \otimes (XX'), \\ s_{\gamma\xi}(\omega, \gamma, P) &= \sum_t \frac{T^t}{P^t} \nabla b^t \otimes X - \sum_t \frac{T^t}{(P^t)^2} DP^t \otimes X \otimes b^{t'}, \\ s_{\xi\xi}(\omega, \gamma, P) &= - \sum_t \frac{T^t}{(P^t)^2} b^t b^{t'}, \end{aligned}$$

all evaluated at $\gamma'X$. The empirical and population gradient information are

$$\mathbb{J}_N(\gamma, P) := -\mathbb{P}_N s_{\xi\xi}(\omega_i, \gamma, P) \geq 0, \quad \mathbb{J}(\gamma, P) := \mathbb{E} \mathbb{J}_N(\gamma, P), \quad \mathbb{J}_o := \mathbb{J}(\gamma_o, P_o),$$

and the design Gram and its mean are

$$\hat{\Sigma}_N(\gamma) := \mathbb{P}_N b^{t_i}(\gamma'X_i) b^{t_i}(\gamma'X_i)', \quad \Sigma(\gamma) := \mathbb{E} \hat{\Sigma}_N(\gamma),$$

with t_i the realized treatment. The information blocks at the truth are $I_{\gamma\gamma} :=$

$\mathbb{E}[s_{\boldsymbol{\gamma}}(\omega, \boldsymbol{\gamma}_o, P_o)^{\otimes 2}]$ and $I_{\boldsymbol{\gamma}_\xi} := \mathbb{E}[s_{\boldsymbol{\gamma}} s_{\boldsymbol{\gamma}}'](\boldsymbol{\gamma}_o, P_o)$, and the sieve efficient information is

$$\tilde{I}_K := I_{\boldsymbol{\gamma}\boldsymbol{\gamma}} - I_{\boldsymbol{\gamma}_\xi} \mathbb{J}_o^+ I_{\boldsymbol{\gamma}_\xi}.$$

Neighborhoods: $\mathcal{N} := \{\boldsymbol{\gamma} : e'_J \boldsymbol{\gamma}^c = e'_J \boldsymbol{\gamma}^n = 0, \|\boldsymbol{\gamma}\|_F \leq 1, \|\boldsymbol{\gamma} - \boldsymbol{\gamma}_o\| \leq C\rho_N\}$, which contains the segment from $\boldsymbol{\gamma}_o$ to $\hat{\boldsymbol{\gamma}}$ with probability tending to one by Theorem 3, and

$$\mathcal{U}_N := \{(\boldsymbol{\gamma}, P_\xi) : \boldsymbol{\gamma} \in \mathcal{N}, \xi \in \Xi(K_N), \mathsf{L}((\boldsymbol{\gamma}, P_\xi), \theta_o) \leq C\rho_N\}.$$

Cell geometry: on the mesh of width $h \asymp K_N^{-1}$, a cell Q is *fully covered* if $Q \subseteq \mathcal{Z}$ up to a set of measure zero, *empty* if $|Q \cap \mathcal{Z}| = 0$, and a *boundary cell* otherwise. The *core* $\mathcal{A}$ consists of the fully covered cells whose mesh neighbors are also fully covered, so $\text{dist}(\Omega_{\mathcal{A}}, \mathcal{Z}^c) \geq h$; the *boundary layer* $\mathcal{W}$ collects the boundary cells and the outer layer of fully covered cells; $\Omega_{\mathcal{A}}, \Omega_{\mathcal{W}}$ are the corresponding regions, and $m_Q := \mathbb{P}(\boldsymbol{\gamma}'_o X \in Q)$. By Assumption 6(a), $\mathcal{Z}$ is connected with finite perimeter, so $|\Omega_{\mathcal{W}}| = O(h)$ and $\mathbb{P}_{\boldsymbol{\gamma}_o}(\Omega_{\mathcal{W}}) \leq \bar{p}|\Omega_{\mathcal{W}}| = O(h) \rightarrow 0$. Here $\mathbb{P}_{\boldsymbol{\gamma}_o}$ denotes the distribution of the index $\boldsymbol{\gamma}'_o X$, with density $p_{\boldsymbol{\gamma}_o}$ bounded above by $\bar{p}$ under Assumption 6(a). Basis functions whose support meets $\mathcal{A}$ are called *interior*; those supported in $\Omega_{\mathcal{W}} \cup \Omega_{\mathcal{V}}$ are called *boundary* basis functions, indexed $\mathcal{C}$.

Rates: $r_N := \sqrt{K_N^2 \log N/N}$ and $\tilde{r}_N := \sqrt{K_N^4 \log N/N}$; throughout, Assumption 6(c), i.e. $K_N = o((N/\log N)^{1/6})$, gives $K_N^6 \log N = o(N)$, hence $\tilde{r}_N \rightarrow 0$, $K_N \rho_N \rightarrow 0$, and $K_N^3 \sqrt{\log N/N} \rightarrow 0$.

Core stability. The core $\Omega_{\mathcal{A}}$ is interior to $\mathcal{Z} = \text{supp}(\boldsymbol{\gamma}'_o X)$ by the mesh setback $(q+1)h$. Since X is bounded, $\text{supp}(\boldsymbol{\gamma}' X)$ lies within $O(\rho_N)$ of $\mathcal{Z}$ for $\boldsymbol{\gamma} \in \mathcal{N}$, and $\rho_N = o(h)$ under Assumption 6(c), with $\rho_N/h \asymp K_N^2 \sqrt{\log N/N} \rightarrow 0$; the setback therefore absorbs the support movement, so $\Omega_{\mathcal{A}} \subseteq \text{supp}(\boldsymbol{\gamma}' X)$ and $p_{\boldsymbol{\gamma}} \in [\underline{p}, \bar{p}]$ on $\Omega_{\mathcal{A}}$ for every $\boldsymbol{\gamma} \in \mathcal{N}$ by Assumption 6(a). Connectedness of $\Omega_{\mathcal{A}}$ follows from that of $\mathcal{Z}$. The boundary layer keeps measure $|\Omega_{\mathcal{W}}| = O(h)$.

Least-favorable direction. Let $m_X(z) := \mathbb{E}[X | Z = z]$, let Π_∞ denote the projection, in the Fisher inner product $\langle g, h \rangle_w := \mathbb{E} \sum_t g^t(Z) h^t(Z) / P_o^t(Z)$, onto the nuisance tangent space $\mathcal{T}$, the closed span of the propensity scores s_ξ , and let $h^{\text{lf}} := \Pi_\infty[(DP_o^c \otimes m_X, DP_o^n \otimes m_X)]$. Write $\epsilon_K^{\text{lf}} := \text{dist}_{L^2}(h^{\text{lf}}, \text{GradSpl}(K))$.

Lemma C.1 (nondegeneracy of the efficient information). *Under Assumptions 4, 5 and 6, $v' I_{\boldsymbol{\gamma}\boldsymbol{\gamma}}^{\text{eff}} v > 0$ for every nonzero $v \in \mathcal{T}_{\boldsymbol{\gamma}_o}$.*

Proof. Write $v \in L$ as a matrix $V \in \mathbb{R}^{\dim(X) \times 2}$, so that the path $\gamma_o + \vartheta V$ shifts the index by $\vartheta V'X$. By construction $v'I_{\gamma\gamma}^{\text{eff}}v = \|v's_\gamma - \Pi_\infty(v's_\gamma)\|_w^2$, the squared Fisher distance from the index score in direction v to the nuisance tangent $\mathcal{T}$, and it vanishes if and only if $v's_\gamma \in \mathcal{T}$. Differentiating the propensities along the path, the index score in direction v is carried by the field $x \mapsto DP_o(\gamma'_o x)V'x$, while $\mathcal{T}$ is the closed span of the propensity perturbations $\nabla\delta G$ obtained by moving the surplus at fixed γ_o , each a gradient field in z alone. Membership $v's_\gamma \in \mathcal{T}$ therefore requires two things of $DP_o(\gamma'_o X)V'X$: that it depend on X only through $Z = \gamma'_o X$, and that as a field in z it be a gradient.

Decompose $V = \gamma_o A + V_\perp$ with $A \in \mathbb{R}^{2 \times 2}$ and the columns of $V_\perp$ orthogonal to γ_o^c , γ_o^n and e_J , which is the decomposition used in Step 4 of Proposition B.3. The two blocks are orthogonal in $\langle \cdot, \cdot \rangle_w$, since Φ_A is Z -measurable while $DP_o(Z)V'_\perp X$ has conditional mean zero given Z , so it suffices to bound each. For $V_\perp$, Assumption 5(e) gives $\mathbb{E}[\text{Var}(V_\perp^t X | Z)] \geq c_3 \|V_\perp^t\|^2$ for each column, so the part of $DP_o(Z)V'_\perp X$ that is not Z -measurable has Fisher norm bounded below by $c_H^2 c_3 \|V_\perp\|_F^2$ with $c_H := \inf_{\bar{z}} \lambda_{\min}(DP_o) > 0$ (Lemma G.3), and no element of $\mathcal{T}$ can match it. For $\gamma_o A$, the induced field is $z \mapsto DP_o(z)A'z = \Phi_A(z)$ in the notation of Lemma B.9, which is Z -measurable, so only the gradient requirement binds. Lemma B.9 gives $\inf_{g \in \mathcal{G}} \int_{\mathcal{Z}_\eta} \|\Phi_A - g\|^2 \geq \underline{\lambda} \|A\|_F^2$ whenever $\text{vec}(A) \perp \text{vec}(I_2)$, the infimum running over gradient fields, and $\mathcal{T}$ restricted to $\mathcal{Z}_\eta$ consists of gradient fields. The Lebesgue bound of Lemma B.9 transfers to $\langle \cdot, \cdot \rangle_w$ because p_Z is bounded away from zero on $\mathcal{Z}_\eta$ by Lemma B.8(iii) and $P_o^t \leq 1$. Hence Φ_A is matchable within $\mathcal{T}$ only when $A \propto I_2$, that is only when $V \propto \gamma_o$.

Collecting the two blocks, $v's_\gamma \in \mathcal{T}$ forces $V_\perp = 0$ and $A \propto I_2$, so $v \in \text{span}(\text{vec}(\gamma_o)) \oplus \text{span}\{e_J \otimes u : u \in \mathbb{R}^2\}$, the radial direction together with the per-judge intercept-mean directions. Both families are annihilated by Π_{γ_o} and lie outside $\mathcal{T}_{\gamma_o}$. For nonzero $v \in \mathcal{T}_{\gamma_o}$ the residual is therefore nonzero, and $v'I_{\gamma\gamma}^{\text{eff}}v > 0$. $\square$

Lemma C.2 (maximal inequality for sieve-indexed classes). *Let $\mathcal{F}_N$ be a class of functions of ω indexed by $(\gamma, \xi) \in \mathcal{N} \times \Xi(K_N)$ (or a subset), with envelope F_N and uniform entropy $\sup_Q \log N(\epsilon \|F_N\|_{L^2(Q)}, \mathcal{F}_N, L^2(Q)) \leq Cd_{K_N} \log(A_N/\epsilon)$ for a constant C and $A_N \leq N^C$. Then*

$$\mathbb{E} \sup_{\mathcal{F}_N} |(\mathbb{P}_N - \mathbb{P})f| \lesssim F_N \left(\sqrt{\frac{d_{K_N} \log N}{N}} + \frac{d_{K_N} \log N}{N} \right) \lesssim F_N K_N \sqrt{\frac{\log N}{N}},$$

the second bound holding when K_N is polynomial in N and $d_{K_N} \log N = O(N)$. Moreover, every class used below is of the form $f_{\gamma, \xi}(\omega) = \sum_t T^t w(P_\xi^t(\gamma' X)) \varphi_\xi^t(\gamma' X) \psi(X)$ with w Lipschitz on $[\kappa, 1 - \kappa]$, ψ bounded, and φ_ξ^t lying in a fixed $O(d_{K_N})$ -dimensional linear space with envelope F_N and Lipschitz constant in z bounded by a polynomial of K_N ; any such class satisfies the entropy bound.

Proof. The moment bound is the standard maximal inequality via the uniform entropy integral, or Chernozhukov et al. (2014, Theorem 5.2) with $\sigma = \|F_N\|$. For the entropy: for fixed γ , the map $\xi \mapsto f_{\gamma, \xi}$ factors through the finite-dimensional space containing (φ_ξ^t, P_ξ^t) , and products and Lipschitz transforms of VC-subgraph classes with bounded envelopes retain uniform entropy of order $d_{K_N} \log(1/\epsilon)$ up to constants (Van Der Vaart and Wellner, 1996, Lemma 2.6.15, Theorem 2.6.7, and the permanence properties of Section 2.10). Covering the fixed-dimensional $\mathcal{N}$ costs $O(\log(L_N/\epsilon))$ with L_N the polynomial Lipschitz modulus in γ , which is absorbed into A_N . $\square$

Lemma C.3 (cap reduction). *For every γ , every $\xi \in \Xi(K_N)$, and every realization, $\hat{\Sigma}_N(\gamma) \leq \mathbb{J}_N(\gamma, P_\xi) \leq \kappa^{-2} \hat{\Sigma}_N(\gamma)$, and the three matrices share their range.*

Proof. For each i exactly one $T_i^t = 1$, so $\sum_t T_i^t b^t b^{t'} = b^{t_i} b^{t'_i}$, and the weight satisfies $T_i^t \leq T_i^t / (P_\xi^t)^2 \leq \kappa^{-2} T_i^t$ pathwise. Summing positive semidefinite rank-one terms gives the two-sided bound; the common range is $\text{span}\{b^{t_i}(\gamma' X_i)\}$. $\square$

Lemma C.4 (population conditioning on the core). *By core stability there are $0 < c_1 \leq C_1$ such that, for all $\gamma \in \mathcal{N}$ and all a supported on $\Omega_{\mathcal{A}}$, $c_1 \|a\|_{\mathcal{A}}^2 \leq a' \Sigma(\gamma) a \leq C_1 \|a\|_{\mathcal{A}}^2$.*

Proof. Since $\kappa \leq P_o^t \leq 1 - \kappa$ and $\sum_t P_o^t = 1$, $\Sigma(\gamma) \asymp \int \sum_t b^t b^{t'} d\mathbb{P}_\gamma$ up to the constants $\kappa, 1$.

Upper bound. With the density ceiling $\bar{p}$ on $\bar{\mathcal{Z}}$ (Assumption 6(a) and the pushforward bound), $a' \Sigma(\gamma) a \leq \bar{p} \|\nabla s_a\|_{L^2(\bar{\mathcal{Z}})}^2$. For a supported on $\Omega_{\mathcal{A}}$, every nonzero B-spline coefficient of the order- $(q-1)$ spline ∇s_a is anchored at a core cell, so cellwise de Boor L^2 -stability gives $\|\nabla s_a\|_{L^2(\bar{\mathcal{Z}})}^2 \leq C(q) \|\nabla s_a\|_{L^2(\Omega_{\mathcal{A}})}^2 = C(q) \|a\|_{\mathcal{A}}^2$.

Lower bound. On each core cell Q , core stability gives the pointwise floor $p_\gamma \geq \underline{p}$, so $\int_Q \|\nabla s_a\|^2 d\mathbb{P}_\gamma \geq \underline{p} \int_Q \|\nabla s_a\|^2 dz$; summing over core cells gives $a' \Sigma(\gamma) a \geq \kappa \underline{p} \|a\|_{\mathcal{A}}^2$. $\square$

Lemma C.5 (boundary layer remainder, contracted form). *Let $m(\cdot)$ be any integrand with $\sup \|m\| \leq \bar{m}$, and form $\xi := \mathbb{P}_N[w b m']$ and $J := \mathbb{P}_N[w b b']$ with weights $w_i \in [\underline{w}, \bar{w}] \subset (0, \infty)$, $b = (b^{t_i}(\gamma' X_i))$. Then $\xi' J^+ \xi = \mathbb{P}_N[w(\Pi m)(\Pi m)']$ with Π the*

$L^2(\mathbb{P}_N, w)$ -projection onto $\text{span}(b)$, and the boundary block satisfies, uniformly over $\gamma \in \mathcal{N}$,

$$\xi'_C J_C^+ \xi_C \leq \mathbb{P}_N [w m m' \mathbb{1}\{\gamma' X_i \in \Omega_{\mathcal{W}}\}] \leq \bar{w} \bar{m}^2 \mathbb{P}_N [\mathbb{1}\{\gamma' X_i \in \Omega_{\mathcal{W}}\}] = O_P(h).$$

Proof. The identity is $\mu' \Sigma^+ \mu = \|\Pi g\|^2$ applied componentwise. Ordering the boundary splines first in the Schur decomposition, the boundary component is $\|\Pi_C m\|^2$; every function in $\text{span}(b_C)$ is supported on $\Omega_{\mathcal{W}} \cup \Omega_{\mathcal{V}}$, so $\Pi_C m = \Pi_C(m \mathbb{1}_{\Omega_{\mathcal{W}} \cup \Omega_{\mathcal{V}}})$, and a projection being a contraction, $\|\Pi_C m\|^2 \leq \bar{w} \bar{m}^2 \mathbb{P}_N [\mathbb{1}\{\gamma' X \in \Omega_{\mathcal{W}}\}]$ (empty cells carry no observations). The indicator average is a triangular array: the sets $\{\gamma' X \in \Omega_{\mathcal{W}, N}\}$, $\gamma \in \mathcal{N}$, are preimages of one region under a fixed-dimensional family of linear maps, of bounded VC dimension, with means $\mathbb{P}_\gamma(\Omega_{\mathcal{W}, N}) \leq \bar{p} |\Omega_{\mathcal{W}, N}| = O(h)$; a uniform Bernstein bound gives $\sup_{\mathcal{N}} \mathbb{P}_N [\mathbb{1}\{\gamma' X \in \Omega_{\mathcal{W}, N}\}] = O(h) + O_P(\sqrt{h/N}) = O_P(h)$. $\square$

Lemma C.6 (leverage and slow variation). *There are constants depending only on q and the mesh ratio such that (i) $\sup_{z \in \Omega_{\mathcal{A}}} b^t(z)' L_{\mathcal{A}}^{-1} b^t(z) \leq C_2 K_N^2$ for $t \in \{c, n\}$, with $L_{\mathcal{A}}^{-1}$ the pseudo-inverse on the interior subspace; (ii) for every unit direction v in γ -space and $\gamma \in \mathcal{N}$, $\|\Sigma(\gamma)^{-1/2} \partial_v \Sigma(\gamma) \Sigma(\gamma)^{-1/2}\|_{\text{op}} \leq c_4 K_N$ on the core subspace.*

Proof. (i) The quantity equals $\sup_a (\partial_{z^t} s_a(z))^2 / \|a\|_{\mathcal{A}}^2$. By de Boor L^∞ - and L^2 -stability in dimension two, $|\partial_{z^t} s_a(z)| \leq C(q) h^{-1} \|\partial_{z^t} s_a\|_{L^2(\Omega_{\mathcal{A}})} \leq C(q) h^{-1} \|a\|_{\mathcal{A}}$. (ii) With $\dot{b}^t := \partial_v [b^t(\gamma' X)]$, Cauchy–Schwarz in the inner product $\langle f, g \rangle = \mathbb{E} \sum_t P_o^t f_t g_t$ gives $|a' \partial_v \Sigma a| \leq 2(a' \Sigma a)^{1/2} \langle \dot{b}^t a, \dot{b}^t a \rangle^{1/2}$, and $|\dot{b}^t a| \leq R_X \|D^2 s_a(\gamma' X)\|$, so by the ceiling, the spline Bernstein inequality $\|D^2 s_a\|_{L^2} \leq C(q) h^{-1} \|\nabla s_a\|_{L^2}$, and Lemma C.4, $\langle \dot{b}^t a, \dot{b}^t a \rangle \lesssim K_N^2 a' \Sigma a$. $\square$

Lemma C.7 (relative convergence of the design Gram). *Under core stability and $K_N^2 \log N = o(N)$, $\sup_{\gamma \in \mathcal{N}} \|\Sigma(\gamma)^{-1/2} (\hat{\Sigma}_N(\gamma) - \Sigma(\gamma)) \Sigma(\gamma)^{-1/2}\|_{\text{op}} = O_P(r_N)$ on the core subspace.*

Proof. Precondition by the fixed $W := L_{\mathcal{A}}^{-1/2}$ and set $\zeta_i(\gamma) := W b^{t_i}(\gamma' X_i)$, $B_N(\gamma) := \mathbb{P}_N \zeta_i \zeta_i'$, $\Gamma(\gamma) := \mathbb{E} B_N(\gamma)$. By Lemma C.4, $c_1 I \leq \Gamma(\gamma) \leq C_1 I$, and by Lemma C.6(i), $\|\zeta_i\|^2 \leq C_2 K_N^2 =: M_N$. The matrix Bernstein inequality (Tropp, 2015, Theorem 6.1.1) with variance proxy $C_1 M_N / N$ gives, for fixed γ and $t \in (0, 1)$, $\mathbb{P}(\|B_N - \Gamma\|_{\text{op}} \geq t) \leq 2|\mathcal{A}| \exp(-c_3 N t^2 / K_N^2)$. The map $\gamma \mapsto B_N - \Gamma$ is Lipschitz with polynomial constant $L_N \lesssim K_N^3$ (differentiating ζ_i costs h^{-2} by de Boor stability and Markov, and $\|\partial_v \Gamma\|_{\text{op}} \lesssim K_N$ by Lemma C.6(ii)). A net of $\mathcal{N}$ at resolution r_N / L_N has log-cardinality

$O(\log N)$; the union bound with $t = Ar_N$ and $Nt^2/K_N^2 = A^2 \log N$ vanishes for A large, and the off-net oscillation adds r_N . Conjugating back by $\|\Sigma^{-1/2}L_{\mathcal{A}}^{1/2}\|_{\text{op}}^2 \leq c_1^{-1}$ completes the proof. $\square$

Lemma C.8 (uniform restricted eigenvalue). *With probability tending to one, for all $\gamma \in \mathcal{N}$, $\xi \in \Xi(K_N)$, and a supported on $\Omega_{\mathcal{A}}$, $c\|a\|_{\mathcal{A}}^2 \leq a'\mathbb{J}_N(\gamma, P_{\xi})a \leq C\|a\|_{\mathcal{A}}^2$ with $0 < c \leq C < \infty$ not depending on (γ, ξ) . Consequently $\|L_{\mathcal{A}}^{1/2}\mathbb{J}_N(\gamma, P_{\xi})^+L_{\mathcal{A}}^{1/2}\|_{\text{op}} = O_{\text{P}}(1)$ uniformly, on the core subspace.*

Proof. Lemmas C.7 and C.4 give $a'\hat{\Sigma}_N(\gamma)a = (1 + o_{\text{P}}(1))a'\Sigma(\gamma)a \asymp \|a\|_{\mathcal{A}}^2$ uniformly; Lemma C.3 transfers the band to $\mathbb{J}_N$ uniformly over ξ . $\square$

Lemma C.9 (relative convergence and continuity of the gradient information). *Under Assumption 6(c):*

- (i) $\sup_{(\gamma, P) \in \mathcal{U}_N} \|\mathbb{J}(\gamma, P)^{-1/2}(\mathbb{J}_N(\gamma, P) - \mathbb{J}(\gamma, P))\mathbb{J}(\gamma, P)^{-1/2}\|_{\text{op}} = O_{\text{P}}(\tilde{r}_N) = o_{\text{P}}(1)$ on the core subspace;
- (ii) for every $(\gamma, P) \in \mathcal{U}_N$, $\|P - P_o\|_{L^\infty(\Omega_{\mathcal{A}})} \lesssim K_N \rho_N$;
- (iii) $\sup_{(\gamma, P) \in \mathcal{U}_N} \|\mathbb{J}_o^{-1/2}(\mathbb{J}(\gamma, P) - \mathbb{J}_o)\mathbb{J}_o^{-1/2}\|_{\text{op}} \lesssim K_N \rho_N = o(1)$, and hence

$$\sup_{\mathcal{U}_N} \|\mathbb{J}_o^{-1/2}\mathbb{J}_N(\gamma, P)\mathbb{J}_o^{-1/2} - \mathbf{I}\|_{\text{op}} = o_{\text{P}}(1)$$

on the core subspace.

Proof. (i) As in Lemma C.7 with the bounded weights $T^t/(P^t)^2 \in [\underline{w}, \bar{w}]$ folded in; the one new feature is that the net must also cover P over the d_{K_N} -dimensional $\Xi(K_N) \cap \mathcal{U}_N$, costing $\log M \lesssim K_N^2 \log N$ and hence the rate $\tilde{r}_N$; the weight derivative $\partial_{\xi}(P^t)^{-2} = -2(P^t)^{-3}b^t$ has polynomial modulus by the cap. (ii) Inside $\mathcal{U}_N$, $\|P - P_o\|_{L^2(\Omega_{\mathcal{A}}, dz)} \lesssim \rho_N$ by the definition of $\mathbf{L}$, the index-shift control $2MR_X\|\gamma - \gamma_o\|$, and the floor from core stability. Interposing the de Boor–Fix approximant P_{ξ° and applying the inverse inequality $\|s\|_{L^\infty} \leq DK_N\|s\|_{L^2}$ to the spline $P - P_{\xi^\circ}$ on core-anchored cells gives the claim. (iii) Split $\mathbb{J}(\gamma, P) - \mathbb{J}_o$ into the weight change, bounded through the cap by $C\|P - P_o\|_{L^\infty(\Omega_{\mathcal{A}})} \lesssim K_N \rho_N$ using (ii), and the index shift, bounded by $cK_N\|\gamma - \gamma_o\| \lesssim K_N \rho_N$ by integrating Lemma C.6(ii) along the segment; combine with (i). $\square$

Write $\hat{\xi}_N(\gamma) := \arg \max_{\xi \in \Xi(K_N)} \mathcal{L}_N(\gamma, P_\xi)$ for the inner sieve MLE and $\mathcal{L}_N^p(\gamma) := \mathcal{L}_N(\gamma, P_{\hat{\xi}_N(\gamma)})$ for the profile criterion. The inner criterion is concave in ξ on the polytope $\Xi(K_N)$, so $P_{\hat{\xi}_N(\gamma)}(\gamma' X_i)$ is unique at every observation and $\mathcal{L}_N^p$ is well defined; a measurable selection of $\hat{\xi}_N(\gamma)$ is fixed throughout.

Lemma C.10 (uniform inner rate). *For every $C' > 0$ there is C such that, with probability tending to one, $\sup\{\mathbb{L}((\gamma, P_{\hat{\xi}_N(\gamma)}), \theta_o) : \|\gamma - \gamma_o\| \leq C' \rho_N, \gamma \in \mathcal{N}\} \leq C \rho_N$; in particular $(\gamma, P_{\hat{\xi}_N(\gamma)}) \in \mathcal{U}_N$ for every such γ , and $\mathcal{U}_N$ contains $(\tilde{\gamma}, P_{\hat{\xi}_N(\tilde{\gamma})})$ for every point $\tilde{\gamma}$ on the segment joining γ_o and $\hat{\gamma}$.*

Proof. Apply the rate argument of Proposition B.1 to the whole family $\theta(\gamma) := (\gamma, P_{\hat{\xi}_N(\gamma)})$ simultaneously. Three inputs are uniform in γ .

Curvature to the truth: by correct specification, $-\mathbb{E}[l_i(\theta) - l_i(\theta_o)] = \text{KL}(\theta_o \parallel \theta) \geq \text{H}^2(\theta, \theta_o)$ for every $\theta \in \Theta_N$, so the criterion is curved in the distance to θ_o itself.

Shared modulus: the centered empirical process is controlled over the entire $\Gamma \times \Xi(K_N)$ by the modulus ϕ_N of Proposition B.1, which therefore bounds the fluctuation at every $\theta(\gamma)$ at once.

Approximate maximization with uniform slack: inner optimality gives $\mathcal{L}_N(\gamma, P_{\hat{\xi}_N(\gamma)}) \geq \mathcal{L}_N(\gamma, P_{\xi^\circ})$, and the anchor satisfies, uniformly over $\|\gamma - \gamma_o\| \leq C' \rho_N$,

$$\text{KL}(\theta_o \parallel (\gamma, P_{\xi^\circ})) \lesssim (\|P_o - P_{\xi^\circ}\|_\infty + \|P_o(\gamma' \cdot) - P_o(\gamma_o' \cdot)\|_{L^2})^2 \lesssim (K_N^{-k_o} + \rho_N)^2 \lesssim \rho_N^2,$$

the index shift bounded by $\|DP_o\|_\infty R_X \|\gamma - \gamma_o\|$. Theorem 3.4.1 of Van Der Vaart and Wellner (1996), applied on the shells of $\text{H}(\cdot, \theta_o)$ with these three inputs, yields $\sup_\gamma \text{H}(\theta(\gamma), \theta_o) = O_P(\rho_N)$, and $\mathbb{L} \leq 2\text{H}$. The segment statement follows since $\|\tilde{\gamma} - \gamma_o\| \leq \|\hat{\gamma} - \gamma_o\| = O_P(\rho_N)$ by Theorem 3. $\square$

Lemma C.11 (interiority and the inner first-order condition). *With probability tending to one, for every γ as in Lemma C.10: $P_{\hat{\xi}_N(\gamma)}^t \in (\kappa, 1 - \kappa)$ on Ω_A , the constraints of $\Xi(K_N)$ restricted to interior coordinates are slack, and the inner first-order condition $\mathbb{P}_N s_\xi(\omega_i, \gamma, P_{\hat{\xi}_N(\gamma)}) = 0$ holds on the interior subspace, together with*

$$\frac{\partial \hat{\xi}_N(\gamma)}{\partial \gamma} = -(\mathbb{P}_N s_{\xi\xi})^+ \mathbb{P}_N s_{\xi\gamma}, \quad \nabla \mathcal{L}_N^p(\gamma) = \mathbb{P}_N s_\gamma(\omega_i, \gamma, P_{\hat{\xi}_N(\gamma)}),$$

the derivative and envelope formulas on that subspace.

Proof. By Assumption 5(b) and compactness, $P_o^t \in [\kappa + \eta_0, 1 - \kappa - \eta_0]$ on $\bar{\mathcal{Z}}$ for some $\eta_0 > 0$, and similarly the Jacobian constraints hold with fixed margins by Lemma G.3 and the strict substitutability margin. Lemma C.9(ii) gives $\sup_{\boldsymbol{\gamma}} \|P_{\hat{\xi}_N(\boldsymbol{\gamma})} - P_o\|_{L^\infty(\Omega_{\mathcal{A}})} = O_P(K_N \rho_N) = o_P(1)$, which is eventually below every margin, so no constraint involving core values binds. The inner FOC and the implicit-function formula then follow from concavity and Lemma C.8 (invertibility on the interior subspace); the envelope formula is the envelope theorem for the concave inner problem. $\square$

Lemma C.12 (sphere-constrained expansion). *Let $\Pi_o := \mathbf{I}_L - \text{vec}(\boldsymbol{\gamma}_o)\text{vec}(\boldsymbol{\gamma}_o)'$ be the projection onto $\mathcal{T}_{\boldsymbol{\gamma}_o}$, and let $\bar{H}_N := \int_0^1 \mathbf{I}_L \nabla^2 \mathcal{L}_N^p(\boldsymbol{\gamma}_\vartheta) \mathbf{I}_L d\vartheta$ with $\boldsymbol{\gamma}_\vartheta := \boldsymbol{\gamma}_o + \vartheta(\hat{\boldsymbol{\gamma}} - \boldsymbol{\gamma}_o)$. Then*

$$\Pi_o \bar{H}_N \Pi_o \sqrt{N} (\text{vec}(\hat{\boldsymbol{\gamma}}) - \text{vec}(\boldsymbol{\gamma}_o)) = -\sqrt{N} \Pi_o \mathbf{I}_L \nabla \mathcal{L}_N^p(\boldsymbol{\gamma}_o) + o_P(1),$$

provided $\sqrt{N} \rho_N^2 \rightarrow 0$, $\|\bar{H}_N\|_{\text{op}} = O_P(1)$, and $\mathbb{P}_N s_{\boldsymbol{\gamma}}(\omega_i, \boldsymbol{\gamma}_o, P_{\hat{\xi}_N(\boldsymbol{\gamma}_o)}) = O_P(N^{-1/2} + \rho_N)$.

Proof. Every $\boldsymbol{\gamma}_\vartheta$ lies in $\mathcal{N}$ (indices remain in the convex $\bar{\mathcal{Z}}$), so $\mathcal{L}_N^p$ is twice differentiable along the segment by Lemma C.11, and the fundamental theorem of calculus gives, coordinate-free, $\mathbf{I}_L \nabla \mathcal{L}_N^p(\hat{\boldsymbol{\gamma}}) = \mathbf{I}_L \nabla \mathcal{L}_N^p(\boldsymbol{\gamma}_o) + \bar{H}_N (\text{vec}(\hat{\boldsymbol{\gamma}}) - \text{vec}(\boldsymbol{\gamma}_o))$. The constrained first-order condition at $\hat{\boldsymbol{\gamma}}$ reads $\mathbf{I}_L \nabla \mathcal{L}_N^p(\hat{\boldsymbol{\gamma}}) = \hat{\lambda} \text{vec}(\hat{\boldsymbol{\gamma}})$ with $\hat{\lambda} = \text{vec}(\hat{\boldsymbol{\gamma}})' \mathbf{I}_L \nabla \mathcal{L}_N^p(\hat{\boldsymbol{\gamma}})$. Here $\hat{\lambda}$ is the multiplier for the constraint $\|\boldsymbol{\gamma}\|_F = 1$, unrelated to the judge shares λ_j of the main text. Multiply by Π_o . On the right, $\Pi_o \text{vec}(\hat{\boldsymbol{\gamma}}) = \Pi_o (\text{vec}(\hat{\boldsymbol{\gamma}}) - \text{vec}(\boldsymbol{\gamma}_o))$, so the Lagrange term is bounded by $|\hat{\lambda}| \|\hat{\boldsymbol{\gamma}} - \boldsymbol{\gamma}_o\|$; since $\hat{\lambda} = O_P(N^{-1/2} + \rho_N)$ by the envelope formula and the score bound, this term is $O_P((N^{-1/2} + \rho_N) \rho_N) = o_P(N^{-1/2})$. On the left, replace $\text{vec}(\hat{\boldsymbol{\gamma}}) - \text{vec}(\boldsymbol{\gamma}_o)$ by $\Pi_o (\text{vec}(\hat{\boldsymbol{\gamma}}) - \text{vec}(\boldsymbol{\gamma}_o))$ at the cost of the radial component, which equals $-\frac{1}{2} \|\hat{\boldsymbol{\gamma}} - \boldsymbol{\gamma}_o\|^2 \text{vec}(\boldsymbol{\gamma}_o)$ plus higher order by the normalization $\|\hat{\boldsymbol{\gamma}}\|_F = 1$; with $\|\bar{H}_N\|_{\text{op}} = O_P(1)$ this costs $O_P(\rho_N^2) = o_P(N^{-1/2})$. Multiplying by $\sqrt{N}$ collects the display. $\square$

By Lemma C.11 and the chain rule, on the relevant event,

$$\nabla^2 \mathcal{L}_N^p(\boldsymbol{\gamma}) = \mathbb{P}_N s_{\boldsymbol{\gamma}\boldsymbol{\gamma}} - \mathbb{P}_N s_{\boldsymbol{\gamma}\xi} (\mathbb{P}_N s_{\xi\xi})^+ \mathbb{P}_N s_{\xi\boldsymbol{\gamma}}, \quad \text{all blocks at } (\boldsymbol{\gamma}, P_{\hat{\xi}_N(\boldsymbol{\gamma})}). \quad (\text{C.1})$$

The lemmas of this part bound each block uniformly over $\mathcal{U}_N$, so that (C.1) evaluated along the segment converges to $-\tilde{I}_{K_N}$.

Lemma C.13 (the score is uniformly small). $\sup_{(\boldsymbol{\gamma}, P) \in \mathcal{U}_N} \|\mathbb{P}_N s_{\boldsymbol{\gamma}}(\omega_i, \boldsymbol{\gamma}, P)\| = O_P(r_N + \rho_N) = o_P(1)$.

Proof. The class $\{s_{\gamma}(\cdot, \gamma, P)\}$ has the bounded envelope $2MR_X\kappa^{-1}$ and the structure of Lemma C.2, so the centered part is $O_P(r_N)$. For the mean, $\sum_t DP^t \equiv 0$ gives the exact identity $\mathbb{E}s_{\gamma}(\omega, \gamma, P) = -\mathbb{E}\sum_t \frac{\delta_{\gamma, P}^t}{P^t} DP^t \otimes X$ with $\delta_{\gamma, P}^t(x) := P^t(\gamma'x) - P_o^t(\gamma'_ox)$, whence $\|\mathbb{E}s_{\gamma}\| \lesssim \mathbb{L}((\gamma, P), \theta_o) \leq C\rho_N$ on $\mathcal{U}_N$. $\square$

Lemma C.14 (second-derivative block).

$$\sup_{(\gamma, P) \in \mathcal{U}_N} \left\| \mathbb{P}_N \sum_t \frac{T_i^t}{P^t(\gamma'X_i)} D^2 P^t(\gamma'X_i) \otimes (X_i X'_i) \right\| = o_P(1).$$

Proof. Center at $\mathbb{E}[T^t | X] = P_o^t(Z)$ and split. The centered part is an empirical process with envelope $\lesssim K_N$ (Lemma B.3(ii) gives $\|D^2 P^t\|_{\infty} \lesssim MK_N$) and the structure of Lemma C.2, hence $O_P(K_N \cdot K_N \sqrt{\log N/N}) = O_P(K_N^2 \sqrt{\log N/N}) = o_P(1)$. The mean part is bounded by $K_N \mathbb{E}\sum_t |P_o^t(\gamma'_o X) - P^t(\gamma'X)| \lesssim K_N \rho_N = o(1)$ on $\mathcal{U}_N$. $\square$

Lemma C.15 (outer-product block).

$$\sup_{(\gamma, P) \in \mathcal{U}_N} \left\| \mathbb{P}_N \sum_t \frac{T_i^t}{(P^t)^2} DP^t DP^{t'} \otimes (X_i X'_i) - I_{\gamma\gamma} \right\| = o_P(1).$$

Combined with Lemma C.14, $\sup_{\mathcal{U}_N} \|\mathbb{P}_N s_{\gamma\gamma} + I_{\gamma\gamma}\| = o_P(1)$.

Proof. The centered part is $O_P(r_N)$ by Lemma C.2 with bounded envelope (Lemma B.3(i)). For the population drift, write the mean at (γ, P) minus the mean at (γ_o, P_o) as a weight change plus a derivative change; the cap bounds the weight change by $C \mathbb{E}|P^t(\gamma'X) - P_o^t(\gamma'_o X)| \lesssim \rho_N$, and the derivative change by $C \mathbb{E}\|DP^t(\gamma'X) - DP_o^t(\gamma'_o X)\|$. Split the latter as $D(P^t - P_o^t)(\gamma'X)$ plus $DP_o^t(\gamma'X) - DP_o^t(\gamma'_o X)$; the second piece is $\lesssim \|D^2 P_o\|_{\infty} R\rho_N$. For the first piece split the integral over the core and the boundary layer: on Ω_A , interpose P_{ξ_o} and apply the Markov–Bernstein inverse inequality on full cells, giving $\lesssim K_N \rho_N + K_N^{1-k_o}$; on Ω_W , the Jacobian bound of $\Xi(K_N)$ bounds the integrand and $\mathbb{P}_{\gamma}(\Omega_W) = O(h)$, giving $O(\sqrt{h})$ after Cauchy–Schwarz. All pieces are $o(1)$. Finally, at the truth, mutual exclusivity of the treatment indicators ($T^t T^{t'} = T^t \delta_{tt'}$) gives exactly $\mathbb{E}\sum_t \frac{1}{P_o^t} DP_o^t DP_o^{t'} \otimes X X' = \mathbb{E}[s_{\gamma}^{\otimes 2}] = I_{\gamma\gamma}$. $\square$

Lemma C.16 (prefactored cross block).

$$\sup_{(\gamma, P) \in \mathcal{U}_N} \left\| \mathbb{J}_o^{+/2} (\mathbb{P}_N s_{\xi\gamma}(\omega_i, \gamma, P) - \mathbb{P}_{s_{\xi\gamma}}(\omega_i, \gamma_o, P_o)) \right\| = O_P \left(K_N^2 \sqrt{\log N/N} + K_N \rho_N + \sqrt{h} \right) = o_P(1)$$

on the interior subspace.

Proof. Decompose $s_{\xi\gamma} = \text{I} + \text{II}$ with $\text{I} := \sum_t (T^t/P^t) \nabla b^t \otimes X$ and $\text{II} := -\sum_t (T^t/(P^t)^2) b^t \otimes DP^{t'} \otimes X$, and bound each part after prefactoring.

Envelope in the prefactored metric. For a on the interior subspace, $\nabla b^t(z)'a = \nabla \partial_{z^t} s_a(z)$ is a second derivative of the spline s_a , so de Boor stability followed by Markov–Bernstein gives $|D^2 s_a(z)| \leq Ch^{-2} \|\nabla s_a\|_{L^2(\Omega_{\mathcal{A}})}$, i.e. $\sup_{z \in \Omega_{\mathcal{A}}} \|\mathbb{J}_o^{+/2} \nabla b^t(z)\| \lesssim K_N^2$, and by Lemma C.6(i), $\sup_{z \in \Omega_{\mathcal{A}}} \|\mathbb{J}_o^{+/2} b^t(z)\| \lesssim K_N$.

Stochastic parts. The centered processes of I and II, prefactored by $\mathbb{J}_o^{+/2}$, have envelopes $\lesssim K_N^2$ and $\lesssim K_N$ and the structure of Lemma C.2, hence are $O_P(K_N^2 \cdot K_N \sqrt{\log N/N}) = O_P(K_N^3 \sqrt{\log N/N}) = o_P(1)$.

Bias parts. Centering T^t leaves multipliers $P_o^t(\gamma'_o X) - P^t(\gamma' X)$. For I, the second-derivative Bernstein inequality gives, on the interior subspace, $\mathbb{P} \sum_t \nabla b^t \nabla b^{t'} \leq CK_N^2 \mathbb{J}(\gamma, P)$, so Cauchy–Schwarz in the data measure bounds the prefactored mean by $\kappa^{-1} \mathbb{L}((\gamma, P), \theta_o) \cdot (CK_N^2)^{1/2} \cdot \|\mathbb{J}_o^{+/2} \mathbb{J}(\gamma, P)^{1/2}\|_{\text{op}} \lesssim K_N \rho_N$, using Lemma C.9(iii) for the last factor. For II the same argument with the envelope K_N gives $\lesssim K_N \rho_N$. The drift from (γ, P) to (γ_o, P_o) in the remaining population terms is bounded exactly as in Lemma C.15, contributing $K_N \rho_N + \sqrt{h}$.

Collar. □

Lemma C.17 (profile Hessian limit).

$$\sup_{(\gamma, P) \in \mathcal{U}_N} \left\| \mathbb{P}_N s_{\gamma\gamma} - \mathbb{P}_N s_{\gamma\xi} (\mathbb{P}_N s_{\xi\xi})^+ \mathbb{P}_N s_{\xi\gamma} + \tilde{I}_{K_N} \right\| = o_P(1),$$

and consequently $\bar{H}_N = -\mathbf{I}_L \tilde{I}_{K_N} \mathbf{I}_L + o_P(1)$ and $\|\bar{H}_N\|_{\text{op}} = O_P(1)$.

Proof. By Lemma C.15, $\mathbb{P}_N s_{\gamma\gamma} \rightarrow -I_{\gamma\gamma}$ uniformly. For the projection term write $\Pi_N := \xi_N' \mathbb{J}_N^+ \xi_N$ with $\xi_N := \mathbb{P}_N s_{\xi\gamma}$, so that $-\mathbb{P}_N s_{\gamma\xi} (\mathbb{P}_N s_{\xi\xi})^+ \mathbb{P}_N s_{\xi\gamma} = \Pi_N$. Split by the boundary–first Schur decomposition: the boundary layer block is $O_P(h)$ by Lemma C.5. On the interior block, conjugate: $\Pi_N^{\text{core}} = (\mathbb{J}_o^{+/2} \xi_N)' \left(\mathbb{J}_o^{1/2} \mathbb{J}_N^+ \mathbb{J}_o^{1/2} \right) (\mathbb{J}_o^{+/2} \xi_N)$. Lemma C.9(iii) makes the middle factor converge in operator norm to the orthogonal projection onto the interior range, and Lemma C.16 gives $\mathbb{J}_o^{+/2} \xi_N \rightarrow \mathbb{J}_o^{+/2} I_{\xi\gamma}$. Hence $\Pi_N \rightarrow I_{\gamma\xi} \mathbb{J}_o^+ I_{\xi\gamma}$ uniformly, and the display follows from the definition of $\tilde{I}_{K_N}$. Along the segment, Lemma C.10 places $(\gamma_\vartheta, P_{\hat{\xi}_N(\gamma_\vartheta)})$ in $\mathcal{U}_N$, so (C.1) integrates to $\bar{H}_N = -\mathbf{I}_L \tilde{I}_{K_N} \mathbf{I}_L + o_P(1)$; boundedness holds since $\tilde{I}_{K_N} \leq I_{\gamma\gamma}$. □

Lemma C.18 (pseudo-true nuisance). *Let $\xi_N^* := \arg\max_{\xi \in \Xi(K_N)} \mathbb{E} l_i(\gamma_o, P_\xi)$ and write $P^* := P_{\xi_N^*}$, $\delta := P^* - P_o$. Then, for N large: (i) ξ_N^* satisfies the first-order condition $\mathbb{E} \sum_t (\delta^t / P^{*t}) a' b^t(Z) = 0$ for every a on the interior subspace; (ii) $\|\delta\|_{L^2} \lesssim K_N^{-k_o}$ and $\|D\delta\|_{L^2} \lesssim K_N^{1-k_o}$; (iii) $\|P^* - P_o\|_{L^\infty(\Omega_{\mathcal{A}})} \lesssim K_N^{1-k_o}$ and $(\gamma_o, P^*) \in \mathcal{U}_N$.*

Proof. (i) Interiority at the pseudo-true value follows as in Lemma C.11 once (iii) is established, and the FOC is the stationarity of the concave population inner problem; using $\sum_t b^t \equiv 0$ and $\mathbb{E}[T^t | X] = P_o^t(Z)$ it takes the displayed exact form. (ii) Write $P_o = P_{\xi^o} + R$ with the de Boor–Fix approximant, $\|R\|_\infty \lesssim K_N^{-k_o}$, $\|DR\|_\infty \lesssim K_N^{1-k_o}$. Subtracting the analogous exact identity at P_o from (i) yields the normal equation $G(\xi_N^* - \xi^o) = \eta$ with $\mathbb{Q} := \mathbb{E} \sum_t b^t b^{t'}/P^{*t}$ and $\eta := \mathbb{E} \sum_t b^t R^t/P^{*t}$. By the cap and Lemma C.4, $\mathbb{Q} \asymp L_{\mathcal{A}}$ on the interior subspace, and for any a , Cauchy–Schwarz gives $|a'\eta| \leq \kappa^{-1/2} (a'Ga)^{1/2} (\mathbb{E} \sum_t (R^t)^2)^{1/2}$, so $\|\xi_N^* - \xi^o\|_G = \|\mathbb{Q}^{+1/2}\eta\| \lesssim K_N^{-k_o}$. The gradient seminorm $\|\cdot\|_{\mathcal{A}}$ controls the L^2 propensity gap, $\|P^* - P_{\xi^o}\|_{L^2(\Omega_{\mathcal{A}})} \asymp \|\xi_N^* - \xi^o\|_{\mathcal{A}} \lesssim K_N^{-k_o}$, and the triangle inequality with R gives (ii); the derivative bound uses the inverse inequality on the spline part and $\|DR\|_\infty$. (iii) The inverse inequality costs one factor K_N on the spline part, giving the sup-norm rate; $\mathbf{L}((\gamma_o, P^*), \theta_o) \lesssim K_N^{-k_o} \leq \rho_N$ places (γ_o, P^*) in $\mathcal{U}_N$. $\square$

Lemma C.19 (inner estimation error at the truth). $\|\hat{\xi}_N(\gamma_o) - \xi_N^*\|_{\mathcal{A}} = O_P(K_N/\sqrt{N})$, and

$$\hat{\xi}_N(\gamma_o) - \xi_N^* = \mathbb{J}_N(\gamma_o, P_{\hat{\xi}_N})^+ g_N, \quad g_N := (\mathbb{P}_N - \mathbb{P}) s_\xi(\omega_i, \gamma_o, P^*),$$

for a mean-value point $\tilde{\xi}_N$ on the segment, which lies in $\Xi(K_N) \cap \mathcal{U}_N$.

Proof. Comparing the empirical FOC at $\hat{\xi}_N(\gamma_o)$ with the centered score at ξ_N^* and expanding coordinatewise gives the displayed identity; the segment point lies in $\Xi(K_N)$ by convexity and in $\mathcal{U}_N$ by convexity of $\mathbf{L}(\cdot, \theta_o)$ along it. The variance of the score is the information: by mutual exclusivity of the treatment indicators, $\text{Var}(s_\xi(\omega, \gamma_o, P^*)) = \mathbb{J}(\gamma_o, P^*)$, so $\mathbb{E} \|\mathbb{J}(\gamma_o, P^*)^{+1/2} g_N\|^2 = \text{rank}(\mathbb{J}(\gamma_o, P^*))/N \lesssim K_N^2/N$, the rank counting core and boundary layer coordinates alike. Lemma C.8 bounds $\|L_{\mathcal{A}}^{1/2} \mathbb{J}_N(\gamma_o, P_{\hat{\xi}_N})^+ L_{\mathcal{A}}^{1/2}\|_{\text{op}}$ on the interior subspace, and $L_{\mathcal{A}} \asymp \mathbb{J}(\gamma_o, P^*)$ there, so $\|\hat{\xi}_N(\gamma_o) - \xi_N^*\|_{\mathcal{A}} \lesssim \|\mathbb{J}(\gamma_o, P^*)^{+1/2} g_N\| = O_P(K_N/\sqrt{N})$. $\square$

Lemma C.20 (reduction to the pseudo-true score).

$$\begin{aligned} & \mathbf{I}_L \nabla \mathcal{L}_N^p(\boldsymbol{\gamma}_o) \\ &= \mathbf{I}_L \left(\mathbb{P}_N s_{\boldsymbol{\gamma}}(\omega_i, \boldsymbol{\gamma}_o, P^*) - \mathbb{P} s_{\boldsymbol{\gamma}\xi}(\boldsymbol{\gamma}_o, \xi_N^*) \mathbb{P} s_{\xi\xi}(\boldsymbol{\gamma}_o, \xi_N^*)^+ \mathbb{P}_N s_{\xi}(\omega_i, \boldsymbol{\gamma}_o, P^*) \right) + o_P(N^{-1/2}). \end{aligned}$$

Proof. By the envelope formula and a mean-value expansion of $\mathbb{P}_N s_{\boldsymbol{\gamma}}(\omega_i, \boldsymbol{\gamma}_o, \cdot)$ around ξ_N^* , $\nabla \mathcal{L}_N^p(\boldsymbol{\gamma}_o) = \mathbb{P}_N s_{\boldsymbol{\gamma}}(\omega_i, \boldsymbol{\gamma}_o, P^*) + \mathbb{P}_N s_{\boldsymbol{\gamma}\xi}(\omega_i, \boldsymbol{\gamma}_o, \tilde{\xi}_N') [\hat{\xi}_N(\boldsymbol{\gamma}_o) - \xi_N^*]$ for a segment point $\tilde{\xi}_N'$, and Lemma C.19 substitutes the error. Two swaps remain: the empirical blocks at segment points against the population blocks at ξ_N^* . For the cross block, Lemma C.16 gives $\|\mathbb{J}_o^{+/2}(\mathbb{P}_N s_{\xi\boldsymbol{\gamma}}(\boldsymbol{\gamma}_o, \tilde{\xi}_N') - \mathbb{P} s_{\xi\boldsymbol{\gamma}}(\boldsymbol{\gamma}_o, \xi_N^*))\| = O_P(K_N^3 \sqrt{\log N/N} + K_N \rho_N)$, and paired with the factor $\|\mathbb{J}_o^{+/2} g_N\| = O_P(K_N/\sqrt{N})$ from Lemma C.19, the swap cost is $O_P((K_N^3 \sqrt{\log N/N} + K_N \rho_N) \cdot K_N/\sqrt{N}) = o_P(N^{-1/2})$ under $K_N^6 \log N = o(N)$ and $K_N^2 \rho_N \rightarrow 0$. For the inverse block, Lemma C.9 gives the relative error $O_P(\tilde{r}_N + K_N \rho_N)$ of $\mathbb{J}_N(\boldsymbol{\gamma}_o, P_{\xi_N^*})$ against $\mathbb{J}(\boldsymbol{\gamma}_o, P^*)$, and the identity $A^{1/2}(B^+ - A^+)A^{1/2} = -(A^{1/2}B^+A^{1/2})(A^{-1/2}(B - A)A^{-1/2})$ converts it into the same paired form; the boundary layer contributions are $O_P(h \cdot K_N/\sqrt{N}) = o_P(N^{-1/2})$ by Lemma C.5. $\square$

Lemma C.21 (centering of the profiled score). $\|\mathbb{P} s_{\boldsymbol{\gamma}}(\omega_i, \boldsymbol{\gamma}_o, P^*)\| \leq 2MR_X \kappa^{-1} \sqrt{3} \|\delta\|_{L^2} \lesssim K_N^{-k_o}$, and hence $\sqrt{N} \|\mathbb{P} s_{\boldsymbol{\gamma}}(\omega_i, \boldsymbol{\gamma}_o, P^*)\| = O(\sqrt{N} K_N^{-k_o}) = o(1)$ by Assumption 6(c).

Proof. Since $\sum_t DP^{*t} \equiv 0$ and $\mathbb{E}[T^t | X] = P_o^t(Z)$, conditioning on Z gives the exact identity

$$\mathbb{P} s_{\boldsymbol{\gamma}}(\omega_i, \boldsymbol{\gamma}_o, P^*) = -\mathbb{E} \sum_t \frac{\delta^t(Z)}{P^{*t}(Z)} DP^{*t}(Z) \otimes m_X(Z),$$

and the claim follows from the cap, $\|DP^{*t}\| \leq 2M$ (Lemma B.3(i)), $\|m_X\| \leq R_X$, the Cauchy–Schwarz inequality, and $\|\delta\|_{L^2} \lesssim K_N^{-k_o}$ (Lemma C.18(ii)). $\square$

Lemma C.22 (CLT for the profiled score). *Let*

$$\psi_{Ni} := s_{\boldsymbol{\gamma}}(\omega_i, \boldsymbol{\gamma}_o, P^*) - \mathbb{P} s_{\boldsymbol{\gamma}\xi}(\boldsymbol{\gamma}_o, \xi_N^*) \mathbb{P} s_{\xi\xi}(\boldsymbol{\gamma}_o, \xi_N^*)^+ s_{\xi}(\omega_i, \boldsymbol{\gamma}_o, P^*)$$

and $Q_N := N^{-1/2} \sum_i \Pi_o \mathbf{I}_L \psi_{Ni}$. Then $\text{Var}(Q_N) = \Pi_o \mathbf{I}_L \tilde{I}_{K_N} \mathbf{I}_L \Pi_o + o(1)$ and, for every fixed v , $v'Q_N - \mathbb{E}v'Q_N \rightarrow_d N(0, \lim_N v' \Pi_o \mathbf{I}_L \tilde{I}_{K_N} \mathbf{I}_L \Pi_o v)$ along subsequences where the limit exists, with $\mathbb{E}Q_N \rightarrow 0$ by Lemma C.21.

Proof. Variance. The projection summand has mean zero by the pseudo-true FOC. Expanding the square, $\text{Var}(\psi_{N1}) = \mathbb{E}[s_{\boldsymbol{\gamma}}(\cdot, P^*)^{\otimes 2}] + \mathbb{P} s_{\boldsymbol{\gamma}\xi} \mathbb{P} s_{\xi\xi}^+ \mathbb{P} s_{\xi\boldsymbol{\gamma}} + o(1)$, where the

collapse uses the information equalities $\mathbb{E}[s_{\gamma}s'_{\xi}] = -\mathbb{P}s_{\gamma\xi}$ and $\mathbb{E}[s_{\xi}s'_{\xi}] = -\mathbb{P}s_{\xi\xi}$, exact at P_o and holding at ξ_N^* up to $O(K_N^{1-k_o})$ by Lemma C.18(iii) and the cap, and the dropped $(\mathbb{E}\psi_{N1})^{\otimes 2} = O(K_N^{2-4k_o})$ is negligible. Replacing P^* by P_o in each block costs $O(K_N^{1-k_o}) = o(1)$ in operator norm after prefactoring (Lemma C.9(iii)), which identifies the limit as $\tilde{I}_{K_N} + o(1)$.

Lindeberg. It suffices that $\|\psi_{N1}\|^2$ be uniformly integrable. The score part converges in L^2 to $s_{\gamma}(\cdot, \gamma_o, P_o)$ since s_{γ} depends on the nuisance through DP^t/P^t and $\|D(P^* - P_o)\|_{L^2} \rightarrow 0$; the projection part is an $L^2(\mathbb{P})$ -projection of s_{γ} onto $\text{span}(s_{\xi})$, bounded uniformly in N as a projection. Uniform integrability follows, and Lindeberg–Feller applies to the triangular array. $\square$

Lemma C.23 (from sieve to semiparametric efficiency). $\tilde{I}_{K_N} = I_{\gamma\gamma}^{\text{eff}} + o(1)$.

Proof. Let Π_K denote the $L^2(\mathbb{P})$ -projection onto $\text{span}(s_{\xi})$, the sieve score span, and Π_{∞} the projection onto the closure of the full nuisance tangent space, so that $\tilde{I}_K = \mathbb{E}[(s_{\gamma} - \Pi_K s_{\gamma})^{\otimes 2}]$ and $I_{\gamma\gamma}^{\text{eff}} = \mathbb{E}[(s_{\gamma} - \Pi_{\infty} s_{\gamma})^{\otimes 2}]$. Write $s_{\gamma} - \Pi_K s_{\gamma} = (s_{\gamma} - \Pi_{\infty} s_{\gamma}) + (\Pi_{\infty} - \Pi_K)s_{\gamma}$. The first summand is orthogonal to the full tangent space, which contains $(\Pi_{\infty} - \Pi_K)s_{\gamma}$, so $\tilde{I}_K = I_{\gamma\gamma}^{\text{eff}} + \mathbb{E}[(\Pi_{\infty} - \Pi_K)s_{\gamma}]^{\otimes 2}$. Moreover $\Pi_K(s_{\gamma} - \Pi_{\infty} s_{\gamma}) = 0$, since the efficient residual is orthogonal to the sieve span, whence $(\Pi_{\infty} - \Pi_K)s_{\gamma} = (I - \Pi_K)\Pi_{\infty}s_{\gamma}$, the sieve approximation error of the least-favorable score, of L^2 norm at most $C\epsilon_K^{\text{lf}}$. Finally $\epsilon_{K_N}^{\text{lf}} \rightarrow 0$: h^{lf} lies in $\mathcal{T}$; the propensity perturbations $\nabla\delta G$ from smooth surplus δG are dense in $\mathcal{T}$, and the de Boor–Fix approximant of a smooth surplus places its gradient within any ϵ of the sieve for K large. $\square$

Proof of Theorem 4. Combine Lemma C.12 with Lemmas C.17, C.20, C.21, and C.22:

$$\Pi_o \mathbf{I}_L \tilde{I}_{K_N} \mathbf{I}_L \Pi_o \sqrt{N}(\text{vec}(\hat{\gamma}) - \text{vec}(\gamma_o)) = Q_N + o_P(1), \quad Q_N \rightarrow_d N(0, \Pi_o I_{\gamma\gamma}^{\text{eff}} \Pi_o),$$

the limit variance by Lemma C.23. By Lemma C.1 and Lemma C.23, $\Pi_o \mathbf{I}_L \tilde{I}_{K_N} \mathbf{I}_L \Pi_o$ is invertible on $\mathcal{T}_{\gamma_o}$ for N large with uniformly bounded inverse, and $\sqrt{N}(\text{vec}(\hat{\gamma}) - \text{vec}(\gamma_o))$ is supported on $\mathcal{T}_{\gamma_o}$ up to $o_P(1)$ (the intercept-mean components vanish and the radial component is $O_P(\sqrt{N}\rho_N^2) = o_P(1)$). Inverting on $\mathcal{T}_{\gamma_o}$ and applying Slutsky,

$$\sqrt{N}(\text{vec}(\hat{\gamma}) - \text{vec}(\gamma_o)) = (I_{\gamma\gamma}^{\text{eff}})^+ \frac{1}{\sqrt{N}} \sum_{i=1}^N s_{\gamma}^{\text{eff}}(T_i, X_i) + o_P(1) \rightarrow_d N(0, (I_{\gamma\gamma}^{\text{eff}})^+),$$

where $s_{\gamma}^{\text{eff}} = \Pi_o(s_{\gamma} - \Pi_{\infty}s_{\gamma})$ and the substitution of ψ_{Ni} by $s_{\gamma,i}^{\text{eff}}$ costs $o_P(1)$ in L^2 by the convergences established in Lemmas C.22 and C.23. This is the asymptotically linear representation of the theorem, and efficiency in the sense of Definition 1 follows by Proposition 2. $\square$

D Second-Stage Rates with Generated Regressors

This subsection proves Theorem 5. Fix the pair (t, s) and the remaining treatment r ; all objects refer to the t -subsample. Write $S_i = (P_i^t, \tilde{Z}_i^{t,r})$, $\hat{S}_i = (\hat{P}_i^t, \hat{\tilde{Z}}_i^{t,r})$, and $\tilde{d} = (\tilde{K}_N + \tilde{q})^2$; let $\tilde{b}(\cdot) = \tilde{b}_{\tilde{K}_N}(\cdot) \in \mathbb{R}^{\tilde{d}}$ be the order- $\tilde{q}$ tensor B-spline basis on the quasi-uniform mesh of width $\tilde{h} \asymp \tilde{K}_N^{-1}$ covering $\text{supp}(S_i)$. The regression basis is $\mathbf{b}_i := W_i \otimes \tilde{b}(S_i)$ and its generated version $\hat{\mathbf{b}}_i := W_i \otimes \tilde{b}(\hat{S}_i)$; the estimator solves the least squares problem of Section 3.3, with solution $\hat{\delta} = (\hat{\delta}_p)_p$ and fitted coefficients $\hat{\beta}_{ts,p}(\cdot) = \tilde{b}(\cdot)' \hat{\delta}_p$. Throughout, $\mathcal{S}$ is the fixed compact interior region of Assumption 7, $\mathcal{S}^+ \supset \mathcal{S}$ a slightly larger compact set on which the density bounds of Assumption 7(c) still hold, and all uniform statements about fitted functions are on $\mathcal{S}$, controlled by basis functions whose supports meet $\mathcal{S}$ and hence lie in $\mathcal{S}^+$ for $\tilde{K}_N$ large. Only observations whose regressors activate the retained basis functions enter the estimates below, and for N large these have $S_i \in \mathcal{S}^+$, where the interior passthrough of Assumption 7(e) applies. By that condition and boundedness,

$$\Delta_N := \max_{i \leq N: S_i \in \mathcal{S}^+} \|\hat{S}_i - S_i\| = O_P(K_N \rho_N). \quad (\text{D.1})$$

Lemma D.1 (approximation). *Under Assumption 7(b) there are $\delta_p^* \in \mathbb{R}^{\tilde{d}}$ with, for every component p , $\|\beta_{ts,p} - \tilde{b}(\cdot)' \delta_p^*\|_{L^\infty(\mathcal{S}^+)} \lesssim \tilde{K}_N^{-\chi}$ and $\|\partial_1 \beta_{ts,p} - \partial_1 \tilde{b}(\cdot)' \delta_p^*\|_{L^\infty(\mathcal{S}^+)} \lesssim \tilde{K}_N^{-(\chi-1)}$.*

Proof. Apply the de Boor–Fix quasi-interpolant argument of Lemma B.4 to each $\beta_{ts,p} \in C^\chi$ with $\tilde{q} \geq \chi + 1$; the simultaneous bounds give the level rate $\tilde{K}_N^{-\chi}$ and the first-derivative rate $\tilde{K}_N^{-(\chi-1)}$ on the mesh covering $\mathcal{S}^+$. $\square$

Lemma D.2 (design). *Let $\mathbb{Q}_N := \mathbb{P}_N[T_i^t \mathbf{b}_i \mathbf{b}_i']$ and $\mathbb{Q} := \mathbb{E}[T^t \mathbf{b} \mathbf{b}']$, restricted to coefficients of basis functions meeting $\mathcal{S}$. Under Assumption 7(a),(c),(d): (i) $c \tilde{h}^2 \mathbf{I} \leq \mathbb{Q} \leq C \tilde{h}^2 \mathbf{I}$; (ii) $\|\mathbb{Q}^{-1/2}(\mathbb{Q}_N - \mathbb{Q})\mathbb{Q}^{-1/2}\|_{\text{op}} = O_P(\tilde{K}_N \sqrt{\log N/N}) = o_P(1)$; (iii) with generated regressors, $\|\mathbb{Q}^{-1/2}(\hat{\mathbb{Q}}_N - \mathbb{Q}_N)\mathbb{Q}^{-1/2}\|_{\text{op}} = O_P(\tilde{K}_N^3 \Delta_N) = o_P(1)$, where $\hat{\mathbb{Q}}_N :=$*

$$\mathbb{P}_N[T_i^t \hat{\mathbf{b}}_i \hat{\mathbf{b}}_i'].$$

Proof. (i) For a coefficient vector $c = (c_p)_p$,

$$c' \mathbb{Q}_C = \mathbb{E} \left[T^t \left(\sum_p W_p \tilde{b}' c_p \right)^2 \right] = \mathbb{E}[T^t g_c(S)' \mathbb{E}[WW' | T^t = 1, S] g_c(S)]$$

with $g_{c,p} := \tilde{b}' c_p$; Assumption 7(a) floors the middle matrix on $\mathcal{S}$ and boundedness of W caps it, while the two-sided density of S on $\mathcal{S}^+$ and de Boor L^2 -stability give $\|g_{c,p}\|_{L^2(dF_S)}^2 \asymp \tilde{h}^2 \|c_p\|^2$.

(ii) Matrix Bernstein as in Lemma C.7: the normalized summands $\mathbb{Q}^{-1/2} T^t \mathbf{b} \mathbf{b}' \mathbb{Q}^{-1/2} / N$ have operator norm $\lesssim \tilde{d}^{\text{lev}} / N$ with leverage $\tilde{d}^{\text{lev}} \lesssim \tilde{K}_N^2$ (tensor splines in two dimensions with a density floor), giving the tail $\exp(-cNt^2 / \tilde{K}_N^2)$ and the stated rate after the union bound over the $O(\tilde{K}_N^2)$ -dimensional index; $\tilde{K}_N^2 \log N = o(N)$ is Assumption 7(d).

(iii) Each entry of $\tilde{b}(\hat{S}_i) - \tilde{b}(S_i)$ is bounded by $\min(1, C\tilde{K}_N \Delta_N)$ and only $O(1)$ entries are active per observation, so $\|\hat{\mathbf{b}}_i - \mathbf{b}_i\| \lesssim \tilde{K}_N \Delta_N$ and $\|\mathbf{b}_i\| \lesssim 1$. The two factors of $\mathbb{Q}^{-1/2} \asymp \tilde{h}^{-1}$ contribute $\tilde{h}^{-2} = \tilde{K}_N^2$, so the relative perturbation is $O_P(\tilde{K}_N^2 \cdot \tilde{K}_N \Delta_N) = O_P(\tilde{K}_N^3 \Delta_N)$, and $\tilde{K}_N^3 \Delta_N = O_P(\tilde{K}_N^3 K_N \rho_N) = o_P(1)$ by the coupling $\tilde{K}_N^3 K_N \rho_N \rightarrow 0$ in Assumption 7(d). $\square$

Lemma D.3 (oracle rates). *Let $\tilde{\delta}$ solve the least squares problem with the true regressors S_i . Then, for every p ,*

$$\begin{aligned} \|\tilde{b}(\cdot)' \tilde{\delta}_p - \beta_{ts,p}\|_{L^\infty(\mathcal{S})} &= O_P \left(\tilde{K}_N \sqrt{\log N / N} + \tilde{K}_N^{-\chi} \right), \\ \|\partial_1 \tilde{b}(\cdot)' \tilde{\delta}_p - \partial_1 \beta_{ts,p}\|_{L^\infty(\mathcal{S})} &= O_P \left(\tilde{K}_N^2 \sqrt{\log N / N} + \tilde{K}_N^{-(\chi-1)} \right). \end{aligned}$$

Proof. Write $Y_i = \mathbf{b}_i' \delta^* + a_i + \varepsilon_i$ on $T_i^t = 1$, with $a_i := W_i'(\beta_{ts}(S_i) - (\tilde{b}' \delta_p^*)_p(S_i))$ the approximation error, $|a_i| \lesssim \tilde{K}_N^{-\chi}$ on $\mathcal{S}^+$. Then $\tilde{\delta} - \delta^* = \mathbb{Q}_N^+ \mathbb{P}_N[T^t \mathbf{b}(\varepsilon + a)]$. For the variance part, fix $\zeta \in \mathcal{S}$: the fitted value $\tilde{b}(\zeta)'(\tilde{\delta}_p - \delta_p^*)$ is a linear contrast with $\text{Var} \lesssim \sigma^2 \tilde{b}(s)' [\mathbb{Q}^{-1}]_{pp} \tilde{b}(s) / N \lesssim \tilde{K}_N^2 / N$ by Lemma D.2(i) and (ii); a Bernstein tail plus a union bound over an N^{-C} -net of $\mathcal{S}$ (the fitted spline is $\tilde{K}_N$ -Lipschitz up to scale) gives the $\sqrt{\log N}$ factor and the sup-norm variance rate $\tilde{K}_N \sqrt{\log N / N}$. The bias part contributes $\|\tilde{b}' \mathbb{Q}_N^+ \mathbb{P}_N[T^t \mathbf{b} a]\|_\infty \lesssim \tilde{K}_N^{-\chi}$ by the same contrast bound and $|a_i| \lesssim \tilde{K}_N^{-\chi}$. The derivative rates follow by the Markov inequality for splines,

$\|\partial_1 g\|_{L^\infty} \leq C\tilde{K}_N\|g\|_{L^\infty}$ applied to the spline error, together with the derivative approximation bound of Lemma D.1.

□

Lemma D.4 (generated-regressor perturbation).

$$\begin{aligned} \max_p \|\tilde{b}(\cdot)'(\hat{\delta}_p - \tilde{\delta}_p)\|_{L^\infty(\mathcal{S})} &= O_P\left(K_N\rho_N + \tilde{K}_N\sqrt{\log N/N} \cdot o_P(1)\right) \\ &= O_P(K_N\rho_N) + o_P\left(\tilde{K}_N\sqrt{\log N/N}\right). \end{aligned}$$

Proof. The generated problem regresses Y_i on $\hat{\mathbf{b}}_i$. Decompose $Y_i = \hat{\mathbf{b}}_i'\delta^* + \hat{a}_i + b_i + \varepsilon_i$ with $\hat{a}_i$ the approximation error at $\hat{S}_i$ and $b_i := W_i'(\beta_{ts}(S_i) - \beta_{ts}(\hat{S}_i))$, $|b_i| \lesssim \Delta_N$ by the Lipschitz continuity of β_{ts} on $\mathcal{S}^+$. Then $\hat{\delta} - \tilde{\delta} = \hat{\mathbb{Q}}_N^+ \mathbb{P}_N[T^t \hat{\mathbf{b}}(\varepsilon + \hat{a} + b)] - \mathbb{Q}_N^+ \mathbb{P}_N[T^t \mathbf{b}(\varepsilon + a)]$, and the difference splits into (i) the mean-model error b : a contrast bounded by $C\Delta_N = O_P(K_N\rho_N)$; (ii) the design perturbation: by Lemma D.2(iii) a relative $o_P(1)$ change, which multiplies the oracle variance rate; (iii) the noise cross term of the form $(\hat{\mathbb{Q}}_N^+ \mathbb{P}_N[T^t(\hat{\mathbf{b}} - \mathbf{b})\varepsilon])$: the class $\{\tilde{b}_j(\cdot + \eta) : \|\eta\| \leq CK_N\rho_N\}$ is a fixed-dimensional Lipschitz-parametrized spline class, so the centered process is $O_P(\tilde{K}_N\sqrt{\log N/N})$ times the modulus $\min(1, \tilde{K}_N K_N\rho_N) = o_P(1)$; (iv) the approximation-error shift $\hat{a} - a$, bounded by $\tilde{K}_N^{-\chi}$ -level terms already counted. Summing gives the claim. □

Proof of Theorem 5. Combining Lemmas D.3 and D.4,

$$\|\hat{\beta}_{ts,p} - \beta_{ts,p}\|_{L^\infty(\mathcal{S})} = O_P\left(\tilde{K}_N\sqrt{\log N/N} + \tilde{K}_N^{-\chi} + K_N\rho_N\right),$$

and applying the spline Markov inequality to the estimation error plus Lemma D.1 to the bias, the derivative rate carries one extra factor $\tilde{K}_N$ in all three terms, with the passthrough term $\tilde{K}_N K_N\rho_N$. For the functionals: at a point (z, w) whose arguments lie in $\mathcal{S}$, $\hat{\mu}_{ts}(z, w) - \mu_{ts}(z, w)$ is a fixed linear combination, with bounded weights, of $\hat{\beta}_{ts}(\hat{P}^t(z), \tilde{z}^{t,r}) - \beta_{ts}(P_o^t(z), \tilde{z}^{t,r})$ and the corresponding derivative difference weighted by $\hat{P}^t(z)$. Each difference splits into the estimation error of the fitted function at the estimated argument, bounded by the uniform rates above, and the movement of the argument, bounded by the Lipschitz continuity of β_{ts} and $\partial_1 \beta_{ts}$ times $|\hat{P}^t(z) - P_o^t(z)| = O_P(K_N\rho_N)$; the derivative difference dominates. The same holds for $\hat{\tau}_{ts}$ with the roles of (t, s) reversed and the correction terms included, giving the stated uniform rates at the derivative order. The Lipschitz continuity of $\partial_1 \beta_{ts}$ uses $\chi \geq 2$, and the bounded

weights use $|\hat{P}^t| \leq 1$ together with boundedness of w ; the evaluation of $\hat{P}^t(z)$ at nonrandom z invokes the interior uniform rate of Theorem 3 on the region carrying the suprema. $\square$

E Additional Results and Discussion

E.1 Semiparametric Identification of Aggregate Effects

To see what our specification retains, expand the selected-outcome regression formally in covariate interactions of increasing order:

$$m_{ts}(P_i^t, Z_i^{t,r}, W_i) = W_i' \beta_{ts}(P_i^t, \tilde{Z}_i^{t,r}) + \sum_{q=2}^{\infty} (W_i^{\otimes q})' \beta_{ts}^{(q)}(P_i^t, \tilde{Z}_i^{t,r}), \quad (\text{E.1})$$

where $W_i^{\otimes q}$ is formed by degree- q monomials in the nonconstant covariates and $\beta_{ts}^{(q)}(\cdot)$ denotes the associated vector of coefficient functions. Suppose that second and higher-order terms are orthogonalized to the first term. The specification (7) retains the first-order term, while retaining all terms would amount to fully nonparametric estimation, which the dimensionality of W_i rules out. The omitted terms need not vanish, so (7) can be viewed as a truncation of (E.1) rather than an exact relationship.

Under this interpretation, we can define $\beta_{ts}(\cdot)$ as the projection of the outcome onto the first-order function class, that is, the function minimizing $\mathbb{E}[(Y_i - W_i' g(P_i^t, \tilde{Z}_i^{t,r}))^2 | T_i = t]$ over $g(\cdot)$, characterized by the orthogonality conditions

$$\mathbb{E}[(Y_i - W_i' \beta_{ts}(P_i^t, \tilde{Z}_i^{t,r})) W_i h(P_i^t, \tilde{Z}_i^{t,r}) | T_i = t] = 0 \quad \text{for all functions } h(\cdot). \quad (\text{E.2})$$

The analogous construction on the s -treatment subsample defines $\beta_{st}(\cdot)$. When (7) holds exactly, the projection above yields the true coefficient functions.

Proposition 1 concerns the true selected-outcome regressions, so replacing them with projections requires an assumption on the error. Define the projection error $\nu_{ts}(p, z^{t,r}, w) = m_{ts}(p, z^{t,r}, w) - w' \beta_{ts}(p, \tilde{z}^{t,r})$, where $\tilde{z}^{t,r} = z^{t,r} - \mathbb{E}[Z_i^{t,r} | W_i = w]$ is the value taken by $\tilde{Z}_i^{t,r}$.

Assumption A.1. (Propensity-Invariant Projection Error). *For each ordered pair (t, s) , the projection error does not vary with the propensity argument, i.e., $\nu_{ts}(p, z^{t,r}, w) = \nu_{ts}(z^{t,r}, w)$ for all $(p, z^{t,r}, w)$ in the support.*

Assumption A.1 states that dependence of the selected-outcome regression on the propensity is captured entirely by the first-order block. Higher-order covariate interactions may be present, but they shift the level of the regression rather than its response to the treatment propensity. In terms of the expansion (E.1), the assumption holds when the higher-order coefficient functions $\beta_{ts}^{(q)}(\cdot)$, $q \geq 2$, do not vary with the propensity argument. The assumption is also testable: augmenting the specification with second-order covariate interactions whose coefficients vary with $(P_i^t, \tilde{Z}_i^{t,r})$ and testing whether the propensity-varying components of the added coefficients are zero provides a direct check.

Under Assumption A.1, the derivative identity (8) continues to hold under the projection interpretation of $\beta_{ts}(\cdot)$, and similarly on the (s, t) margin. The correction terms in (3) are therefore consistently estimable even when the first-order projection misstates the level of the regression.

The remaining discrepancy is the level error $\nu_{ts}(\cdot)$, which the aggregation implicit in our parameters of interest can average out. Substituting the projection into Proposition 1 yields the plug-in quantity

$$\mu_{ts}^{\text{vc}}(z, w) = [w' \beta_{ts}(p, a) + pw' \partial_p \beta_{ts}(p, a)]_{p=P^t(z), a=\tilde{z}^{t,r}}.$$

Proposition E.1 (Semiparametric Identification of Covariate-Weighted Average Effects). *Suppose the assumptions of Proposition 1 and Assumption A.1 hold, and let $\omega(W_i) = \lambda' W_i$ for a fixed vector λ . Then, provided the moments exist,*

$$\mathbb{E}[\omega(W_i) \mu_{ts}^{\text{vc}}(Z_i, W_i)] = \mathbb{E}[\omega(W_i) \mu_{ts}(Z_i, W_i)],$$

and consequently the same equality holds for the margin-specific effect function $\tau_{ts} = \mu_{ts} - \mu_{st}$. Moreover, the equality extends to weights of the form $\lambda' W_i h(P_i^t, \tilde{Z}_i^{t,r})$ for functions $h(\cdot)$.

The proof follows from the orthogonality conditions and Assumption A.1.⁷ Choosing $\omega(W_i)$ to be a binary case characteristic, or the constant, shows that cell-level average effects can be identified even when the varying-coefficient functions misspecify

⁷Under Assumption A.1, the error of the plug-in quantity reduces to $-\nu_{ts}$, and by the law of iterated expectations, the population expectation of the weighted error is an expectation over the t -treatment subsample after reweighting by $1/P^t(Z_i)$, a function of the propensity argument, so (E.2) applies.

the level. Proposition E.1 thus delimits what the second stage identifies without further functional-form commitments. Covariate-weighted averages of margin-specific effects, including the cell averages reported in Section 6.3, remain identified when (7) is read as a first-order projection, whereas effects at individual covariate values require the specification to hold exactly. The empirical conclusions of Section 6 rest on the former.

E.2 Two-Step versus Joint Estimation

The joint likelihood of (Y, T) given X factors as $l(y, t | x) = l(y | t, x)l(t | x)$, where $l(y | t, x)$ is the outcome factor and $l(t | x)$ is the treatment factor. The index coefficients γ and the link $P(\cdot)$ parametrize the treatment factor $l(t | x)$. Whether joint estimation sharpens γ beyond the first-stage efficiency bound in Theorem 4 relies on what is maintained for the outcome mean, and we distinguish two cases.

Suppose first that $\mathbb{E}[Y | T, W, Z]$ is left unrestricted. Then the outcome factor carries no information about γ , for two reasons. First, the only γ -score the outcome carries is the mean shift γ induces. Under a free mean this is a perturbation of the mean itself, an element of the outcome nuisance tangent. Residualizing against that tangent therefore annihilates it. Second, the treatment-factor score $s_\gamma = \partial_\gamma \log l(T | X)$ is orthogonal to the whole outcome nuisance tangent. Every outcome score a is mean-zero given (T, X) , so $\mathbb{E}[s_\gamma a] = \mathbb{E}[s_\gamma \mathbb{E}[a | T, X]] = 0$, and the outcome-nuisance projection leaves s_γ intact. The efficient score for γ thus reduces to its first-stage form, two-step and joint share the same influence function, and the SMLE is efficient in the joint model.

Now impose the maintained varying-coefficient model

$$\mathbb{E}[Y | T = t, W, Z] = W' \beta_{ts}(P^t(\gamma' X), \tilde{Z}^{t,r}), \quad t \in \{d, c, n\},$$

with an error satisfying $\mathbb{E}[\varepsilon | T, W, Z] = 0$. The index now enters the outcome non-redundantly, and the restriction over-identifies γ . Both arguments of the varying coefficients carry γ : the focal propensity $P^t(\gamma' X)$ and the relative index residual $\tilde{Z}^{t,r} = \tilde{Z}^t - \tilde{Z}^r$. Conditional on $T = t$, the outcome score for γ runs through the mean shift

$$h = \partial_\gamma \left[W' \beta_{ts}(P^t, \tilde{Z}^{t,r}) \right] = W' \left[\partial_P \beta_{ts} \partial_\gamma P^t(\gamma' X) + \partial_Z \beta_{ts} \partial_\gamma \tilde{Z}^{t,r} \right],$$

with $\partial_Z \beta_{ts}$ the partial in the relative-index residual argument. Each channel is bilinear in (W, X) . h is therefore quadratic in W . The error takes up $h - \mathbb{E}[h|T = t, W, Z]$, the part that depends on the judge beyond the index residual. The varying coefficients $\beta_{ts}(\cdot)$ take up the linear-in- W part of $\mathbb{E}[h|T = t, W, Z]$ and the discrete square $W_b^{\otimes 2}$ collapses to a linear function of W_b . The remainder is reached by neither and is the over-identifying content, living in the continuous quadratic block $W_a^{\otimes 2}$ and $W_a \otimes W_b$. The SMLE is efficient for γ within the first-stage model but relies on limited information in the full system. The joint bound for γ is then weakly tighter, strictly so when this block is non-degenerate. Even then, the gain is not available to the two-step procedure itself. The second stage fits the varying coefficients by least squares, which does not use this over-identifying content. A least-squares objective would deliver it only under the exact conditional mean with homoscedastic errors, and in general realizing the gain requires the one-step correction below or full joint estimation.

This verdict takes the varying-coefficient specification as the exact conditional mean of the outcome given (T, W, Z) . Under the projection interpretation of Section E.1, the outcome mean is unrestricted and $\beta_{ts}(\cdot)$ is defined by the orthogonality conditions (E.2). The decomposition of h then changes. The error can absorb only the part of h orthogonal to the function class $\{w'g(P^t, \tilde{Z}^{t,r})\}$, which leaves the projection of h onto this class, and the varying coefficients take up that projection entirely. No component of the outcome score for γ survives orthogonalization against the nuisance tangent. The over-identifying content, and with it any efficiency gain, is therefore specific to taking the specification as an exact conditional mean. Under the projection interpretation the outcome factor carries no information about γ , and two-step and joint estimation coincide asymptotically, as in the unrestricted case above.

This block is degenerate, and any gain vanishes, when the varying coefficients are constant in their arguments or the continuous covariates W_a are absent. Whether a gain transmits to the structural estimands $\tau_{ts}(\cdot)$ depends on how they load on this direction, and where they do not the two-step remains efficient for them. Write s_γ^{out} for the outcome efficient score for γ , the outcome-factor score residualized against the outcome nuisance tangent. Regardless of its form, s_γ^{out} is mean-zero given (T, X) and hence orthogonal to the first-stage efficient score s_γ^{eff} . The joint efficient score is therefore $s_\gamma^{\text{eff}} + s_\gamma^{\text{out}}$, and the joint information is $I_{\gamma\gamma}^{\text{eff}} + \text{Var}(s_\gamma^{\text{out}})$. The link P enters the outcome only as an argument of the unrestricted functions. Its outcome-factor score is therefore absorbed, and it contributes nothing beyond the first stage. Since

$\hat{\gamma}$ already zeroes the empirical first-stage score, the one-step Newton update to the first-stage estimate is

$$\hat{\gamma}^+ = \hat{\gamma} + \left(I_{\gamma\gamma}^{\text{eff}} + \text{Var}(s_{\gamma}^{\text{out}}) \right)^+ \frac{1}{N} \sum_{i=1}^N s_{\gamma}^{\text{out}}(Y_i, T_i, X_i),$$

renormalized to Γ , which attains the joint bound. The update is finite-dimensional. The cost lies in forming s_{γ}^{out} , which requires the nonparametric outcome nuisances. The correction is therefore semiparametric and costs about what joint estimation costs.

The case for the sequential procedure is primarily computational efficiency. Joint estimation maximizes a single likelihood over γ , the link $P(\cdot)$, the varying coefficients $\beta_{ts}(\cdot)$, and the error distribution at once, coupling one finite-dimensional and three infinite-dimensional objects in a nonconvex program. The two-step procedure splits this into two smaller and separately well-posed problems, each with its own tuning and diagnostics. The SMLE recovers (γ, P) on its own, and the varying coefficients follow from a flexible regression of Y on $(P^t, \tilde{Z}^{t,r}, W)$ within the t -treatment subgroup. Estimating γ from the first stage alone does not lean on the outcome's functional form. The two-step therefore remains consistent for γ even when the varying-coefficient model is misspecified, whereas the one-step and joint estimators, built on it, need not be. Relaxing it to an unrestricted mean leaves no over-identifying content and returns the first case, where two-step and joint estimation coincide asymptotically.

F Additional Empirical Results

Figures 5 and 6 report the cell averages of Section 6.3 with the second stage estimated on the interactive-effects logit first stage.

G Technical Lemmas

Lemma G.1. *The following inequalities hold for all $\theta, \tilde{\theta} \in \Theta$:*

$$\mathbb{L}^2(\theta, \tilde{\theta}) \leq 4\mathbb{H}^2(\theta, \tilde{\theta}) \leq 4\text{KL}(\theta \parallel \tilde{\theta}).$$

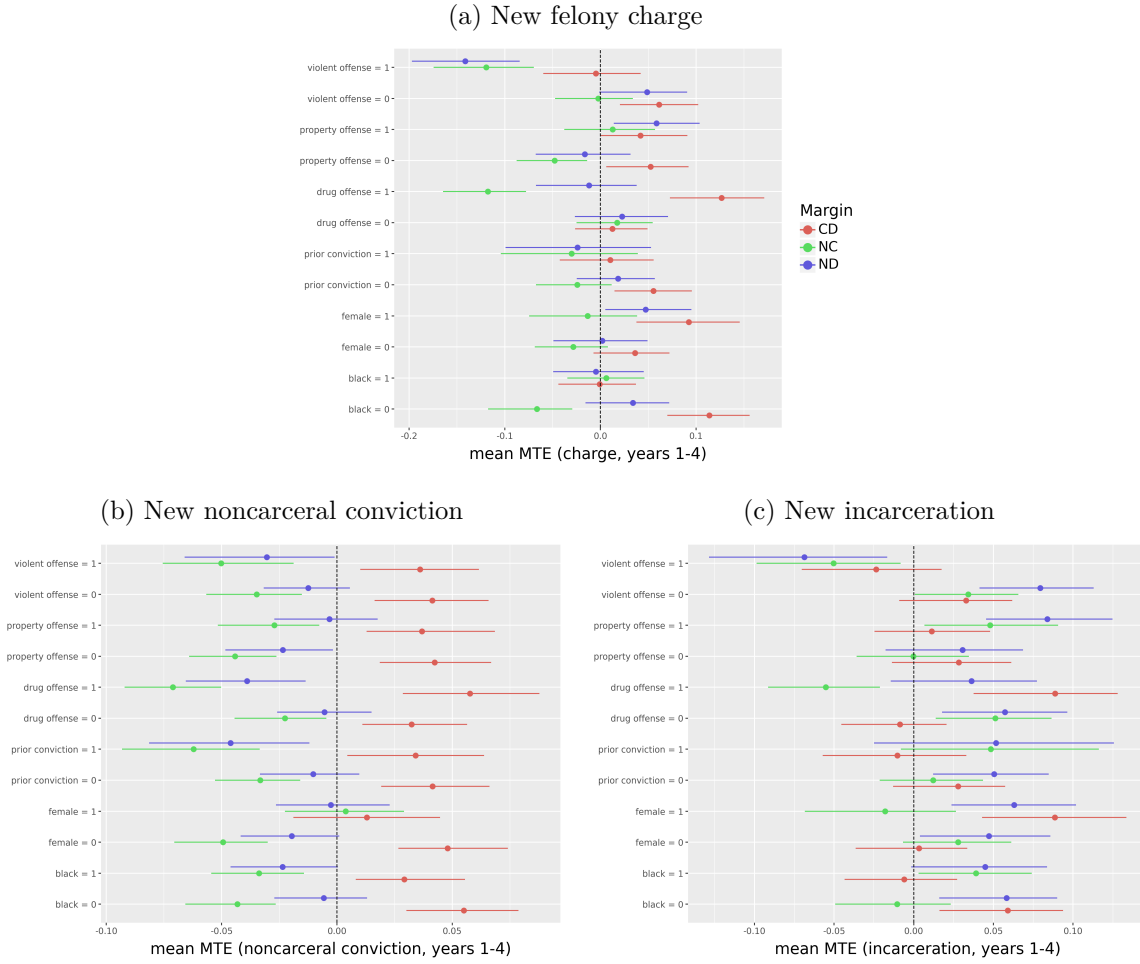
Proof of Lemma G.1. First inequality. Let $\theta = (\gamma, f)$ and $\tilde{\theta} = (\tilde{\gamma}, \tilde{f})$. Since $P^t(\cdot | f)$ is

Table 6: Judge Randomization: Correlation of Judge-Specific Coefficients with Case Characteristics

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
<i>Panel A: Conviction margin</i>									
(0)	0.005	0.004	-0.013	0.005	0.012	0.003	-0.002	-0.009	0.005
(1)	-0.003	0.008	0.006	0.007	-0.011	0.002	0.002	-0.002	0.000
(2)	0.005	-0.004	-0.001	-0.007	0.011	0.017	-0.001	0.008	0.017
(3)	-0.000	-0.002	0.009	0.001	-0.009	-0.012	0.011	-0.008	-0.007
(4)	-0.002	0.007	-0.007	0.005	-0.006	-0.022	-0.008	-0.011	-0.021
(5)	-0.004	0.001	-0.001	0.000	-0.003	-0.010	-0.002	-0.008	-0.008
(6)	-0.002	-0.003	0.001	-0.008	0.010	0.003	0.002	-0.001	0.002
(7)	-0.001	0.007	-0.004	0.015	-0.013	-0.006	0.007	0.007	-0.005
(8)	-0.010	0.010	-0.004	0.007	-0.020	-0.032	-0.010	-0.016	-0.035
(9)	0.000	0.002	-0.001	0.007	-0.010	-0.009	-0.003	-0.001	-0.010
<i>Panel B: Incarceration margin</i>									
(0)	0.009	-0.008	0.002	-0.006	0.019	0.030	0.010	0.026	0.028
(1)	-0.005	0.005	-0.002	0.002	-0.007	-0.012	-0.005	-0.011	-0.015
(2)	0.009	-0.005	-0.007	-0.016	0.028	0.033	-0.011	0.004	0.031
(3)	0.004	-0.002	0.000	0.004	-0.001	-0.006	0.005	-0.006	-0.004
(4)	0.004	0.001	-0.000	0.005	0.002	0.012	-0.004	0.005	0.009
(5)	0.005	-0.001	0.005	0.010	-0.007	0.018	0.006	0.003	0.018
(6)	0.002	0.000	-0.001	-0.000	0.004	0.007	0.005	0.002	0.006
(7)	-0.001	0.002	-0.002	0.002	-0.001	-0.002	0.003	0.006	-0.004
(8)	-0.005	0.004	0.001	0.007	-0.017	-0.022	-0.005	-0.002	-0.022
(9)	0.001	-0.004	0.002	0.002	-0.004	-0.004	0.000	0.008	-0.003

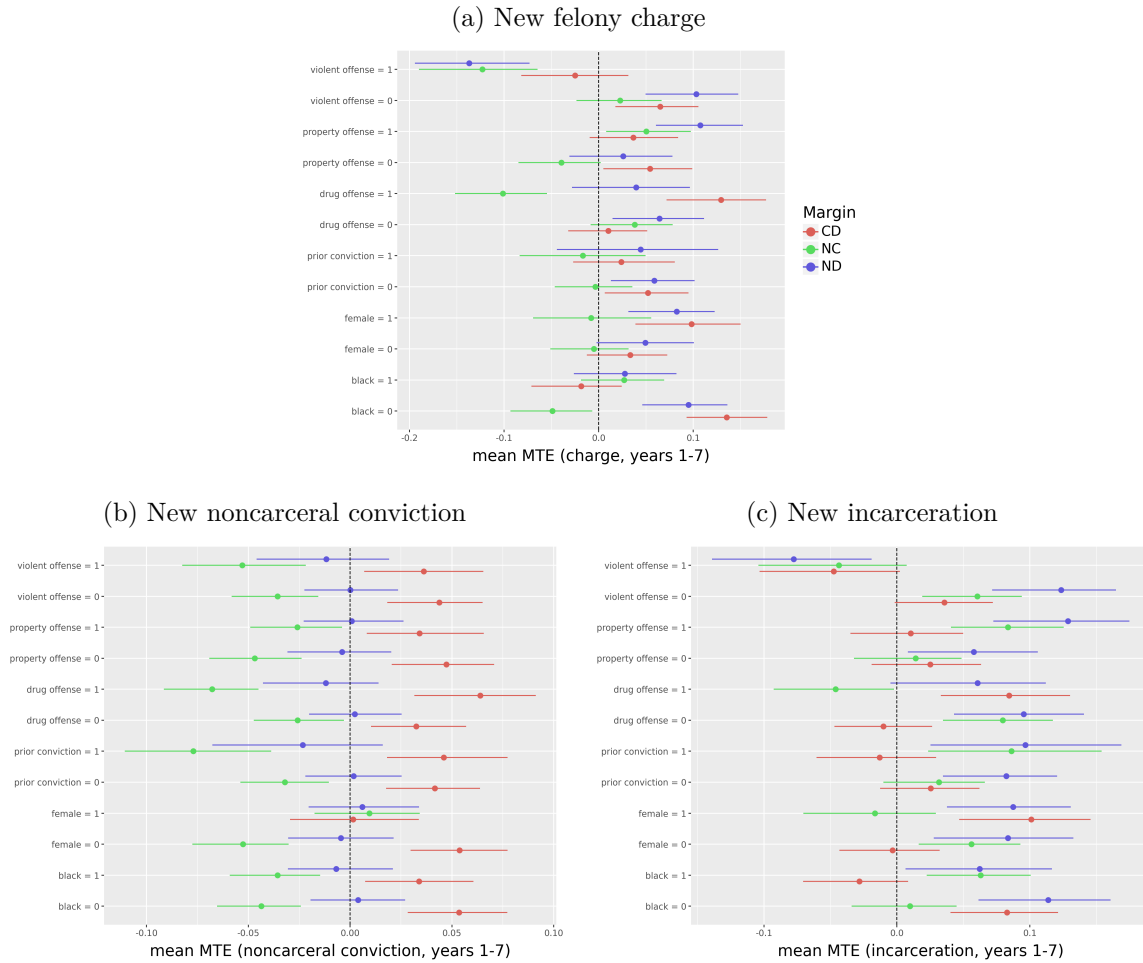
Notes: Each entry is the sample correlation, across the 256,475 cases, between a judge-specific coefficient (row) and a case characteristic (column). Circuit fixed effects are partialled out of both the coefficients and the characteristics, so the correlations are within circuit. Rows and columns share the same indexing: (1) Female; (2) Black; (3) Any violent charge; (4) Any property charge; (5) Any drug charge; (6) Prior conviction; (7) Number of charges; (8) Misdemeanor charges; (9) Time since prior conviction. Row (0) is the intercept, which has no column counterpart.

Figure 5: Cell Average Effects under the Interactive-Effects Logit First Stage: Cumulative Years 1 to 4



Notes: As in Figure 3 but with the second stage estimated on the interactive-effects logit first stage. Point estimates only.

Figure 6: Cell Average Effects under the Interactive-Effects Logit First Stage: Cumulative Years 1 to 7



Notes: As in Figure 4 but with the second stage estimated on the interactive-effects logit first stage. Point estimates only.

uniformly bounded by 1, it holds that

$$\begin{aligned}
& \sum_{t \in \{d, c, n\}} (P^t(\gamma' X | f) - P^t(\tilde{\gamma}' X | \tilde{f}))^2 \\
&= \sum_{t \in \{d, c, n\}} (\sqrt{P^t(\gamma' X | f)} - \sqrt{P^t(\tilde{\gamma}' X | \tilde{f})})^2 (\sqrt{P^t(\gamma' X | f)} + \sqrt{P^t(\tilde{\gamma}' X | \tilde{f})})^2 \\
&\leq 4 \sum_{t \in \{d, c, n\}} (\sqrt{P^t(\gamma' X | f)} - \sqrt{P^t(\tilde{\gamma}' X | \tilde{f})})^2.
\end{aligned}$$

Taking expectation on both sides with respect to the distribution of X yields that $\mathbb{L}(\theta, \tilde{\theta})^2 \leq 4\mathbb{H}(\theta, \tilde{\theta})^2$.

Second inequality. This direction follows from the standard inequality $\mathbb{H}^2(p, q) \leq \text{KL}(p||q)$ for any two distributions p and q . $\square$

For our purposes, it is convenient to think of the mapping $f \mapsto P^t(z | f)$, $t \in \{d, c, n\}$ as defined on L^1 , which gives rise to the signed extension of $P^t(z | f)$ for f that are not necessarily nonnegative or integrating to one:

$$\begin{aligned}
P^d(z | f) &= \int_{-\infty}^{-z^c} \int_{-\infty}^{-z^n} f(u^c, u^n) du^c du^n, \\
P^c(z | f) &= \int_{-z^c}^{\infty} \int_{-\infty}^{z^c - z^n + u^c} f(u^c, u^n) du^n du^c, \\
P^n(z | f) &= \int_{-z^n}^{\infty} \int_{-\infty}^{z^n - z^c + u^n} f(u^c, u^n) du^c du^n.
\end{aligned} \tag{G.1}$$

The following lemma provides analytic expressions for the first and second derivatives of $P^t(z | f)$ with respect to z .

Lemma G.2. *Let f be an integrable function on $\mathbb{R}^2$ satisfying $|f(u)| \leq C(1 + \|u\|)^{-1-\epsilon}$ for all $u \in \mathbb{R}^2$ and some constants $C > 0$ and $\epsilon > 0$. Then the extended treatment propensities are continuously differentiable and admit the following partial derivative*

formulas for $t \in \{d, c, n\}$:

$$\begin{aligned} D_1 P^d(z|f) &= - \int_{-\infty}^{-z^n} f(-z^c, u^n) du^n, \\ D_2 P^d(z|f) &= - \int_{-\infty}^{-z^c} f(u^c, -z^n) du^c, \\ D_2 P^c(z|f) &= - \int_{-z^c}^{\infty} f(u^c, z^c - z^n + u^c) du^c, \\ D_1 P^n(z|f) &= - \int_{-z^n}^{\infty} f(z^n - z^c + u^n, u^n) du^n. \end{aligned}$$

The remaining derivatives can be obtained by the identity $P^d(z|f) + P^c(z|f) + P^n(z|f) = \int_{\mathbb{R}^2} f(u) du$. Moreover, if f is continuously differentiable and $\|\nabla f(u)\| \leq C(1 + \|u\|)^{-1-\epsilon}$, then the extended treatment propensities are twice continuously differentiable and admit the following second-order partial derivative formulas for $t \in \{d, c, n\}$:

$$\begin{aligned} D_{11} P^d(z|f) &= \int_{-\infty}^{-z^n} D_1 f(-z^c, u^n) du^n, \\ D_{12} P^d(z|f) &= f(-z^c, -z^n), \\ D_{22} P^d(z|f) &= \int_{-\infty}^{-z^c} D_2 f(u^c, -z^n) du^c, \\ D_{12} P^c(z|f) &= -f(-z^c, -z^n) - \int_{-z^c}^{\infty} D_2 f(u^c, z^c - z^n + u^c) du^c, \\ D_{22} P^c(z|f) &= \int_{-z^c}^{\infty} D_2 f(u^c, z^c - z^n + u^c) du^c, \\ D_{11} P^n(z|f) &= \int_{-z^n}^{\infty} D_1 f(z^n - z^c + u^n, u^n) du^n, \\ D_{12} P^n(z|f) &= -f(-z^c, -z^n) - \int_{-z^n}^{\infty} D_1 f(z^n - z^c + u^n, u^n) du^n. \end{aligned}$$

Proof of Lemma G.2. Throughout, write $\alpha = 1 + \epsilon$, so the hypotheses read $|f(u)| \leq C(1 + \|u\|)^{-\alpha}$ and $|D_l f(u)| \leq C(1 + \|u\|)^{-\alpha}$ for $l = 1, 2$. Both bounds give absolutely integrable integrands in every formula below, justifying the Leibniz interchange.

First-order derivatives. Start from the iterated representation (G.1) of P^d :

$$P^d(z|f) = \int_{-\infty}^{-z^c} F(u^c, z^n) du^c, \quad F(u^c, z^n) = \int_{-\infty}^{-z^n} f(u^c, u^n) du^n.$$

Only the upper limit depends on z^c , so by Leibniz's rule

$$D_1 P^d(z|f) = -F(-z^c, z^n) = -\int_{-\infty}^{-z^n} f(-z^c, u^n) du^n.$$

For $D_2 P^d$, only the upper limit of the inner integral F depends on z^n . The decay bound on f produces an integrable majorant in u^c uniformly in a neighborhood of z^n , so a second application of Leibniz gives

$$D_2 P^d(z|f) = \int_{-\infty}^{-z^c} \partial_{z^n} F(u^c, z^n) du^c = -\int_{-\infty}^{-z^c} f(u^c, -z^n) du^c.$$

The formulas for $D_2 P^c$ and $D_1 P^n$ follow from the same argument applied to the iterated representations of P^c and P^n in (G.1).

Second-order derivatives. Differentiating the first-order formulas a second time, in each case the integrand depends on the differentiation variable both through the upper limit of integration and through the argument of f . For $D_{11} P^d$, only the argument depends on z^c :

$$D_{11} P^d(z|f) = D_1 \left(-\int_{-\infty}^{-z^n} f(-z^c, u^n) du^n \right) = \int_{-\infty}^{-z^n} D_1 f(-z^c, u^n) du^n,$$

the chain rule contributing the sign change. For $D_{12} P^d$, only the upper limit of the integral for $D_2 P^d$ depends on z^c , contributing a boundary term:

$$D_{12} P^d(z|f) = D_1 \left(-\int_{-\infty}^{-z^c} f(u^c, -z^n) du^c \right) = f(-z^c, -z^n).$$

The remaining displayed identities (for $D_{22} P^d$, $D_{12} P^c$, $D_{22} P^c$, $D_{11} P^n$, $D_{12} P^n$) follow from the same combination of an argument-derivative term and a boundary term. Twice continuous differentiability of $P^t(\cdot|f)$ follows from continuity of f and ∇f together with the dominated-convergence theorem applied to each Leibniz step. $\square$

Lemma G.3. *Let $f \in \mathcal{F}$. Then $G(\cdot|f)$ is C^2 on $\mathbb{R}^2$, with*

$$\nabla_z G(z|f) = (P^c(z|f), P^n(z|f)) = P(z|f), \quad \nabla_z^2 G(z|f) = DP(z|f),$$

where $DP(z|f)$ is the Jacobian matrix of the propensity function $z \mapsto P(z|f)$. Moreover, $DP(z|f)$ is symmetric and positive definite for all $z \in \mathbb{R}^2$.

Proof of Lemma G.3. We split the argument into the gradient identity, the Hessian identity together with its symmetry, and the positive-definiteness claim.

Step 1 (Gradient): For each fixed $u \in \mathbb{R}^2$, the integrand

$$\Phi(z, u) = \max_{t \in \{d, c, n\}} (z^t + u^t)$$

is convex and piecewise linear in z . Off the boundary where the argmax changes, whose Lebesgue-measure is zero in u at any fixed z , $\Phi(\cdot, u)$ is differentiable with $\partial_{z^c} \Phi(z, u) = \mathbb{1}\{z^c + u^c > z^t + u^t, t \neq c\}$. Because f is absolutely continuous with respect to Lebesgue measure, the boundary set has f -measure zero. The difference quotient $h^{-1}[\Phi(z + he_c, u) - \Phi(z, u)]$ is bounded by 1 in absolute value uniformly in u and converges, as $h \rightarrow 0$, to $\partial_{z^c} \Phi(z, u)$ for f -almost every u . Dominated convergence with the constant majorant 1 yields

$$\frac{\partial G(z|f)}{\partial z^c} = \int_{\mathbb{R}^2} \mathbb{1}\{z^c + u^c > z^t + u^t, t \neq c\} f(u) du = P^c(z|f),$$

and analogously $\partial G(z|f)/\partial z^n = P^n(z|f)$. Hence $\nabla_z G(z|f) = (P^c(z|f), P^n(z|f))$ on $\mathbb{R}^2$.

Step 2 (Hessian and symmetry): By definition of $\mathcal{F}$, every $f \in \mathcal{F}$ satisfies the decay bound of Lemma G.2. Consequently, $(P^c(\cdot|f), P^n(\cdot|f))$ is C^1 on $\mathbb{R}^2$ with bounded derivatives. Combined with $\nabla_z G(\cdot|f) = (P^c(\cdot|f), P^n(\cdot|f))$, this implies that $G(\cdot|f)$ is C^2 with $\nabla_z^2 G(z|f)$ equal to $DP(z|f)$. Symmetry of $DP(z|f)$ follows from the equality of mixed partials of the C^2 function G .

Step 3 (Positive definiteness):

Using $P^c = 1 - P^d - P^n$ and Lemma G.2, we have

$$\begin{aligned} DP_{11}(z|f) &= D_1 P^c(z|f) = -D_1 P^d(z|f) - D_1 P^n(z|f) \\ &= \int_{-\infty}^{-z^n} f(-z^c, u^n) du^n + \int_{-z^n}^{\infty} f(z^n - z^c + u^n, u^n) du^n. \end{aligned}$$

The change of variable $u^n = z^n - z^c + u^c$ recasts the second integral as

$$\int_{-z^c}^{\infty} f(u^c, z^c - z^n + u^c) du^c = -D_2 P^c(z|f) = |DP_{12}(z|f)|,$$

where the last equality uses that $DP_{12} = D_2P^c \leq 0$ by Lemma G.2. Combining,

$$DP_{11}(z|f) - |DP_{12}(z|f)| = \int_{-\infty}^{-z^n} f(-z^c, u^n) du^n > 0,$$

the strict positivity following from $f > 0$ almost everywhere on $\mathbb{R}^2$. The symmetric argument gives $DP_{22}(z|f) - |DP_{21}(z|f)| > 0$. Hence each Gershgorin disc of $DP(z|f)$, centered at the diagonal DP_{ii} in each i th row with radius $\sum_{j \neq i} |DP_{ij}|$, lies strictly in the open right half-plane of $\mathbb{C}$. By the Gershgorin's circle theorem, every eigenvalue of $DP(z|f)$ is contained in one of these discs and is therefore real (by symmetry of DP) and strictly positive. Consequently $DP(z|f)$ is positive definite at every $z \in \mathbb{R}^2$. $\square$

Lemma G.4 (Injectivity of treatment propensities). *Let $f \in \mathcal{F}$. Then the induced propensity function $z \mapsto (P^c(z|f), P^n(z|f))$ is injective on $\mathbb{R}^2$.*

Proof of Lemma G.4. By Lemma G.3, $G(\cdot|f)$ is C^2 on $\mathbb{R}^2$ with Hessian $\nabla_z^2 G(z|f) = DP(z|f)$ positive definite at every z . A C^2 function whose Hessian is positive definite everywhere is strictly convex, and the strict convexity of $G(\cdot|f)$ in turn implies strict monotonicity of its gradient: for any $x, y \in \mathbb{R}^2$ with $x \neq y$,

$$(x - y)'(\nabla_z G(x|f) - \nabla_z G(y|f)) > 0.$$

Since $\nabla_z G(z|f) = (P^c(z|f), P^n(z|f))$ by Lemma G.3, $z \mapsto (P^c(z|f), P^n(z|f))$ is injective on $\mathbb{R}^2$. $\square$

Lemma G.5 (P_o is an open map). *The propensity map P_o is an open map on $\mathbb{R}^2$ and a homeomorphism onto its image. Consequently, for any $A_1, A_2 \subseteq \mathbb{R}^2$,*

$$P_o(\text{Int}A_1 \cap \text{Int}A_2) = \text{Int}P_o(A_1) \cap \text{Int}P_o(A_2).$$

Proof of Lemma G.5. By Lemma G.3, the Jacobian $DP_o(z) = DP(z|f_o)$ is positive definite, hence nonsingular, at every $z \in \mathbb{R}^2$. The inverse function theorem then makes P_o a local diffeomorphism at each point, so it maps open sets to open sets and is therefore an open map. Being injective by Lemma G.4, P_o is a continuous open bijection onto its image, hence a homeomorphism onto it, and so preserves interiors: $P_o(\text{Int}A) = \text{Int}P_o(A)$ for every $A \subseteq \mathbb{R}^2$. Injectivity also gives $P_o(\text{Int}A_1 \cap \text{Int}A_2) = P_o(\text{Int}A_1) \cap P_o(\text{Int}A_2)$, and combining the two displays proves the claim. $\square$

Lemma G.6 (Positive semidefiniteness and injectivity over $\mathcal{P}$). *For every $P \in \mathcal{P}$, the Jacobian $DP(z)$ is symmetric and positive semidefinite at every $z \in \bar{\mathcal{Z}}$, and is positive definite at every z at which it is nonsingular. Moreover, if $DP(z_0) > 0$ at a point $z_0 \in \bar{\mathcal{Z}}$, then $P(z_0) \neq P(z)$ for every $z \in \bar{\mathcal{Z}}$ with $z \neq z_0$; in particular P is injective on $\{z \in \bar{\mathcal{Z}} : DP(z) > 0\}$.*

Proof of Lemma G.6. Fix $P \in \mathcal{P}$ and $z \in \bar{\mathcal{Z}}$. The cross-partial equality in the definition of $\mathcal{P}$ makes $DP(z)$ symmetric, with $DP_{12}(z) = DP_{21}(z)$. The sign and dominance conditions give $DP_{12}(z) \leq 0$, $DP_{11}(z) \geq |DP_{12}(z)|$, and $DP_{22}(z) \geq |DP_{21}(z)|$, so each Gershgorin disc of $DP(z)$, centered at the nonnegative diagonal entry $DP_{ii}(z)$ with radius $\sum_{l \neq i} |DP_{il}(z)|$, lies in the closed right half-plane of $\mathbb{C}$. By the Gershgorin circle theorem and the symmetry of $DP(z)$, every eigenvalue is real and nonnegative, so $DP(z)$ is positive semidefinite. A symmetric positive semidefinite matrix with no zero eigenvalue is positive definite, which gives the second claim. For the separation claim, fix z_0 with $DP(z_0) \geq 0$ and any $z \in \bar{\mathcal{Z}}$ with $v := z - z_0 \neq 0$. Since $\bar{\mathcal{Z}}$ is convex, $z_0 + sv \in \bar{\mathcal{Z}}$ for every $s \in [0, 1]$, and the fundamental theorem of calculus gives

$$v'(P(z) - P(z_0)) = \int_0^1 v' DP(z_0 + \vartheta v) v d\vartheta.$$

The integrand is nonnegative because $DP \geq 0$ on $\bar{\mathcal{Z}}$, is continuous in s , and equals $v' DP(z_0) v > 0$ at $s = 0$, so the integral is strictly positive. Hence $P(z) \neq P(z_0)$, and applying this with z_0 ranging over the positive-definite set gives injectivity of P there. $\square$